Video Electronics Standards Association

VESA DisplayPort Alt Mode on USB Type-C Standard

Version 1.0a

05 August, 2015

www.vesa.org

Licensed to Jay Lin (linsp@realtek.com)

Purpose

The purpose of this document is to specify the use of DisplayPort™ as an Alternate Mode for the USB Type-C™ connector.

Summary

This Standard describes the use of the DisplayPort protocol between devices that support the USB Type-C connector using USB Type-C to USB Type-C Cable Assemblies. It describes Cable assemblies that adapt the USB Type-C connector to DisplayPort connectors. It describes Cable assemblies and Adaptors to adapt DisplayPort on USB Type-C to other video protocols. It describes the simultaneous use of USB Enhanced SuperSpeed and the DisplayPort protocol on the USB Type-C connector for docking applications.

This Standard describes the USB Type-C connector pin assignments and DisplayPort signaling rates to be used in various scenarios when using the above Cable assemblies and Adaptors. It describes how the USB-PD Alternate Mode Discovery and Entry processes are applied to DisplayPort on the USB Type-C connector.

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Contents

Purpose **1**

Summary **1**

Intellectual Property 9
Trademarks 9
Patents 10
Support for this Standard 10
Acknowledgements 11
Revision History 11

Section 1 Introduction **12**

1.1 Purpose 12
1.2 Scope 12
1.3 Document Organization 13
1.4 Terms and Abbreviations 14
1.5 Conventions 17
1.5.1 Precedence 17
1.5.2 Keywords 17
1.5.3 Numbering 17
1.6 Reference Documents 18

Section 2 Overview **19**

2.1 Introduction 19
2.2 USB Type-C Receptacles, Plugs, and Cables 19
2.3 Signaling and Cable Types 20
2.4 Multi-function Systems 21

Section 3 Pin Assignments and Descriptions **22**

3.1 Pin Assignment Overview 23
3.2 USB Type-C DisplayPort Pin Assignments 24
3.2.1 Pin Assignment Support Requirements 27
3.3 DFP_D Pin Assignment A 30
3.4 DFP_D Pin Assignment B 32
3.5 DFP_D Pin Assignment C 34
3.6 DFP_D Pin Assignment D 36
3.7 DFP_D Pin Assignment E 38
3.8 DFP_D Pin Assignment F 40
3.9 UFP_D Pin Assignment A 42
3.10 UFP_D Pin Assignment B 44
3.11 UFP_D Pin Assignment C 46
3.12 UFP_D Pin Assignment D 48
3.13 UFP_D Pin Assignment E 50
3.14 USB Gen 2 PHY Layer Signaling for DisplayPort (GEN2_BR) 52

Licensed to Jay Lin (insp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.15 AUX and HPD Signaling 52
3.15.1 AUX Signaling 52
3.15.2 HPD Signaling 57
3.16 DisplayPort Connector CONFIG1 and CONFIG2 Pins 61

Section 4 USB Type-C Cables Used for DisplayPort 62
4.1 Scenario 1 USB Type-C Cable 62
4.1.1 Scenario 1 USB Type-C Cable Using GEN2_BR 63
4.1.2 Scenario 1 USB Type-C Cable using DP_BR 67
4.2 Scenarios 2a and 2b USB Type-C to DisplayPort Cables 71
4.2.1 Scenario 2a USB Type-C DFP_D to DisplayPort UFP_D 71
4.2.2 Scenario 2b DisplayPort DFP_D to USB Type-C UFP_D 75
4.2.3 USB Type-C to DisplayPort Cable Electrical Properties 79
4.2.4 USB Type-C to DisplayPort Cable Functionality 83
4.3 Scenarios 3a and 3c USB Type-C DFP_D to Protocol Converter 86
4.3.1 HDMI Protocol Converters 87
4.3.2 VGA Protocol Converters 87
4.3.3 DVI Protocol Converters 87
4.4 Scenario 4a USB Type-C DFP_D to a Multi-function Dock 88
4.4.1 Scenario 4a USB Type-C Cable Using GEN2_BR Multi-function 89
4.4.2 Scenario 4a USB Type-C Cable Using DP_BR Multi-function 93
4.5 Scenario 4b USB Type-C DFP_D to a *DockPort v1.0a* Adaptor 97
4.5.1 USB Type-C DFP_D to a *DockPort v1.0a* Adaptor Electrical Properties . . . 101
4.6 Scenarios 4c and 4d USB Type-C DFP_D to Multi-function Cradle/Captive Cable Dock 102
4.6.1 Scenarios 4c and 4d USB Type-C DFP_D to Multi-function Cradle/Captive Cable Dock Using GEN2_BR 103
4.6.2 Scenarios 4c and 4d USB Type-C DFP_D to Multi-function Cradle/Captive Cable Dock using DP_BR 104

Section 5 Discovery and USB-PD 105
5.1 Connection Discovery 105
5.2 DisplayPort Mode Discovery, Entry, and Exit 106
5.2.1 Structured VDM Header 107
5.2.2 DisplayPort Enter Mode 110
5.2.3 DisplayPort Status Update 110
5.2.4 DisplayPort Configure 112
5.2.5 Attention 114
5.2.6 DisplayPort Exit Mode 114
5.3 VDM Flows 115

Section 6 Miscellaneous 128
6.1 VCONN and VBUS 128

Licensed to Jay Lin (jmlin@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

6.2 Low-power Operation 128
6.2.1 Display Sleep 128
6.2.2 USB Suspend General Properties (Informative) 129
6.2.3 USB Type-C DFP_U 129
6.2.4 USB Type-C UFP_U that Supports a VCONN-Powered Adaptor 130
6.3 Equivalent USB Functionality 130
6.4 USB Billboard URL 130

Appendix A DisplayPort Mode Examples 131
A.1 Purpose 131
A.1.1 DisplayPort Mode Examples 131
A.2 USB Type-C USB Host to DisplayPort Video Adaptor Example 131
A.2.1 DisplayPort Video Adaptor Assumptions 132
A.2.2 Block Diagrams 133
A.2.3 Solution 135
A.3 USB Dock with DisplayPort Example 138
A.3.1 USB Dock with DisplayPort Assumptions 138
A.3.2 Block Diagram 139
A.3.3 Solution 140

Appendix B Main Contribution History 143

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Tables

Table 1: Patents 10
Table 2: Main Contributors to *DP Alt Mode on USB Type-C v1.0a* 11
Table 3: Revision History 11

Table 1-1: Terms and Abbreviations Referenced from USB Specifications 14
Table 1-2: Terms and Abbreviations 16
Table 1-3: Keywords 17
Table 1-4: Numbering 17
Table 1-5: Reference Documents 18

Table 2-1: USB Type-C to USB Type-C DisplayPort Signaling 20

Table 3-1: USB Type-C Receptacle DFP_D Pin Assignment Summary – Normal Plug Orientation . . . 24
Table 3-2: USB Type-C Receptacle DFP_D Pin Assignment Summary – Flipped Plug Orientation . . . 25
Table 3-3: USB Type-C Receptacle UFP_D Pin Assignment Summary – Normal Plug Orientation . . . 26
Table 3-4: USB Type-C Receptacle UFP_D Pin Assignment Summary – Flipped Plug Orientation . . . 26
Table 3-5: USB Type-C Receptacle DFP_D Pin Assignment Requirements 27
Table 3-6: USB Type-C Receptacle UFP_D Pin Assignment Requirements 27
Table 3-7: USB Type-C Plug DFP_D Pin Assignment Requirements (for UFP_D) 28
Table 3-8: USB Type-C Plug UFP_D Pin Assignment Requirements (for DFP_D) 28
Table 3-9: DFP_D Pin Assignment A – Normal Plug Orientation 30
Table 3-10: DFP_D Pin Assignment A – Flipped Plug Orientation 31
Table 3-11: DFP_D Pin Assignment B – Normal Plug Orientation 32
Table 3-12: DFP_D Pin Assignment B – Flipped Plug Orientation 33
Table 3-13: DFP_D Pin Assignment C – Normal Plug Orientation 34
Table 3-14: DFP_D Pin Assignment C – Flipped Plug Orientation 35
Table 3-15: DFP_D Pin Assignment D – Normal Plug Orientation 36
Table 3-16: DFP_D Pin Assignment D – Flipped Plug Orientation 37
Table 3-17: DFP_D Pin Assignment E – Normal Plug Orientation 38
Table 3-18: DFP_D Pin Assignment E – Flipped Plug Orientation 39
Table 3-19: DFP_D Pin Assignment F – Normal Plug Orientation 40
Table 3-20: DFP_D Pin Assignment F – Flipped Plug Orientation 41
Table 3-21: UFP_D Pin Assignment A – Normal Plug Orientation 42
Table 3-22: UFP_D Pin Assignment A – Flipped Plug Orientation 43
Table 3-23: UFP_D Pin Assignment B – Normal Plug Orientation 44
Table 3-24: UFP_D Pin Assignment B – Flipped Plug Orientation 45
Table 3-25: UFP_D Pin Assignment C – Normal Plug Orientation 46
Table 3-26: UFP_D Pin Assignment C – Flipped Plug Orientation 47
Table 3-27: UFP_D Pin Assignment D – Normal Plug Orientation 48
Table 3-28: UFP_D Pin Assignment D – Flipped Plug Orientation 49
Table 3-29: UFP_D Pin Assignment E – Normal Plug Orientation 50
Table 3-30: UFP_D Pin Assignment E – Flipped Plug Orientation 51

Licensed to Jay Lin (mail@jaylin.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 4-1: USB Type-C DFP_D to DisplayPort UFP_D Use 74
Table 4-2: DisplayPort DFP_D to USB Type-C UFP_D Use 78
Table 4-3: DisplayPort Connector AUX States 83
Table 4-4: DisplayPort Connector HPD and AUX States When Disconnected or in Adaptor DFP_D Configuration 85
Table 4-5: DisplayPort Connector HPD and AUX States in Adaptor UFP_D Configuration. 85
Table 4-6: USB Type-C DFP_D to *DockPort v1.0a* Receptacle 100
Table 4-7: USB Type-C Cable DisplayPort Use (Multi-function Cradle/Captive Cable Dock) GEN2_BR 103
Table 4-8: USB Type-C Cable DisplayPort Use (Multi-function Cradle/Captive Cable Dock) DP_BR 104

Table 5-1: VDM Header 107
Table 5-2: DisplayPort Capabilities (VDO in Responder Discover Modes VDM). 108
Table 5-3: DisplayPort Status 111
Table 5-4: DisplayPort Configurations 113

Table B-1: Main Contributors to *DP Alt on USB Type-C v1.0*. 143

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figures

Figure 2-1: USB Type-C Receptacle Interface (Front View) (Informative) 19
Figure 2-2: USB Type-C Plug Interface (Front View) (Informative) 19

Figure 3-1: AUX Signaling Using USB Type-C to USB Type-C Cables 53
Figure 3-2: AUX Signaling Using USB Type-C to DisplayPort Adaptors 54
Figure 3-3: AUX Signaling Using DisplayPort to USB Type-C Adaptors 55
Figure 3-4: AUX Signaling Using USB Type-C to Protocol Converters 56
Figure 3-5: HPD Signaling Architecture 57
Figure 3-6: HPD Signaling Using USB Type-C to USB Type-C Cables 60
Figure 3-7: HPD Signaling Using USB Type-C to DisplayPort Adaptors 60
Figure 3-8: HPD Signaling Using DisplayPort to USB Type-C Adaptors 61
Figure 3-9: HPD Signaling Using USB Type-C to Protocol Converters 61

Figure 4-1: Scenario 1 62
Figure 4-2: Scenario 1 GEN2_BR Cable Wiring – Normal Plug Orientation 63
Figure 4-3: Scenario 1 GEN2_BR Cable Wiring – DFP_D Plug Inverted 64
Figure 4-4: Scenario 1 GEN2_BR Cable Wiring – UFP_D Plug Inverted 65
Figure 4-5: Scenario 1 GEN2_BR Cable Wiring – DFP_D and UFP_D Plug Inverted 66
Figure 4-6: Scenario 1 DP_BR Cable Wiring – Normal Plug Orientation 67
Figure 4-7: Scenario 1 DP_BR Cable Wiring – DFP_D Plug Inverted 68
Figure 4-8: Scenario 1 DP_BR Cable Wiring – UFP_D Plug Inverted 69
Figure 4-9: Scenario 1 DP_BR Cable Wiring – DFP_D and UFP_D Plug Inverted 70
Figure 4-10: Scenario 2a 71
Figure 4-11: Scenario 2a Cable Wiring – Normal Plug Orientation 72
Figure 4-12: Scenario 2a Cable Wiring – DFP_D Plug Inverted 73
Figure 4-13: Scenario 2b 75
Figure 4-14: Scenario 2b Cable Wiring – Normal Plug Orientation 76
Figure 4-15: Scenario 2b Cable Wiring – UFP_D Plug Inverted 77
Figure 4-16: Insertion Loss Fit at Nyquist for HBR 80
Figure 4-17: Insertion Loss Fit at Nyquist for HBR2 80
Figure 4-18: Insertion Loss Fit at Nyquist for HBR3 81
Figure 4-19: IMR as a Function of ILFitAtNq at HBR3 81
Figure 4-20: IRL as a Function of ILFitAtNq at HBR3 82
Figure 4-21: Scenario 3a 86
Figure 4-22: Scenario 3c 86
Figure 4-23: Scenario 4a 88
Figure 4-24: Scenario 4a GEN2_BR MF Cable Wiring – Normal Plug Orientation 89
Figure 4-25: Scenario 4a GEN2_BR MF Cable Wiring – DFP_D Plug Inverted 90
Figure 4-26: Scenario 4a GEN2_BR MF Cable Wiring – UFP_D Plug Inverted 91
Figure 4-27: Scenario 4a GEN2_BR MF Cable Wiring – DFP_D and UFP_D Plug Inverted 92
Figure 4-28: Scenario 4a DP_BR MF Cable Wiring – Normal Plug Orientation 93
Figure 4-29: Scenario 4a DP_BR MF Cable Wiring – DFP_D Plug Inverted 94
Figure 4-30: Scenario 4a DP_BR MF Cable Wiring – UFP_D Plug Inverted 95

Licensed to Ly Lin (lin@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 4-31: Scenario 4a DP_BR MF Cable Wiring – DFP_D and UFP_D Plug Inverted 96
Figure 4-32: Scenario 4b 97
Figure 4-33: Scenario 4b *DockPort v1.0a* Receptacle Adaptor Wiring – Normal Plug Orientation 98
Figure 4-34: Scenario 4b *DockPort v1.0a* Receptacle Adaptor Wiring – Flipped Plug Orientation 99
Figure 4-35: Scenario 4c 102
Figure 4-36: Scenario 4d 102

Figure 5-1: Connect Initial Actions 115
Figure 5-2: DFP_U DisplayPort Mode Discovery and Entry 117
Figure 5-3: DFP_U DisplayPort Configuration 118
Figure 5-4: DFP_U DisplayPort Operation and Re-configuration 119
Figure 5-5: DFP_U Re-configure to USB Configuration 120
Figure 5-6: DFP_U Exit Mode 121
Figure 5-7: UFP_U DisplayPort Mode Discovery and Entry 122
Figure 5-8: UFP_U USB Configuration Wait State 124
Figure 5-9: UFP_U DisplayPort Configuration 125
Figure 5-10: UFP_U DisplayPort Operation 126
Figure 5-11: UFP_U Exit Mode 127
Figure 5-12: Hard Reset and Disconnect 127

Figure A-1: Host System Block Diagram 133
Figure A-2: DisplayPort Video Adaptor Cable Block Diagram 134
Figure A-3: Example USB Dock with DisplayPort Block Diagram 139

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Preface

Intellectual Property

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Excerpts from *USB r3.1* are Copyright © 2014, Hewlett-Packard Company, Intel Corporation, Microsoft Corporation, Renesas Corporation, STMicroelectronics, and Texas Instruments, Inc., and are reproduced with permission.

Excerpts from drafts of *USB Type-C r1.1* are Copyright © 2014 *USB r3.0* Promoter Group and are reproduced with permission.

Excerpts from drafts of *USB PD r2.0 v1.1* are Copyright © 2014, *USB r2.0* Promoter Group and are reproduced with permission.

While every precaution has been taken in the preparation of this Standard, the Video Electronics Standards Association and its contributors assume no responsibility for errors or omissions, and make no warranties, expressed or implied, of functionality or suitability for any purpose.

Trademarks

All trademarks used within this document are the property of their respective owners.

DisplayPort and DockPort are trademarks and VESA is a registered trademark of the Video Electronics Standards Association.

HDMI is a registered trademark of HDMI, LLC.

USB Type-C is a trademark of the Universal Serial Bus Implementers Forum.

Licensed to Jay Lin (insp@realtek.com)

Patents

VESA® draws attention to the fact that it is claimed that compliance with this Standard might involve the use of a patent or other intellectual property right (collectively, "IPR"). VESA takes no position concerning the evidence, validity, and scope of this IPR.

The following holders of this IPR have assured VESA that they are willing to license the IPR on Reasonable and Non-Discriminatory (RAND) terms. The statement of the holder of this IPR is registered with VESA.

Table 1: Patents

Holder Name	Contact Information	Claims Known
Apple, Inc. 1 Infinite Loop MS 169-3IPL Cupertino, CA 95014 408-783-5093	BJ Watrous Senior Standards Counsel (bjwatrous@apple.com)	–

Attention is drawn to the possibility that some of the elements of this VESA Standard might be the subject of IPR other than those identified above. VESA shall not be held responsible for identifying any or all such IPR, and has made no inquiry into the possible existence of any such IPR.

THIS STANDARD IS BEING OFFERED WITHOUT ANY WARRANTY WHATSOEVER, AND IN PARTICULAR, ANY WARRANTY OF NON-INFRINGEMENT IS EXPRESSLY DISCLAIMED. ANY IMPLEMENTATION OF THIS STANDARD SHALL BE MADE ENTIRELY AT THE IMPLEMENTER'S OWN RISK, AND NEITHER VESA, NOR ANY OF ITS MEMBERS OR SUBMITTERS, SHALL HAVE ANY LIABILITY WHATSOEVER TO ANY IMPLEMENTER OR THIRD PARTY FOR ANY DAMAGES OF ANY NATURE WHATSOEVER DIRECTLY OR INDIRECTLY ARISING FROM THE IMPLEMENTATION OF THIS STANDARD.

Licensed to Jay Lin (jay.lin@realtek.com)

Support for this Standard

Clarifications and application notes to support this Standard might be written. To obtain the latest Standard and any support documentation, contact VESA.

If you have a product that incorporates DisplayPort on a USB Type-C connector, ask the company that manufactured your product for assistance. If you are a manufacturer, VESA can assist you with any clarification you might require. Submit all comments or reported errors, in writing, to VESA, using one of the following methods:

Fax: 510-651-5127, direct this fax to "VESA Technical Support"

Email: support@vesa.org

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Acknowledgements

This document would not have been possible without the efforts of VESA's DP on USB Type-C Connector Subgroup. In particular, Table 2 lists the individuals and their companies that contributed significant time and knowledge to this version of the Standard.

Table 2: Main Contributors to *DP Alt Mode on USB Type-C v1.0a*

Company	Name	Contribution
Apple, Inc.	Colin Whitby-Strevens	Subgroup Document Editor
Intel Corporation	Simon Ellis	Subgroup Leader
MegaChips Technology America	Alan Kobayashi	DisplayPort Task Group Chair
Parade Technologies, Ltd.	Craig Wiley	Subgroup Co-leader

Revision History

Table 3: Revision History

Date	Version	Description
August 5, 2015	1.0a	Added proposed changes from the following *DP Alt Mode on USB Type-C v1.0* SCRs: • DisplayPort Alt Mode Editorial Corrections • DisplayPort Alt Mode Plug Requirement Corrections & Protocol clarifications • DisplayPort Alt Mode VConn Low Power • DisplayPort Alt Mode Billboard URL • DP Alt Mode VGA Adaptor Requirement Reduction to 1 Lane at HBR and HBR2 • DisplayPort Alt Mode Capabilities Clarification • DisplayPort Alt Mode Flipped Plug Orientation Additions and Revisions
September 22, 2014	1.0	Initial release of the Standard.

Licensed to Jay Lin (jaylin@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

1 Introduction

USB Type-C r1.1 envisages the support of Alternate Modes using the USB Type-C™ connector and cables, in which various of the pins on the USB Type-C connector may be re-configured to support interfaces outside the scope of USB Type-C.

USB Type-C r1.1 (and *USB PD r2.0 v1.1*, which it leverages) provides a Discovery Process for discovering the support of Alternate Modes in connected devices, including specifying the method for switching into and out of a mode, whereby an Alternate Mode is used on the USB Type-C connector. *USB Type-C r1.1* provides rules by which Alternate Modes must abide.

This Standard defines the use of DisplayPort™ as an Alternate Mode for the USB Type-C connector.

1.1 Purpose

The purpose of this Standard is to enable an ecosystem in which the USB Type-C connector can be used to support DisplayPort and other video protocols (e.g., HDMI® and VGA) in a way that is fully interoperable with existing DisplayPort and other video protocol products, with an emphasis on a small number of cable types to minimize user confusion.

This Standard defines:

- Use of DisplayPort as an Alternate Mode on the USB Type-C to USB Type-C Cable Assembly
- Cable assemblies that adapt the USB Type-C connector to DisplayPort connectors
- Cable assemblies and Adaptors for DisplayPort on USB Type-C to other video protocols
- Simultaneous use of USB Enhanced SuperSpeed and DisplayPort for docking applications

Additionally, this Standard defines pin assignments and DisplayPort signaling rates to be used in various scenarios using the Cable assemblies and Adaptors listed above.

This Standard describes how the USB-PD Discovery Process is applied to DisplayPort on USB Type-C.

1.2 Scope

This Standard is intended to complement *USB Type-C r1.1* and *DP v1.3*.

Normative information is provided to allow interoperability of components designed to this Standard. Informative material, when provided, may illustrate possible design implementations.

Section 1.6 provides a complete list of referenced Specifications and Standards. Nothing in this Standard is intended to change the normative requirements of the referenced USB Specifications. Nothing in this Standard is intended to change the normative requirements of the referenced *DP v1.3* Standard, except where explicitly stated. Extracts are copied from the referenced Specifications/Standards to assist with clarity of description; however, all such material is informative and the referenced Specifications/Standards should be consulted for the normative requirements. In the event of an apparent conflict between the requirements stated in this Standard and similar requirements in the referenced Specifications/Standards, the referenced Specifications/ Standards shall prevail, except where explicitly stated for *DP v1.3*.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

1.3 Document Organization

This Standard is organized into the following sections that define the overall architecture and structure of DisplayPort as an Alternate Mode on the USB Type-C interface:

- Section 1 – Introduction

  This section provides an overview of this Standard. This section also includes a list of acronyms and a glossary of terms that are used within this Standard and references to related standards, particularly *USB Type-C r1.1*.

- Section 2 – Overview

  This section describes the USB Type-C connector pin assignment, cable types, and docking systems that are used for DisplayPort through the USB Type-C connector, and methods for DisplayPort signaling through the USB Type-C connector.

- Section 3 – Pin Assignments and Descriptions

  This section describes the various pin assignments on the USB Type-C connector when sending and receiving DisplayPort signaling. It also describes the techniques used for DisplayPort AUX and HPD signaling.

- Section 4 – USB Type-C Cables Used for DisplayPort

  This section describes the Cable assemblies and Adaptors used for DisplayPort signaling through a USB Type-C connector.

- Section 5 – Discovery and USB-PD

  This section describes the mechanisms for discovery of DisplayPort support on a USB Type-C connector and the negotiation procedures for entering and exiting DisplayPort Mode and establishing DisplayPort Configuration.

- Section 6 – Miscellaneous

  This section describes power management, low-power operation, and equivalent USB functionality.

- Appendix A

  This appendix provides two examples of using DisplayPort Mode.

- Appendix B

  This appendix lists the contributors of past releases of this Standard.

Licensed to Jay Lin (jinee@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

1.4 Terms and Abbreviations

Table 1-1 lists terms and abbreviations that are defined in the USB Specifications referenced in Table 1-5, and are provided here for convenience.

Table 1-1: Terms and Abbreviations Referenced from USB Specifications

Term	Description (Informative)
Active cable	Electronically Marked Cable with additional electronics to condition the data path signals.
Alternate Mode	Operation defined by a vendor or standards organization that is associated with an SVID assigned by the USB-IF. Entry into and exit from an Alternate Mode is controlled by the *USB PD r2.0 v1.1* Structured VDM Enter Mode and Exit Mode commands.
Captive cable	Cable that is terminated on one end with a USB Type-C Plug and has a vendor-specific connect means (hardwired or custom detachable) on the opposite end.
CC	Configuration Channel.
DFP	Downstream Facing Port. Specifically associated with the flow of data in a USB connection. Typically, the ports on a host or the ports on a hub to which devices are connected. In its initial state, the DFP sources VBUS and VCONN, and supports data. A charge-only DFP only sources VBUS. (See DFP_U in Table 1-2.)
DRP	Acronym used in *USB Type-C r1.1* to refer to a USB data port that can operate as either a DFP or UFP. The role that the port offers may be fixed to either a DFP or UFP, or may alternate between the two port states. The port's role may be changed dynamically. ***Note:*** *This term is not to be confused with the terminology in the USB PD r2.0 v1.1, in which "dual-role port" refers to power roles.*
Electronically Marked Cable	USB Type-C Cable that uses USB-PD to provide the cable's characteristics.
Enhanced SuperSpeed	Adjective referring to any valid collection of USB defined features defined for the bus that runs over the SSRx and SSTx differential pairs in a *USB r3.1* system. Used in place of phrases such as SuperSpeed/SuperSpeedPlus.
Gen 1	Adjective used to refer to the Physical layer associated with a 5.0Gbps signaling rate. The original USB SuperSpeed Phy and a Gen 1 Phy refer to the same Phy.
Gen 2	Adjective used to refer to the Physical layer associated with a 10Gbps signaling rate.
Gen X	Generic term used to refer to any of the combinations *USB r3.1* Gen 1, Gen 2 or Gen 1/Gen 2 when the topic is specific to the phy layers but does not need to be specific to either Gen 1 or Gen 2. Examples include Gen X phy/connection.
IFEXT	Integrated Far-end Crosstalk.
ILFitAtNq	Insertion Loss Fit at Nyquist Frequencies.
IMR	Integrated Differential Multi-reflection.
INEXT	Integrated Near-end Crosstalk.
Initiator	Port that initiates a *USB PD r2.0 v1.1* Vendor Defined Message. It is independent of the port's PD role (e.g., Provider, Consumer, Provider/Consumer, or Consumer/Provider).
IRL	Integrated Return Loss.
Passive cable	Cable that does not incorporate any electronics to condition the data path signals. May or may not be electronically marked.
Responder	Port that responds to the Initiator of a *USB PD r2.0 v1.1* Vendor Defined Message. Independent of the port's PD role (e.g., Provider, Consumer, Provider/Consumer, or Consumer/Provider).

Licensed to [illegible]

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 1-1: Terms and Abbreviations Referenced from USB Specifications (Continued)

Term	Description (Informative)
SBU	Sideband Use.
SID	Unique 16-bit unsigned value assigned by the USB-IF for a Standard ID.
SOP	Start of Packet. K-code marker used to delineate the start of a packet. Three start of packet sequences are defined – SOP, SOP', and SOP", with SOP* used when referring to all three together, in place of SOP/SOP'/SOP".
SSRX	USB SuperSpeed receiver differential pair (possibly qualified to indicate positive or negative side, and/or channel).
SSTX	USB SuperSpeed transmitter differential pair (possibly qualified to indicate positive or negative side, and/or channel).
SVID	Generic term for either an SID or VID.
UFP	Upstream Facing Port. Specifically associated with the flow of data in a USB connection. The port on a device or a hub that connects to a host or the DFP of a hub. In its initial state, the UFP sinks VBUS and supports data. A charge-only DFP only sources VBUS. (See UFP_U in Table 1-2.)
USB Full-Featured Type-C Cable	USB Type-C to Type-C Cable that supports *USB r2.0* and *USB r3.1* data operation. Includes SBU wires.
USB Full-Featured Type-C Plug	USB Type-C Plug specifically designed to implement the USB Full-Featured Type-C Cable.
USB Type-C Plug	Plug that conforms to the mechanical and electrical requirements of *USB Type-C r1.1*.
USB Type-C Receptacle	Receptacle that conforms to the mechanical and electrical requirements of *USB Type-C r1.1*.
VCONN-powered accessory	Accessory that is powered from VCONN to operate in an Alternate Mode.
VDM	Vendor Defined Message. Provided to allow vendors to exchange information outside of that defined by *USB PD r2.0 v1.1*.
VDO	Vendor Defined Object, Used to convey vendor-specific information.
VID	Unique 16-bit unsigned value assigned by the USB-IF to a Vendor.

Licensed to Jay Lin (linspore@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 1-2 lists terms and abbreviations that are introduced in this Standard.

Table 1-2: Terms and Abbreviations

Term	Description
Adaptor	Product with a USB Type-C Plug at one end and a non-USB Type-C Plug or Receptacle at the other end. May be implemented as a Cable assembly or may integrate the connectors into a single unit.
Cable assembly	Product implemented as a cable with a plug at each end. May include active electronics for identification and/or signal conditioning.
DFP_D	Port that is a Downstream Facing Port as defined in *DP v1.3* and referred to as "DFP" in that Standard. Generally associated with a DisplayPort Source device, but may also be associated with a DisplayPort Branch device.
DFP_U	Downstream Facing Port, specified as "DFP" in *USB Type-C r1.1*. Typically the ports on a host or the ports on a hub to which devices are connected.
Direct connect	Host's DFP_U is connected directly, either with or without (e.g., thumb drive) a cable, to the device's UFP_U.
DisplayPort	Connector or DisplayPort port presented through a connector defined in *DP v1.3*.
DisplayPort Configuration	Configuration of the USB Type-C interface in which certain signals are associated with DisplayPort functionality, as specified in this Standard.
DisplayPort Mode	Alternate Mode in which, when selected, the USB Type-C interface behaves according to this Standard.
DP_BR	Bit rate and electrical signaling as specified in *DP v1.3* (i.e., one of RBR, HBR, HBR2, and HBR3).
DRP_U	Dual-Role Port, specified as DRP in *USB Type-C r1.1*. A port that can operate as either a DFP_U or a UFP_U. The role that the port offers may be fixed to either a DFP_U or UFP_U or may alternate between the two port states. The port's role may be changed dynamically.
GEN2_BR	Bit rate and electrical signaling as specified in *USB r3.1* using a Gen 2 PHY Layer.
mDP	Mini DisplayPort, as specified in *DP v1.3*.
Multi-function	Configuration on the USB Type-C connector that simultaneously combines *USB r3.1* SuperSpeed signaling with *DP v1.3* signaling.
Protocol Converter	Adaptor that converts from DisplayPort to a non-DisplayPort audio video protocol.
UFP_D	Port that is an Upstream Facing Port as defined in *DP v1.3* and referred to as a "UFP" in that Standard. Generally associated with a DisplayPort Sink device, but may also be associated with a DisplayPort Branch device.
UFP_U	Upstream Facing Port, specified as "UFP" in *USB Type-C r1.1*. The port on a device or a hub that connects to a host or the DFP_U of a hub.
USB Configuration	Configuration of the USB Type-C interface in which all signals have their default USB functionality, as specified in *USB Type-C r1.1*.
USB Mode	Mode in which DisplayPort Mode is not selected.
USB Type-C	Generic reference to a USB Type-C Plug, USB Type-C Receptacle, or USB Type-C Cable assembly, as specified in *USB Type-C r1.1*.

License: (...@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

1.5 Conventions

1.5.1 Precedence

If there is a conflict between text, figures, and tables, the precedence shall be tables, figures, and then text.

1.5.2 Keywords

Table 1-3 lists keywords that differentiate between the levels of requirements and options within this Standard.

Table 1-3: Keywords

Keyword	Definition
May	Indicates a choice with no implied preference.
N/A	Indicates that a field or value is not applicable and has no defined value and shall not be checked or used by the recipient.
Optional	Describes features not mandated by this Standard. However, if an optional feature is implemented, the feature shall be implemented as defined by this Standard (optional normative).
RESERVED	Indicates reserved bits, bytes, words, fields, and code values that are set aside for future standardization. Their use and interpretation may be specified by future extensions to this Standard and, unless otherwise stated, shall not be used or adapted by vendor implementation. A reserved bit, byte, word, or field shall be cleared to 0 by the sender and shall be ignored by the receiver. RESERVED field values shall not be sent by the sender and, if received, shall be ignored by the receiver.
Shall	Indicates a mandatory requirement. Designers are mandated to implement all such requirements to ensure interoperability with other compliant devices.
Should	Indicates flexibility of choice with a preferred alternative. Equivalent to the phrase “it is recommended that.”

1.5.3 Numbering

Table 1-4 lists the different types of numbering used within this Standard.

Table 1-4: Numbering

Number Type	Definition
Binary	Numbers that are immediately followed by a lowercase “b” (e.g., 01b).
Decimal	Numbers that are not immediately followed by either a “b” or “h”.
Hexadecimal	Numbers that are immediately followed by a lowercase “h” (e.g., 3Ah).

Licensed to Jay Lin (linsy@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

1.6 Reference Documents

Table 1-5: Reference Documents

Document	Version/ Revision	Referenced As	Publication Date
VESA DisplayPort (DP) Standard – see www.vesa.org	Version 1.3	*DP v1.3*	September 17, 2014
VESA DockPort Standard – see www.vesa.org	Version 1.0a	*DockPort v1.0a*	January 12, 2015
VESA DisplayPort Dual-Mode (DP++) Standard – see www.vesa.org	Version 1.1	*DP++ v1.1*	February 8, 2013
High-Bandwidth Digital Content Protection System, Mapping HDCP to DisplayPort – see www.digital-cp.com	Version 2.2	*HDCP v2.2*	December 2012
High-Definition Multimedia Interface Specification – see www.hdmi.org	Version 2.0	*HDMI v2.0*	September 2013
USB Implementers Forum, Inc., *Universal Serial Bus 3.1 Specification* – see www.usb.org/developers/docs	Revision 3.1	*USB r3.1*	July 26, 2013
USB Implementers Forum, Inc., *Universal Serial Bus Device Class Definition for Audio/Video Devices – AV Device Class Overview & AVFunction Definition* – see www.usb.org/developers/docs	Release 1.0	*USB AV r1.0*	December 07, 2011
USB Implementers Forum, Inc., *Universal Serial Bus Device Class Definition for Billboard Devices* – see www.usb.org/developers/docs	Revision 1.0	*USB Billboard r1.0*	August 11, 2014
USB Implementers Forum, Inc., *Universal Serial Bus Power Delivery Specification* – see www.usb.org/developers/docs	Revision 2.0, Version 1.1	*USB PD r2.0 v1.1*	May 7, 2015
USB Implementers Forum, Inc., *Universal Serial Bus Type-C Cable and Connector Specification* – see www.usb.org/developers/docs	Revision 1.1	*USB Type-C r1.1*	April 3, 2015

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

2 Overview

This section describes the USB Type-C connector pin assignment, cable types, and docking systems that are used for DisplayPort through the USB Type-C connector, and methods for DisplayPort signaling through the USB Type-C connector.

2.1 Introduction

Figure 2-1 illustrates the USB Type-C Receptacle, as specified in *USB Type-C r1.1*.

A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12
GND	TX1+	TX1-	VBUS	CC1	D+	D-	SBU1	VBUS	RX2-	RX2+	GND
GND	RX1+	RX1-	VBUS	SBU2	D-	D+	CC2	VBUS	TX2-	TX2+	GND
B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1

Figure 2-1: USB Type-C Receptacle Interface (Front View) (Informative)

Figure 2-2 illustrates the comprehensive functional signal plan for the USB Type-C Plug. The notable differences are as follows:

- Only one CC pin is connected through the cable; the "unused" CC pin is re-configured as VCONN (available as a power source for electronics that may be incorporated into USB Type-C Plugs, Cable assemblies, or Adaptors)
- Only one D+/D- pair instead of two

A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1
GND	RX2+	RX2-	VBUS	SBU1	D-	D+	CC	VBUS	TX1-	TX1+	GND
GND	TX2+	TX2-	VBUS	VCONN			SBU2	VBUS	RX1-	RX1+	GND
B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12

Figure 2-2: USB Type-C Plug Interface (Front View) (Informative)

2.2 USB Type-C Receptacles, Plugs, and Cables

This Standard uses the USB Type-C to USB Full-Featured Type-C Cable assembly, as defined in *USB Type-C r1.1*.

In addition, this Standard defines the following USB Type-C Captive cables:

- USB Type-C Plug to DisplayPort/mini DisplayPort plug Cable assembly
- USB Type-C Plug to another video interface (e.g., HDMI, DVI, or VGA) Cable assembly (i.e., with a plug) or Adaptor (i.e., with a receptacle)
- USB Type-C Plug to *DockPort v1.0a* receptacle

Licensed to Jay Lin (jinshi@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

2.3 Signaling and Cable Types

This Standard uses DisplayPort signaling speeds and electrical specifications as specified in *DP v1.3* (denoted as DP_BR). In addition, this Standard anticipates a future version of DisplayPort in which the signaling speeds and electrical specifications use the USB Gen 2 PHY Layer, as specified in *USB r3.1* (i.e., at 10Gbps (nominally, more precisely likely to be 9.99Gbps, i.e., 270Mbps * 37)). This signaling speed is denoted as GEN2_BR.

Adaptors (USB Type-C Plug at one end and DisplayPort plug or *DockPort v1.0a* receptacle at the other end) shall always use DP_BR. (See Section 4.5 for a full description of the *DockPort v1.0a* receptacle-based Adaptor.)

USB Type-C Protocol Converters (USB Type-C Plug on one end and non-DisplayPort video connector at the other end) shall support DP_BR, but may also support GEN2_BR.

The various varieties of the USB Full-Featured Type-C to USB Type-C Cable assemblies support DP_BR and GEN2_BR as specified in Table 2-1, which indicates the maximum number of DisplayPort lanes that can be supported in each case. Each combination of DisplayPort and USB signaling is associated with a specific pin assignment, described in more detail in Section 3.1. A DFP_D or UFP_D may support fewer than the maximum permitted number of lanes, as specified in Section 3.2.1.

Note that Pin Assignment C is mandatory for DFP_Ds and UFP_Ds that support DisplayPort on USB Type-C Receptacles (see Section 3.2.1) – this provides a guarantee of interoperability on passive USB Type-C Cables.

Table 2-1: USB Type-C to USB Type-C DisplayPort Signaling

Cable Type[a]	Pin Assignment[b]			
	A	B	C	D
	DisplayPort and USB Gen X Signaling			
	GEN2_BR (No USB Gen X)	GEN2_BR 1-Channel USB Gen X	DP_BR (No USB Gen X)	DP_BR 1-Channel USB Gen X
	Maximum Number of DisplayPort Lanes			
Gen 1 Passive (2m)	***Not supported***	***Not supported*** See Note[c]	4	2
Gen 2 Passive (1m)	4	2	4	2
Gen 2 Active	2	1	***Not supported***	***Not supported***

a. These cables are specified in USB Type-C r1.1. References to cable capabilities are informative.
b. For pin assignments, see Section 3.1.
c. This cable is capable of supporting one channel of USB Gen 1, but is not capable of supporting GEN2_BR.

Licensed to Jay Hutchins@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

2.4 Multi-function Systems

A USB Type-C interface may support USB SuperSpeed signaling as specified in *USB r3.1*, or DisplayPort signaling, as described in *DP v1.3*, or both simultaneously on separate USB Type-C SuperSpeed data signal pairs.

In some systems (such as docks), it may be desirable to be able to switch between various configurations, depending on what the user connects to the particular system. It may also be desirable to select a particular configuration when the system is first connected (still with the possibility of changing the configuration later). To support this, a mechanism is provided for the system to indicate its preferred configuration to its port partner (see Section 5.2.3).

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3 Pin Assignments and Descriptions

This section describes the various pin assignments on the USB Type-C connector when sending and receiving DisplayPort signaling. It also describes the techniques used for DisplayPort AUX and HPD signaling.

All USB Type-C pin assignments specified in this document define the pin assignments on the USB Type-C Receptacle and are provided both for when the plug is inserted in normal orientation and for when the plug is inserted in inverted ("flipped") orientation. The pin assignments for the corresponding plug are those specified for the receptacle when the plug is inserted into the receptacle in normal orientation. The plug pin assignment does not change with orientation. The system with the USB Type-C Receptacle shall support provide the appropriate pin functions on the receptacle to match the plug pin assignments when the plug is inserted into the receptacle in both normal and flipped orientation. The system design may implement support for plug rotation by multiplexers in the system design or by pin function adaptation within the DisplayPort Source or Sink device's silicon design. The USB Type-C Plugs on Adaptors and USB Type-C to DisplayPort cables always use the normal orientation pin assignment. Pin A5 is always CC and pin B5 is always VConn.

A USB Type-C Receptacle that supports DFP_D functionality (e.g., the receptacle can behave as a DisplayPort Source device or as a DFP_D on a DisplayPort Branch device) shall support one or more DFP_D pin assignments. Likewise, a USB Type-C Receptacle that supports UFP_D (e.g., the receptacle can behave as a DisplayPort Sink device or as the UFP_D on a DisplayPort Branch device) shall support one or more UFP_D pin assignments.

A USB Type-C Plug that is intended to plug directly into a receptacle-based DFP_D (e.g., the plug can behave as a DisplayPort Sink device or as the UFP_D on a DisplayPort Branch device) shall support one or more DFP_D pin assignments. Likewise, a USB Type-C Plug that is intended to plug directly into a receptacle-based UFP_D (e.g., the plug can behave as a DisplayPort Source device or as the DFP_D on a DisplayPort Branch device) shall support one or more UFP_D pin assignments.

Note that port directionality advertised for a USB Type-C Plug is opposite to the directionality of the DisplayPort functionality implemented behind the plug. For example, the Protocol Converter in a USB Type-C to HDMI or VGA Protocol Converter will implement DisplayPort upstream functionality (it is a DisplayPort receiver), but the USB Type-C Plug on the Protocol Converter will advertise a DFP_D pin assignment because it is intended to plug into a USB Type-C DFP_D receptacle.

Pin assignments are advertised and configured as described in Section 5.

All associations of USB Type-C functions with specific pins on the USB Type-C connector are informative. Reference should be made to *USB Type-C r1.1* for normative USB Type-C pin functionality.

Licensed by Lin (lin@cre-altek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.1 Pin Assignment Overview

Six possible pin assignments are defined for USB Type-C when configured to behave as a DFP_D – A, B, C, D, E, or F. Five possible pin assignments are defined for USB Type-C when configured to behave as a UFP_D – A, B, C, D, or E. Application of the various pin assignments within various scenarios is provided in Section 4.

Pin Assignments A, B, C, and D are intended for use in conjunction with USB Type-C to USB Type-C Cables and with Adaptors from USB Type-C to other video standards. The receptacles at each end of the cable shall use pin assignments with the same letter. Pin Assignments E and F are intended for use in conjunction with Adaptors from USB Type-C to DisplayPort plugs or receptacles.

DisplayPort protocol is carried using GEN2_BR when using Pin Assignments A and B. DisplayPort protocol is carried using DP_BR when using Pin Assignments C, D, E, and F.

Pin Assignments A and B are restricted by the *USB Type-C r1.1* Active cable definition. Pin Assignments E and F are restricted by the *DockPort v1.0a* definition. Pin Assignments C and D follow Pin Assignments E and F, respectively, to reduce the number of unique multiplexing configurations. In addition, the USB Type-C to USB Type-C Cable assembly has crossovers of TX1 with RX1 and TX2 with RX2, whereas the DisplayPort Cable assembly has crossovers of Lane 0 to 3, Lane 1 to 2, Lane 2 to 1, and Lane 3 to 0. The different crossover specifications impact the pin assignments for Source and Sink devices when using reversible cables.

Pin Assignments B, D, and F are used as variants of Pin Assignments A, C, and E, respectively, to simultaneously carry one channel of USB Enhanced SuperSpeed protocol with one or two lanes of DisplayPort. This configuration is described as Multi-function.

Pin Assignment B is the most constrained – it supports one lane of DisplayPort and one channel of USB Enhanced SuperSpeed over a USB Type-C Active cable. Thus, SSTX1/SSRX1 is used for USB Enhanced SuperSpeed and SSTX2 is used for one lane of DisplayPort and is assigned to ML0. SSRX2 is assigned to ML1 for use on the equivalent Passive cables to support extra DisplayPort bandwidth.

Pin Assignment F, which uses DisplayPort signaling rates in combination with one channel of USB Enhanced SuperSpeed, is designed to support *DockPort v1.0a*.

Note: *DockPort ™ is defined only for a DisplayPort Source device on the system. There is no Pin Assignment F for a system that integrates a Display (UFP_D).*

Pin Assignment E, which supports four lanes of DisplayPort using DisplayPort signaling rates, is derived from Pin Assignment F and assigns SSTX1 and SSRX1 to ML2 and ML3, respectively. Pin Assignment E also supports *DockPort v1.0a*.

Pin Assignments E and F allow the USB Type-C connector flip to be implemented by DFP_Ds that support pin allocation reversals (i.e., ML0, ML1, ML2, and ML3 can be reversed to ML3, ML2, ML1, and ML0, respectively).

DFP_D Pin Assignments C and D use the same assignments as Pin Assignments E and F to save on further multiplexing.

Licensed to Jay Lin (jinsperanet@...)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Pin Assignment A supports two lanes of DisplayPort over a USB Type-C Active cable. To minimize multiplexing from Pin Assignment B, SSTX2 and SSTX1 are assigned to ML0 and ML1, respectively. To extend this pin assignment for use on equivalent Passive cables, SSRX1 is assigned to ML3 to be compatible with DFP_D Pin Assignment C, leaving SSRX2 to be assigned to ML2.

UFP_D Pin Assignments A, B, C, and D are derived from the corresponding DFP_D pin assignments, taking into account the pin A2 – A3 to B11 – B10, A11 – A10 to B2 – B3, and SBU1 to SBU2 crossovers in the USB Type-C to USB Type-C Cable assembly.

UFP_D Pin Assignment E is derived from DFP_D Pin Assignment E, taking into account the requirement that the USB Type-C to DisplayPort (or mini DisplayPort) Cable assembly should be reversible. In effect, this exchanges ML0, ML1, ML2, and ML3 with ML3, ML2, ML1, and ML0, respectively. UFP_D Pin Assignment E also has polarity inversion on all pin pairs.

3.2 USB Type-C DisplayPort Pin Assignments

Table 3-1 summarizes the DFP_D pin assignments when the attached plug is in normal orientation. Table 3-2 summarizes the DFP_D pin assignments when the attached plug is in flipped orientation.

Table 3-1: USB Type-C Receptacle DFP_D Pin Assignment Summary – Normal Plug Orientation

Receptacle Pin Number	Pin Assignment	A	B	C	D	E	F
	Cable	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to DP	USB Type-C to DP
	DisplayPort Signaling	GEN2_BR	GEN2_BR	DP_BR	DP_BR	DP_BR	DP_BR
A11 – A10		Open[a]/ML2	Open[a]/ML1	ML0	ML0	ML0	ML0
A2 – A3		ML1	SSTX	ML2	SSTX	ML2	SSTX
B11 – B10		Open[a]/ML3	SSRX	ML3	SSRX	ML3	SSRX
B2 – B3		ML0	ML0	ML1	ML1	ML1	ML1
A8		AUX_CH_P	AUX_CH_P	AUX_CH_P	AUX_CH_P	AUX_CH_P	AUX_CH_P
B8		AUX_CH_N	AUX_CH_N	AUX_CH_N	AUX_CH_N	AUX_CH_N	AUX_CH_N

a. Connections marked as optionally Open are not used when Active cables are connected.

Licensed to Jay Lin (jlin@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 3-2: USB Type-C Receptacle DFP_D Pin Assignment Summary – Flipped Plug Orientation

Receptacle Pin Number	Pin Assignment	A	B	C	D	E	F
	Cable	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to DP	USB Type-C to DP
	DisplayPort Signaling	GEN2_BR	GEN2_BR	DP_BR	DP_BR	DP_BR	DP_BR
A11 – A10		Open[a]/ML3	SSRX	ML3	SSRX	ML3	SSRX
A2 – A3		ML0	ML0	ML1	ML1	ML1	ML1
B11 – B10		Open[a]/ML2	Open[a]/ML1	ML0	ML0	ML0	ML0
B2 – B3		ML1	SSTX	ML2	SSTX	ML2	SSTX
A8		AUX_CH_N	AUX_CH_N	AUX_CH_N	AUX_CH_N	AUX_CH_N	AUX_CH_N
B8		AUX_CH_P	AUX_CH_P	AUX_CH_P	AUX_CH_P	AUX_CH_P	AUX_CH_P

a. Connections marked as optionally Open are not used when Active cables are connected.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 3-3 summarizes the UFP_D pin assignments when the attached plug is in normal orientation.
Table 3-4 summarizes the UFP_D pin assignments when the attached plug is in flipped orientation.

Table 3-3: USB Type-C Receptacle UFP_D Pin Assignment Summary – Normal Plug Orientation

Receptacle Pin Number	Pin Assignment	A	B	C	D	E
	Cable	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to USB Type-C or Protocol Converter	DP to USB Type-C
	DisplayPort Signaling	GEN2_BR	GEN2_BR	DP_BR	DP_BR	DP_BR
A11 – A10		ML0	ML0	ML1	ML1	ML3[a]
A2 – A3		Open[b]/ML3	SSTX	ML3	SSTX	ML1[a]
B11 – B10		ML1	SSRX	ML2	SSRX	ML0[a]
B2 – B3		Open[b]/ML2	Open[b]/ML1	ML0	ML0	ML2[a]
A8		AUX_CH_N	AUX_CH_N	AUX_CH_N	AUX_CH_N	AUX_CH_P[c]
B8		AUX_CH_P	AUX_CH_P	AUX_CH_P	AUX_CH_P	AUX_CH_N[c]

a. The polarity on each lane is inverted when Pin Assignment E is used at the UFP_D receptacle.
b. Connections marked as optionally Open are not used when Active cables are connected.
c. AUX_CH_P and AUX_CH_N are inverted with respect to the other pin assignments when Pin Assignment E is used at the UFP_D receptacle.

Table 3-4: USB Type-C Receptacle UFP_D Pin Assignment Summary – Flipped Plug Orientation

Receptacle Pin Number	Pin Assignment	A	B	C	D	E
	Cable	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to USB Type-C or Protocol Converter	DP to USB Type-C
	DisplayPort Signaling	GEN2_BR	GEN2_BR	DP_BR	DP_BR	DP_BR
A11 – A10		ML1	SSRX	ML2	SSRX	ML0[a]
A2 – A3		Open[b]/ML2	Open[b]/ML1	ML0	ML0	ML2[a]
B11 – B10		ML0	ML0	ML1	ML1	ML3[a]
B2 – B3		Open[b]/ML3	SSTX	ML3	SSTX	ML1[a]
A8		AUX_CH_P	AUX_CH_P	AUX_CH_P	AUX_CH_P	AUX_CH_N[c]
B8		AUX_CH_N	AUX_CH_N	AUX_CH_N	AUX_CH_N	AUX_CH_P[c]

a. The polarity on each lane is inverted when Pin Assignment E is used at the UFP_D receptacle.
b. Connections marked as optionally Open are not used when Active cables are connected.
c. AUX_CH_P and AUX_CH_N are inverted with respect to the other pin assignments when Pin Assignment E is used at the UFP_D receptacle.

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Pins A2, A3, B2, and B3 shall be AC-coupled in receptacle-based systems between the port orientation switch or DisplayPort Source/Sink device (when plug rotation support is provided within the DisplayPort silicon) and the connector. Pins A10, A11, B10, and B11 shall be DC-coupled in receptacle-based systems. These requirements match the requirements for the use of these pins for USB Enhanced SuperSpeed operation. Adaptor cables and plug-based systems shall include AC capacitors on pins A10, A11, B10, and B11.

3.2.1 Pin Assignment Support Requirements

Table 3-5 and Table 3-6 list the required pin assignments for a USB Type-C Receptacle that supports DisplayPort. The required pin assignments depend on whether the port also implements a USB host or USB device (including a UFP with accessory support, as described in *USB Type-C r1.1*).

Table 3-5: USB Type-C Receptacle DFP_D Pin Assignment Requirements

	Pin Assignment	A	B	C	D	E	F
	Cable	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to USB Type-C or Protocol Converter	USB Type-C to DP	USB Type-C to DP
	DisplayPort Signaling	GEN2_BR	GEN2_BR	DP_BR	DP_BR	DP_BR	DP_BR
USB Host		Optional	Required if A is supported; otherwise, prohibited	Required	Optional	Required	Optional
USB Device		Optional	Optional	Required	Optional	Required	(Not applicable)

Table 3-6: USB Type-C Receptacle UFP_D Pin Assignment Requirements

	Pin Assignment	A	B	C	D	E
	Cable	USB Type-C to USB Type-C	USB Type-C to USB Type-C	USB Type-C to USB Type-C	USB Type-C to USB Type-C	USB Type-C to DP
	DisplayPort Signaling	GEN2_BR	GEN2_BR	DP_BR	DP_BR	DP_BR
USB Host		Optional	Required if A is supported; otherwise, prohibited	Required	Optional	Required
USB Device		Optional	Optional	Required	Optional	Required

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 3-7 lists the requirements for a UFP_D that is implemented with a USB Type-C Plug and is therefore intended to plug into a DFP_D receptacle. The USB Type-C Plug on the UFP_D therefore supports one or more of the DFP_D receptacle pin assignments.

Table 3-7: USB Type-C Plug DFP_D Pin Assignment Requirements (for UFP_D)

	Pin Assignment	A	B	C	D	E	F
	DisplayPort Signaling	GEN2_BR	GEN2_BR	DP_BR	DP_BR	DP_BR	DP_BR
USB Host		Optional	Optional if A is supported; otherwise, prohibited	Required if E is not supported; otherwise, optional	Optional if C is supported; otherwise, prohibited	Required if C is not supported; otherwise, optional	(Not applicable)
USB Device		Optional	Optional if A is supported; otherwise, prohibited	Required if E is not supported; otherwise, optional	Optional if C is supported; otherwise, prohibited	Required if C is not supported; otherwise, optional	Optional if E is supported; otherwise, prohibited

Table 3-8 lists the requirements for a DFP_D that is implemented with a USB Type-C Plug and is therefore intended to plug into a UFP_D receptacle. The USB Type-C Plug on the DFP_D therefore supports one or more of the UFP_D receptacle pin assignments.

Table 3-8: USB Type-C Plug UFP_D Pin Assignment Requirements (for DFP_D)

	Pin Assignment	A	B	C	D	E
	DisplayPort Signaling	GEN2_BR	GEN2_BR	DP_BR	DP_BR	DP_BR
USB Host		Optional	Required if A is supported; otherwise, prohibited	Required if E is not supported; otherwise, optional	Optional if C is supported; otherwise, prohibited	Required if C is not supported; otherwise, optional
USB Device		Optional	Optional if A is supported; otherwise, prohibited	Required if E is not supported; otherwise, optional	Optional if C is supported; otherwise, prohibited	Required if C is not supported; otherwise, optional

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Systems supporting a DFP_D on a USB Type-C Receptacle shall support HBR.

Systems supporting a DFP_D or a UFP_D on a USB Type-C Receptacle may support one, two, or four lanes of DisplayPort, according to system functionality.

Adaptors and Captive cable docks supporting a DFP_D on a USB Type-C Plug and adapting this to a DisplayPort connector shall be capable of supporting four lanes of DisplayPort.

Adaptors and Captive cable docks supporting a DFP_D on a USB Type-C Plug and adapting this to a non-DisplayPort video interface may support two or four lanes of DisplayPort, according to Adaptor functionality.

Cradle docks supporting DisplayPort on a USB Type-C Plug that are designed to complement a specific product or product family, and are restricted by mechanical constraints may support one, two, or four lanes of DisplayPort, according to the functionality of the complementary system.

Notes: *The lane count and bandwidth requirements for DVI, VGA, and HDMI Protocol Converters are provided in Section 4.3.*

The number of DisplayPort lanes used at any moment in time for a particular connection follows DP v1.3 requirements.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.3 DFP_D Pin Assignment A

Table 3-9 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using GEN2_BR signaling using normal plug orientation. Table 3-10 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using GEN2_BR signaling using flipped plug orientation. This pin assignment is appropriate only for USB Type-C to USB Type-C Cables and UFP_Ds with USB Type-C Plugs. It supports Passive and Active USB Type-C to USB Type-C Gen 2 cables, as follows:

- Four lanes of DisplayPort for Passive Gen 2 cables
- Two lanes of DisplayPort for Active Gen 2 cables

This pin assignment does not support Gen 1 cables because Gen 1 cables do not support GEN2_BR (see *USB Type-C r1.1* for the capabilities of the various USB Type-C to USB Type-C Cables).

See Section 3.14 for further details regarding GEN2_BR signaling.

Table 3-9: DFP_D Pin Assignment A – Normal Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 1	ML1+	B11	Positive half of Lane 3	ML3+[b]
A3	Negative half Lane 1	ML1-	B10	Negative half of Lane 3	ML3-[b]
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_N
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_P	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Negative half of Lane 2	ML2-[b]	B3	Negative half of Lane 0	ML0-
A11	Positive half of Lane 2	ML2+[b]	B2	Positive half of Lane 0	ML0+
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.
*b. Passive Gen 2 cables only, **not** Active Gen 2 cables.*

Licensed to ... (illegible)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 3-10: DFP_D Pin Assignment A – Flipped Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 0	ML0+	B11	Positive half of Lane 2	ML2+[b]
A3	Negative half Lane 0	ML0-	B10	Negative half of Lane 2	ML2-[b]
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_P
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_N	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Negative half of Lane 3	ML3-[b]	B3	Negative half of Lane 1	ML1-
A11	Positive half of Lane 3	ML3+[b]	B2	Positive half of Lane 1	ML1+
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.
*b. Passive Gen 2 cables only, **not** Active Gen 2 cables.*

Licensed to Jay Lin (insp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.4 DFP_D Pin Assignment B

Table 3-11 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using GEN2_BR signaling combined with USB SuperSpeed (Multi-function) using normal plug orientation. Table 3-12 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using GEN2_BR signaling combined with USB SuperSpeed (Multi-function) using flipped plug orientation. This pin assignment is appropriate only for USB Type-C to USB Type-C Cables and UFP_Ds with USB Type-C Plugs. It supports two lanes of DisplayPort for Passive Gen 2 cables, but supports only one lane of DisplayPort for Active Gen 2 cables.

See Section 3.14 for further details regarding GEN2_BR signaling.

Table 3-11: DFP_D Pin Assignment B – Normal Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2		(USB Type-C)	B11		(USB Type-C)
A3		(USB Type-C)	B10		(USB Type-C)
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_N
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_P	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Negative half of Lane 1	ML1-[b]	B3	Negative half of Lane 0	ML0-
A11	Positive half of Lane 1	ML1+[b]	B2	Positive half of Lane 0	ML0+
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.
*b. Passive Gen 2 cables only, **not** Active Gen 2 cables.*

Licensed to Jay Lin (lrsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 3-12: DFP_D Pin Assignment B – Flipped Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 0	ML0+	B11	Positive half of Lane 1	ML1+[b]
A3	Negative half of Lane 0	ML0-	B10	Negative half of Lane 1	ML1-[b]
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_P
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_N	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10		(USB Type-C)	B3		(USB Type-C)
A11		(USB Type-C)	B2		(USB Type-C)
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.
*b. Passive Gen 2 cables only, **not** Active Gen 2 cables.*

Licensed to Jay Lin (insp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.5 DFP_D Pin Assignment C

Table 3-13 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using DP_BR signaling using normal plug orientation. Table 3-14 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using DP_BR signaling using flipped plug orientation. DisplayPort electrical signaling for this pin assignment shall comply with *DP v1.3* for transmission at RBR, HBR, HBR2, or HBR3, as supported by the system. The encoding and higher-level signaling shall be 8b/10b, as defined in *DP v1.3*.

This pin assignment is appropriate only for passive USB Type-C to USB Type-C Cables and UFP_Ds with USB Type-C Plugs.

Table 3-13: DFP_D Pin Assignment C – Normal Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 2	ML2+	B11	Positive half of Lane 3	ML3+
A3	Negative half of Lane 2	ML2-	B10	Negative half of Lane 3	ML3-
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_N
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_P	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Negative half of Lane 0	ML0-	B3	Negative half of Lane 1	ML1-
A11	Positive half of Lane 0	ML0+	B2	Positive half of Lane 1	ML1+
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.

Licensed to Jay Lin (lin_jsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 3-14: DFP_D Pin Assignment C – Flipped Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 1	ML1+	B11	Positive half of Lane 0	ML0+
A3	Negative half of Lane 1	ML1-	B10	Negative half of Lane 0	ML0-
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_P
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_N	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Negative half of Lane 3	ML3-	B3	Negative half of Lane 2	ML2-
A11	Positive half of Lane 3	ML3+	B2	Positive half of Lane 2	ML2+
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.6 DFP_D Pin Assignment D

Table 3-15 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using DP_BR signaling combined with USB SuperSpeed (Multi-function) using normal plug orientation. Table 3-16 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using DP_BR signaling combined with USB SuperSpeed (Multi-function) using flipped plug orientation. DisplayPort electrical signaling for this pin assignment shall comply with *DP v1.3* for transmission at RBR, HBR, HBR2, and HBR3, as supported by the system. The encoding and higher-level signaling shall be 8b/10b, as defined in *DP v1.3*.

This pin assignment is appropriate only for passive USB Type-C to USB Type-C Cables and UFP_Ds with USB Type-C Plugs.

Table 3-15: DFP_D Pin Assignment D – Normal Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2		(USB Type-C)	B11		(USB Type-C)
A3		(USB Type-C)	B10		(USB Type-C)
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_N
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_P	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Negative half of Lane 0	ML0-	B3	Negative half of Lane 1	ML1-
A11	Positive half of Lane 0	ML0+	B2	Positive half of Lane 1	ML1+
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 3-16: DFP_D Pin Assignment D – Flipped Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 1	ML1+	B11	Positive half of Lane 0	ML0+
A3	Negative half of Lane 1	ML1-	B10	Negative half of Lane 0	ML0-
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_P
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_N	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10		(USB Type-C)	B3		(USB Type-C)
A11		(USB Type-C)	B2		(USB Type-C)
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.7 DFP_D Pin Assignment E

Table 3-17 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using DP_BR signaling using normal plug orientation. Table 3-18 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using DP_BR signaling using flipped plug orientation. DisplayPort electrical signaling for this pin assignment shall comply with *DP v1.3* for transmission at RBR, HBR, HBR2, and HBR3, as supported by the system, except that the reference cable model shall be for the USB Type-C to DP cable defined in Section 4.2.3.

This pin assignment is appropriate only for the receptacles that accept plugs on USB Type-C to DisplayPort cables and USB Type-C to *DockPort v1.0a* Adaptors and for the plugs on such cables and Adaptors. It supports Passive and Active cables.

Note: *The reference cable has lower loss than the HBR reference cable in DP v1.3 to permit higher losses in the system design due to the need for signal multiplexing.*

Table 3-17: DFP_D Pin Assignment E – Normal Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 2	ML2+	B11	Positive half of Lane 3	ML3+
A3	Negative half of Lane 2	ML2-	B10	Negative half of Lane 3	ML3-
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_N
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_P	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Negative half of Lane 0	ML0-	B3	Negative half of Lane 1	ML1-
A11	Positive half of Lane 0	ML0+	B2	Positive half of Lane 1	ML1+
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.

Licensed to Jay_Lin (jlsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 3-18: DFP_D Pin Assignment E – Flipped Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 1	ML1+	B11	Positive half of Lane 0	ML0+
A3	Negative half of Lane 1	ML1-	B10	Negative half of Lane 0	ML0-
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_P
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_N	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Negative half of Lane 3	ML3-	B3	Negative half of Lane 2	ML2-
A11	Positive half of Lane 3	ML3+	B2	Positive half of Lane 2	ML2+
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.

Licensed to Jay Lin (insp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.8 DFP_D Pin Assignment F

Table 3-19 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using DP_BR signaling combined with USB Gen 1 signaling (Multi-function) using normal plug orientation. Table 3-20 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using DP_BR signaling combined with USB Gen 1 signaling (Multi-function) using flipped plug orientation. DisplayPort electrical signaling for this pin assignment shall comply with *DP v1.3* for transmission at RBR, HBR, HBR2, and HBR3, as supported by the system.

This pin assignment is appropriate only for Adaptors that adapt DisplayPort on USB Type-C to DisplayPort receptacles that are then connected to DisplayPort Captive cables that support simultaneous DisplayPort and USB Gen 1 signaling, as defined in *DockPort v1.0a*. USB Gen 2 signaling is not supported because this pin assignment is intended for use with a DisplayPort receptacle, which is not specified to support Gen 2 signaling rates.

Table 3-19: DFP_D Pin Assignment F – Normal Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2		(USB Type-C)	B11		(USB Type-C)
A3		(USB Type-C)	B10		(USB Type-C)
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_N
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_P	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Negative half of Lane 0	ML0-	B3	Negative half of Lane 1	ML1-
A11	Positive half of Lane 0	ML0+	B2	Positive half of Lane 1	ML1+
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.

Licensed to Ivy Lin (insp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 3-20: DFP_D Pin Assignment F – Flipped Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 1	ML1+	B11	Positive half of Lane 0	ML0+
A3	Negative half of Lane 1	ML1-	B10	Negative half of Lane 0	ML0-
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_P
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_N	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10		(USB Type-C)	B3		(USB Type-C)
A11		(USB Type-C)	B2		(USB Type-C)
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.9 UFP_D Pin Assignment A

Table 3-21 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using GEN2_BR signaling using normal plug orientation. Table 3-22 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using GEN2_BR signaling using flipped plug orientation.

This pin assignment is appropriate only for USB Type-C to USB Type-C Cables and DFP_Ds with USB Type-C Plugs. It supports Passive and Active USB Type-C to USB Type-C cables, as follows:

- Four lanes of DisplayPort for Passive Gen 2 cables
- Two lanes of DisplayPort for Active Gen 2 cables

This pin assignment does ***not*** support Gen 1 cables because Gen 1 cables do not support GEN2_BR (see *USB Type-C r1.1* for the capabilities of the various USB Type-C to USB Type-C cables).

See Section 3.14 for further details regarding GEN2_BR signaling.

Table 3-21: UFP_D Pin Assignment A – Normal Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 3	ML3+[b]	B11	Positive half of Lane 1	ML1+
A3	Negative half of Lane 3	ML3-[b]	B10	Negative half of Lane 1	ML1-
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_P
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_N	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Negative half of Lane 0	ML0-	B3	Negative half of Lane 2	ML2-[b]
A11	Positive half of Lane 0	ML0+	B2	Positive half of Lane 2	ML2+[b]
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.
*b. Passive cables only, **not** Active cables.*

Licensed to Jay Lin (jaspar@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 3-22: UFP_D Pin Assignment A – Flipped Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 2	ML2+[b]	B11	Positive half of Lane 0	ML0+
A3	Negative half of Lane 2	ML2-[b]	B10	Negative half of Lane 0	ML0-
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_N
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_P	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Negative half of Lane 1	ML1-	B3	Negative half of Lane 3	ML3-[b]
A11	Positive half of Lane 1	ML1+	B2	Positive half of Lane 3	ML3+[b]
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.
*b. Passive cables only, **not** Active cables.*

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.10 UFP_D Pin Assignment B

Table 3-23 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using GEN2_BR signaling combined with USB SuperSpeed (Multi-function) using normal plug orientation. Table 3-24 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using GEN2_BR signaling combined with USB SuperSpeed (Multi-function) using flipped plug orientation. See Section 3.14 for further details regarding GEN2_BR signaling.

This pin assignment is appropriate only for USB Type-C to USB Type-C Cables and DFP_Ds with USB Type-C Plugs. It supports two lanes of DisplayPort for Passive cables, but supports only one lane of DisplayPort for Active cables.

Table 3-23: UFP_D Pin Assignment B – Normal Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2		(USB Type-C)	B11		(USB Type-C)
A3		(USB Type-C)	B10		(USB Type-C)
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_P
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_N	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Negative half of Lane 0	ML0-	B3	Negative half of Lane 1	ML1-[b]
A11	Positive half of Lane 0	ML0+	B2	Positive half of Lane 1	ML1+[b]
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.
*b. Passive cables only, **not** Active cables.*

Licensed to Jay Lin (lin_jsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 3-24: UFP_D Pin Assignment B – Flipped Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 1	ML1+[b]	B11	Positive half of Lane 0	ML0+
A3	Negative half of Lane 1	ML1-[b]	B10	Negative half of Lane 0	ML0-
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_N
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_P	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10		(USB Type-C)	B3		(USB Type-C)
A11		(USB Type-C)	B2		(USB Type-C)
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.
*b. Passive cables only, **not** Active cables.*

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.11 UFP_D Pin Assignment C

Table 3-25 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using DP_BR signaling using normal plug orientation. Table 3-26 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using DP_BR signaling using flipped plug orientation. DisplayPort electrical signaling for this pin assignment shall comply with *DP v1.3* for transmission at RBR, HBR, HBR2, and HBR3, as supported by the system. The encoding and higher-level signaling shall be 8b/10b, as defined in *DP v1.3*.

This pin assignment is appropriate only for passive USB Type-C to USB Type-C Cables and DFP_Ds with USB Type-C Plugs.

Table 3-25: UFP_D Pin Assignment C – Normal Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 3	ML3+	B11	Positive half of Lane 2	ML2+
A3	Negative half of Lane 3	ML3-	B10	Negative half of Lane 2	ML2-
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_P
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_N	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Negative half of Lane 1	ML1-	B3	Negative half of Lane 0	ML0-
A11	Positive half of Lane 1	ML1+	B2	Positive half of Lane 0	ML0+
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.

Licensed to Jay Lin (lin_jsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 3-26: UFP_D Pin Assignment C – Flipped Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 0	ML0+	B11	Positive half of Lane 1	ML1+
A3	Negative half of Lane 0	ML0-	B10	Negative half of Lane 1	ML1-
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_N
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_P	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Negative half of Lane 2	ML2-	B3	Negative half of Lane 3	ML3-
A11	Positive half of Lane 2	ML2+	B2	Positive half of Lane 3	ML3+
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.12 UFP_D Pin Assignment D

Table 3-27 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using DP_BR signaling combined with USB SuperSpeed (Multi-function) using normal plug orientation. Table 3-28 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using DP_BR signaling combined with USB SuperSpeed (Multi-function) using flipped plug orientation. DisplayPort electrical signaling for this pin assignment shall comply with *DP v1.3* for transmission at RBR, HBR, HBR2, and HBR3, as supported by the system. The encoding and higher-level signaling shall be 8b/10b, as defined in *DP v1.3*.

This pin assignment is appropriate only for passive USB Type-C to USB Type-C Cables and DFP_Ds with USB Type-C Plugs.

Table 3-27: UFP_D Pin Assignment D – Normal Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2		(USB Type-C)	B11		(USB Type-C)
A3		(USB Type-C)	B10		(USB Type-C)
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_P
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_N	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Negative half of Lane 1	ML1-	B3	Negative half of Lane 0	ML0-
A11	Positive half of Lane 1	ML1+	B2	Positive half of Lane 0	ML0+
A12	Ground return	GND	B1	Ground return	GND

a. *"(USB Type-C)" denotes no change from USB Type-C r1.1.*

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 3-28: UFP_D Pin Assignment D – Flipped Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 0	ML0+	B11	Positive half of Lane 1	ML1+
A3	Negative half of Lane 0	ML0-	B10	Negative half of Lane 1	ML1-
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_N
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_P	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10		(USB Type-C)	B3		(USB Type-C)
A11		(USB Type-C)	B2		(USB Type-C)
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.13 UFP_D Pin Assignment E

Table 3-29 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using DP_BR signaling using normal plug orientation. Table 3-30 defines use and configuration of the USB Type-C connector pins when re-configured to support DisplayPort using DP_BR signaling using flipped plug orientation. DisplayPort electrical signaling for this pin assignment shall comply with *DP v1.3* for transmission at RBR, HBR, HBR2, and HBR3, as supported by the system, except that the reference cable model shall be for the USB Type-C to DP cable defined in Section 4.2.3.

This pin assignment is appropriate only for the receptacles that accept plugs on DisplayPort to USB Type-C Cables and for the plugs on such cables and Adaptors. It supports Passive and Active cables.

Table 3-29: UFP_D Pin Assignment E – Normal Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Negative half of Lane 1	ML1-	B11	Negative half of Lane 0	ML0-
A3	Positive half of Lane 1	ML1+	B10	Positive half of Lane 0	ML0+
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_N
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_P	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Positive half of Lane 3	ML3+	B3	Positive half of Lane 2	ML2+
A11	Negative half of Lane 3	ML3-	B2	Negative half of Lane 2	ML2-
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 3-30: UFP_D Pin Assignment E – Flipped Plug Orientation

Pin	Description	DisplayPort Function[a]	Pin	Description	DisplayPort Function[a]
A1	Ground return	GND	B12	Ground return	GND
A2	Negative half of Lane 2	ML2-	B11	Negative half of Lane 3	ML3-
A3	Positive half of Lane 2	ML2+	B10	Positive half of Lane 3	ML3+
A4		(USB Type-C)	B9		(USB Type-C)
A5		(USB Type-C)	B8		AUX_CH_P
A6		(USB Type-C)	B7		(USB Type-C)
A7		(USB Type-C)	B6		(USB Type-C)
A8		AUX_CH_N	B5		(USB Type-C)
A9		(USB Type-C)	B4		(USB Type-C)
A10	Positive half of Lane 0	ML0+	B3	Positive half of Lane 1	ML1+
A11	Negative half of Lane 0	ML0-	B2	Negative half of Lane 1	ML1-
A12	Ground return	GND	B1	Ground return	GND

a. "(USB Type-C)" denotes no change from USB Type-C r1.1.

Notes: *The reference cable has lower loss than the HBR reference cable in DP v1.3 to permit higher losses in the system design due to the need for signal multiplexing.*

Inverted use of pins A8/B8 (AUX_CH_P/AUX_CH_N) and inverted polarity of the Main-Link lanes on pins A2 – A3, A10 – A11, B11 – B10, and B3 – B2 compared to the UFP_D Pin Assignments A, B, C, and D.

Licensed to Jay Lin (rtsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.14 USB Gen 2 PHY Layer Signaling for DisplayPort (GEN2_BR)

DisplayPort protocol may be transmitted and received using the electrical specifications defined in *USB r3.1* for the Gen 2 PHY Layer (GEN2_BR) to be capable of being carried over USB Full-Featured Type-C Active Cable assemblies.

This version of this Standard does not provide a GEN2_BR specification. However, it is anticipated that a complete specification will be added as a new PHY Layer in a future revision of *DP v1.3*. This Standard provides the basis on which it is anticipated that the complete specification for GEN2_BR will be developed. In the event of conflict, the new PHY Layer in the anticipated revision of *DP v1.3* takes precedence.

Open issues in the GEN2_BR definition include encoding and training, the alternative being to maintain DisplayPort 8b/10b encoding and DisplayPort training (possibly with some form of Gen 2 PHY Layer training), or use Gen 2 128b/132b encoding with some form of mapping of the DisplayPort protocol.

3.15 AUX and HPD Signaling

The AUX and HPD implementations are designed to support the scenarios described in Section 4, which may use one of the following:

- USB Type-C to USB Type-C Cable assembly,
- USB Type-C to DisplayPort Cable assembly, –or–
- USB Type-C to other video Protocol Converter

See Section 4.2.4.2 for the behavior of the USB Type-C to DisplayPort Adaptors for DisplayPort DFP_D/UFP_D detection.

Equivalent implementations to those illustrated within this section are permissible.

3.15.1 AUX Signaling

AUX shall be supported on the USB Type-C connector when operating in DisplayPort Mode using differential 100Ω-terminated signaling on the receptacle pin A8 and B8 connection. AC-coupling capacitors shall be placed between the AUX_CH_P/AUX_CH_N transceiver and USB Type-C Receptacle. The AUX differential signal levels shall be as specified in *DP v1.3*, with the exception that the eye diagrams are measured for 1UI after a differential signal transition.

Note: *Receptacle pins A8 and B8 are disconnected in the system until the connection is switched into DisplayPort Mode and a configuration is selected. Likewise, plug pins A8 and B8 on Adaptor cables are disconnected until the connection is switched into DisplayPort Mode and a configuration is selected.*

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.15.1.1 AUX Schematics

The tolerance on the resistors shown in Figure 3-1, Figure 3-2, and Figure 3-3 is ±5%.

The tolerance on the capacitors shown in Figure 3-1, Figure 3-2, and Figure 3-3 is ±20%, except for the capacitors on the DisplayPort systems where the minimum and maximum values are shown.

Figure 3-1 illustrates the system design of a DFP_D on a USB Type-C connector connected to a UFP_D on a USB Type-C connector using a USB Type-C to USB Type-C Cable. The 2MΩ pull-down resistors on SBU1 and SBU2 are representative of the leakage of ESD and EMI/RFI components including termination to ensure no floating nodes, and are intended to show compliance with zSBUTermination in *USB Type-C r1.1*.

Note: *These schematics do not show the ESD protection and EMI/RFI components typically included in the system designs.*

The plug orientation switch may be replaced by AUX polarity inversion logic in the DisplayPort transmitter or receiver, controlled by the plug orientation detection mechanism associated with the USB Type-C Receptacle.

Note: *The 3.3V levels in the Adaptors are derived from VCONN because not all DisplayPort UFP_D devices provide DP_PWR.*

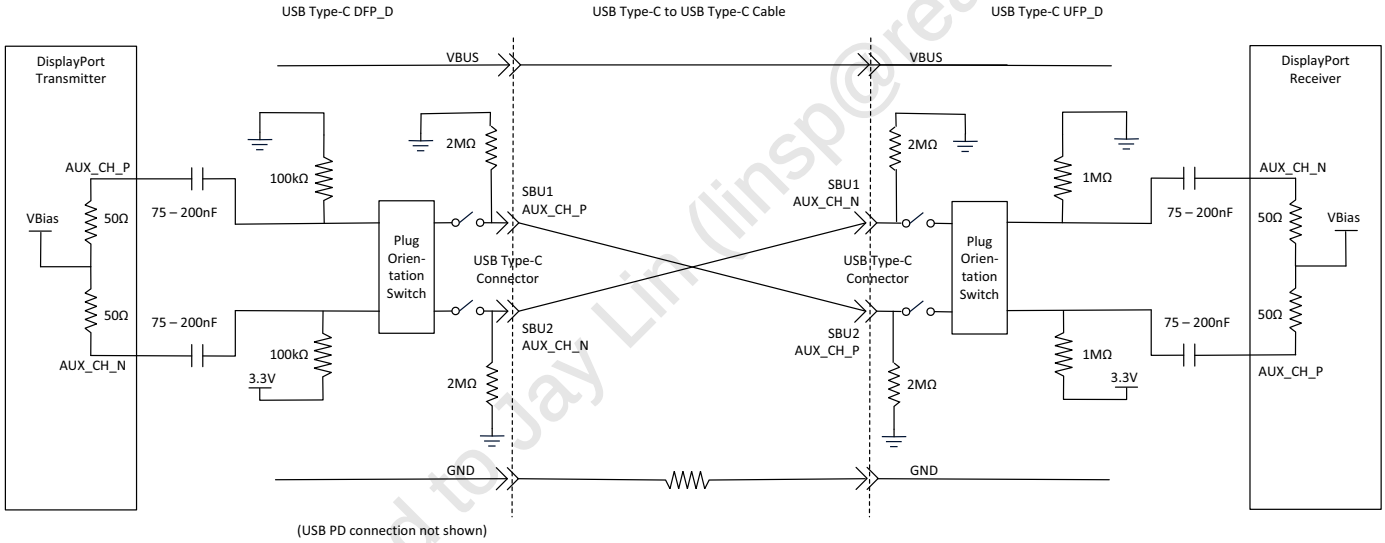


Figure 3-1: AUX Signaling Using USB Type-C to USB Type-C Cables

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 3-2 illustrates the system design of a DFP_D on a USB Type-C connector connected to a UFP_D on a DisplayPort connector using a USB Type-C to DisplayPort Adaptor. Note that the SBU isolation switches are open; however, when connection to a DisplayPort UFP_D is discovered, the AUX pull-up/pull-down resistors that emulate a DisplayPort DFP_D on the AUX single-ended connections are engaged by closing the SBU isolation switches.

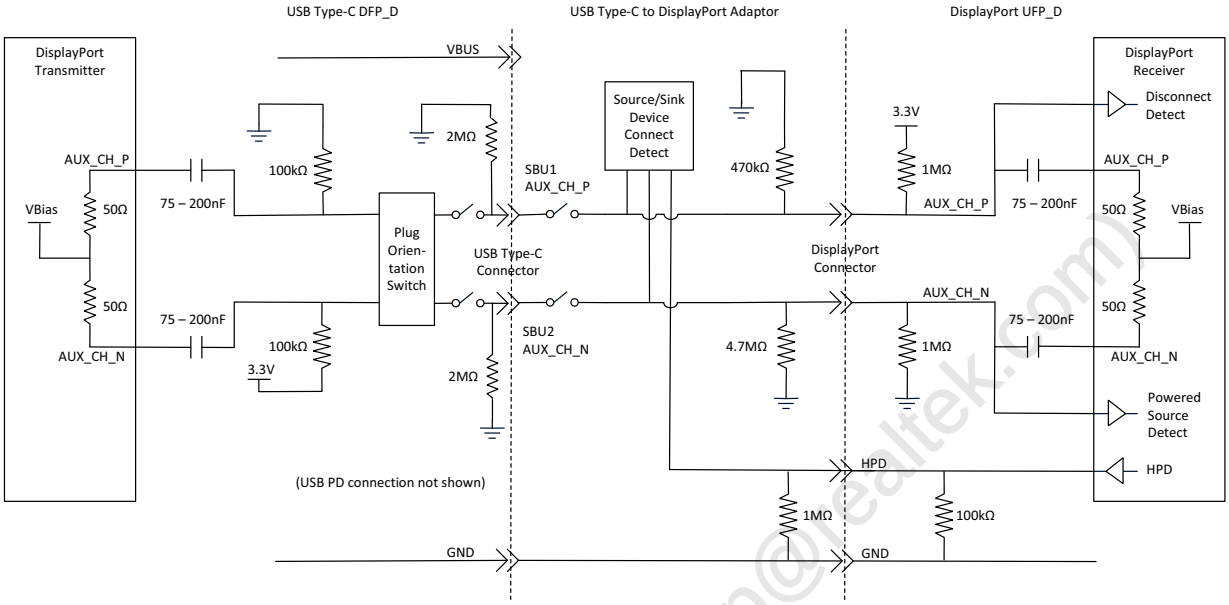


Figure 3-2: AUX Signaling Using USB Type-C to DisplayPort Adaptors

Care should be taken in the system design of the DisplayPort DFP_D/UFP_D Connect Detect logic to be able to detect the weak (nominal 1MΩ) pull-up resistor on AUX_CH_P in the DisplayPort UFP_D, bearing in mind the system leakage that may be present in the connected DisplayPort UFP_D due to its ESD and EMI components. The 470kΩ resistor represents a compromise between providing an adequate pull-down resistor to overcome these losses and the need for a relatively low threshold for the connection detection that this requires. Actual values will depend on the specific system design. The very weak (4.7MΩ) resistor on AUX_CH_N represents circuitry required to avoid a floating conductor when the Adaptor is not connected to a DisplayPort UFP_D without compromising the threshold in the DisplayPort UFP_D used to detect the Adaptor's pull-up resistor.

Licensed to JayLin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 3-3 illustrates the system design of a DFP_D on a DisplayPort connector connected to a UFP_D on a USB Type-C connector using a DisplayPort to USB Type-C Adaptor. Note that this Adaptor is the same Adaptor that is used in the previous example; however, DisplayPort signaling is flowing in the opposite direction.

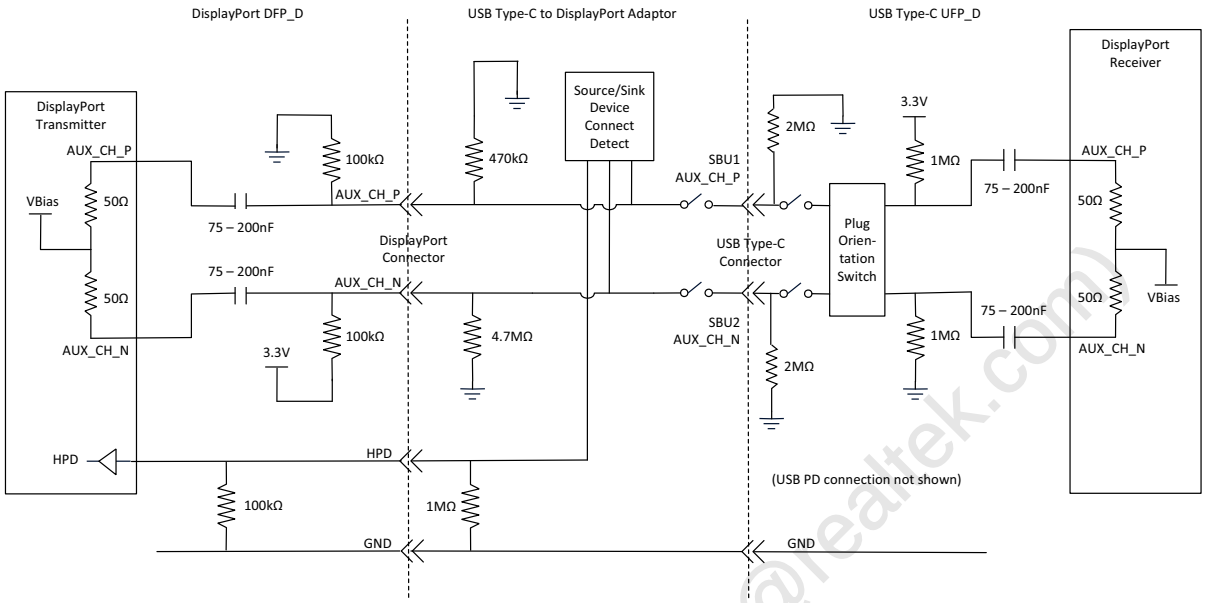


Figure 3-3: AUX Signaling Using DisplayPort to USB Type-C Adaptors

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 3-4 illustrates the system design of a DFP_D on a USB Type-C connector connected to a Protocol Converter. In this design, the Protocol Converter is used to convert DisplayPort protocol to a non-DisplayPort audio-video protocol.

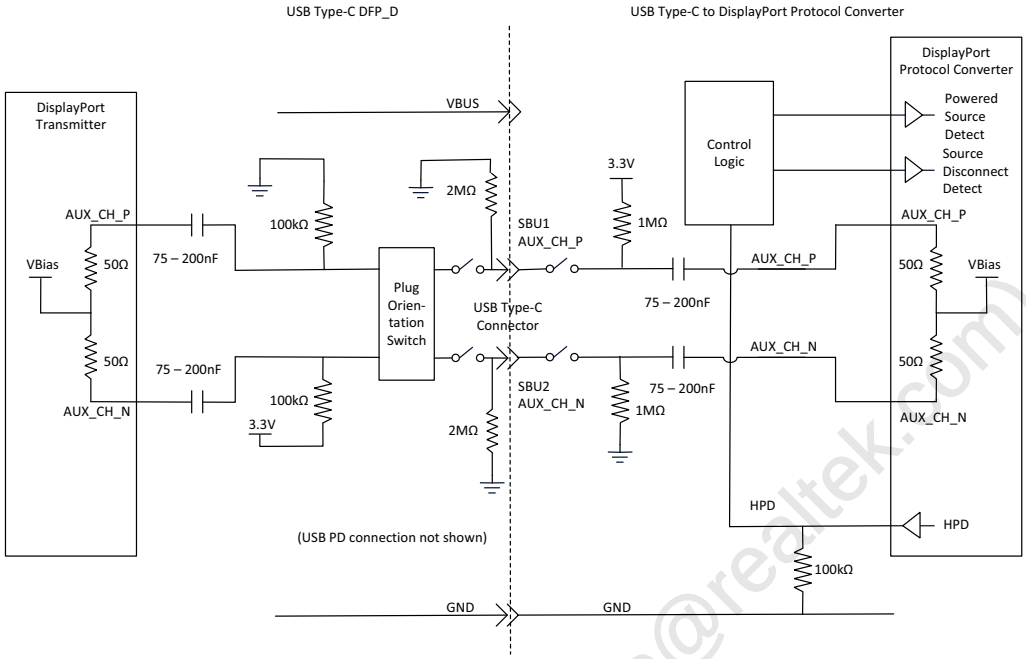


Figure 3-4: AUX Signaling Using USB Type-C to Protocol Converters

If the Protocol Converter is powered when the Adaptor is not in DisplayPort Mode, the following two requirements apply:

- If the Protocol Converter includes a separate Source Disconnect Detect input, that input shall be driven from the Rp/Rd connect detect on the USB Type-C Plug
- If the Protocol Converter includes a separate Powered Source Connect Detect input, that input shall be driven from the state of DisplayPort Mode (i.e., set when the Adaptor is in DisplayPort Mode, and unset when the Adaptor is not in DisplayPort Mode)

If the Protocol Converter is powered only when the Adaptor is in DisplayPort Mode, dynamic control of these inputs is not appropriate and both inputs may be hardwired to the "connected" state.

Licensed to Ray Lin (insp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.15.2 HPD Signaling

HPD shall be supported on the USB Type-C connector when operating in DisplayPort Mode by way of USB-PD messages that carry DisplayPort Status information (see Section 5.2.3 and Section 5.2.5).

Figure 3-5 illustrates the architecture for transmitting the HPD state over a USB-PD connection, and converting between HPD signaling (as specified in *DP v1.3*) and USB-PD messaging. The architecture defines an HPD-to-PD converter and PD-to-HPD converter to perform the necessary conversions. The converters may be integrated within a system or have their HPD driver or detector connected to a DisplayPort connector. See Section 3.15.2.2 for detailed schematics of the various possible applications of this architecture.

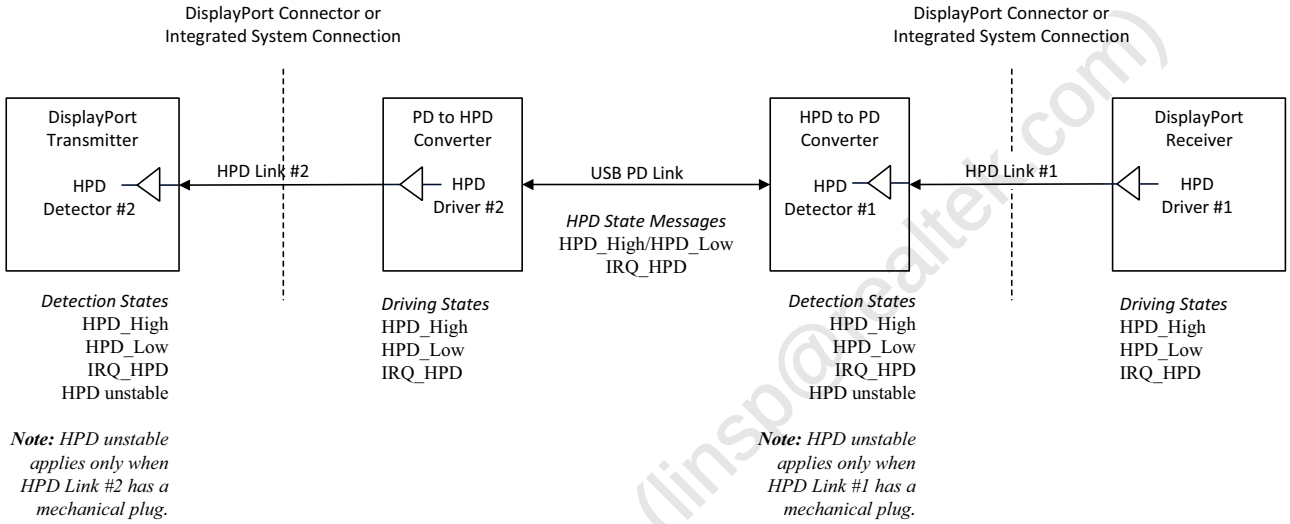


Figure 3-5: HPD Signaling Architecture

Two DisplayPort Status flags are defined to communicate HPD state:

- HPD_State – Indicates whether HPD's logical state is high or low (denoted as HPD_High or HPD_Low, respectively). For the purposes of communicating the HPD state over USB-PD, the logical state of HPD is considered as remaining high while receiving an IRQ_HPD, and is low during the time that HPD is being de-bounced on a new mechanical connection. The logical state of HPD transitions from high to low when a low level on the HPD link has been detected for 2ms (i.e., longer than the maximum IRQ_HPD pulse detection time). The logical state of HPD is unchanged during glitches (as specified in *DP v1.3*) on the HPD link.
- IRQ_HPD – Indicates an IRQ_HPD (i.e., a high-to-low transition on HPD followed by a low-to-high transition was detected between 250us and 2ms later, as specified in *DP v1.3*).

A USB-PD message containing DisplayPort status with the appropriate settings of these flags is transmitted as soon as possible on every HPD logical status change that is detected.

A DFP_U (including a UFP with accessory support, as described in *USB Type-C r1.1*) shall transmit DisplayPort Status updates using the DisplayPort Status Update command (see Section 5.2.3) and a UFP_U shall transmit DisplayPort Status updates using the DisplayPort Attention command (see Section 5.2.5) or as the response to a DisplayPort Status Update command.

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

A *USB PD r2.0 v1.1* engine supporting *USB PD r2.0 v1.1* on a USB Type-C Receptacle or Plug may incorporate an HPD detector on which it monitors the state of an HPD signal as specified in *DP v1.3*, and convert the HPD signal received at this receiver into USB-PD messages carrying DisplayPort Status updates across the USB Type-C connector (an HPD-to-PD converter), or may provide an integrated implementation for generating the USB-PD messages.

A *USB PD r2.0 v1.1* engine supporting *USB PD r2.0 v1.1* on a USB Type-C Receptacle or Plug may incorporate an HPD driver, and convert the PD messages carrying DisplayPort Status updates received across the USB Type-C connector into HPD signals as specified in *DP v1.3* (PD-to-HPD converter), or may provide an integrated implementation for communicating the HPD state.

The HPD drivers and/or detectors in these converters shall comply with the HPD specifications defined in *DP v1.3*, Section 3.3.

A UFP_D on a USB Type-C connector may implement the HPD-to-PD converter (as described in this section) or directly generate the DisplayPort status updates.

A DFP_D on a USB Type-C connector may implement the PD-to-HPD converter (as described in this section) or may directly process the DisplayPort status updates. It is recommended that the HPD receiver in the DFP_D does not implement the de-bounce mechanism and glitch filtering described in *DP v1.3*, Section 3.3 to avoid unnecessary latency in detecting a new connection signaled by HPD going high.

A USB Type-C to DisplayPort Adaptor shall implement a converter able to convert between the HPD signal and PD messages in either direction.

An HPD-to-PD converter that receives a DisplayPort HPD signal through an external connector shall implement the HPD de-bounce recommendation and glitch filtering provided in *DP v1.3*, Section 3.3, on its HPD detector. It shall not indicate HPD_High in a DisplayPort Status message during the time that HPD is being de-bounced (i.e., it indicates HPD_High only after de-bouncing has been applied and was completed by HPD being detected high for 100ms).

A USB Type-C to non-DisplayPort audio-video Protocol Converter may implement the PD-to-HPD converter (as described in this section) or directly process the DisplayPort status updates.

On first connection, HPD shall be assumed low.

It is required by USB Type-C that Alternate Mode messages be transmitted only during the time that the appropriate Alternate Mode is enabled. When DisplayPort Mode has not been enabled, an HPD-to-PD converter shall not transmit any Status update messages. When DisplayPort Mode is enabled, the HPD-to-PD converter shall transmit the current HPD state as part of the initial DisplayPort Status Update message exchange. The HPD detector may monitor the HPD state outside the time that DisplayPort Mode is enabled, so that the detector can perform HPD de-bouncing and track HPD's current state.

A PD-to-HPD converter shall drive a low level on its HPD driver at all times that DisplayPort Configuration on the USB Type-C interface is not enabled. It shall only indicate HPD's current state after the SBU isolation switches have been closed to connect the AUX channel. On disabling the DisplayPort Configuration, the PD-to-HPD converter shall drive a low level on its HPD driver before opening the SBU isolation switches to disconnect the AUX channel.

Uncontrolled Jay Lin (jay_lin_tw@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.15.2.1 Message Queuing

It is possible that other USB-PD traffic will cause the HPD-to-PD converter to delay transmission of HPD's current state, and may cause the arrival timing of event indications to not correspond to the original HPD timing. Accordingly, the converters shall observe the following rules:

- HPD-to-PD Converter Message Queuing Rules
- PD-to-HPD Converter Message Queuing Rules

Both are described in the following sub-sections.

3.15.2.1.1 HPD-to-PD Converter Message Queuing Rules

- If an IRQ_HPD is detected while a previous IRQ_HPD is in the queue, the converter shall transmit DisplayPort Status updates for both events. Any IRQ_HPD events detected while two IRQ_HPD events are in the queue may be discarded.
- If HPD_Low is detected while any previous status updates are in the queue (including IRQ_HPD), the previous status updates may be discarded.
- If HPD_High is detected while a previous HPD_Low is in the queue, both shall be transmitted in the order detected.
- If an IRQ_HPD is detected while a previous HPD_High is in the queue, both shall be transmitted, either in one status update (recommended) or separate messages in the order detected.
- While DisplayPort Mode is disabled, HPD state updates are not queued.
- When DisplayPort Mode changes from enabled to disabled, any entries in the queue are discarded.
- When DisplayPort Mode is enabled, the current HPD state is transmitted as part of the initial DisplayPort Status exchange.

Note: *The worst case status buffering required in the HPD-to-PD converter is the sequence HPD_Low, HPD_High, IRQ_HPD, IRQ_HPD.*

3.15.2.1.2 PD-to-HPD Converter Message Queuing Rules

- If a single message represents both a transition to HPD_High and an IRQ_HPD, the converter shall ensure that HPD is taken high for a minimum of 2ms before generating the IRQ_HPD.
- After generating a low-to-high transition on HPD, the converter shall ensure that HPD is maintained high for a minimum of 2ms, deferring taking HPD low on a subsequent HPD_Low or a concurrent or subsequent IRQ_HPD.
- After generating a high-to-low transition on HPD in response to the receipt of HPD_Low, the converter shall ensure that HPD is maintained low for a minimum of 3ms (i.e., the 2ms IRQ pulse detect time plus 1ms margin), deferring taking HPD high on receipt of a subsequent HPD_High event during this time.
- If a converter receives HPD_Low, any intervening HPD_High and/or IRQ_HPD in the queue shall be discarded.
- When DisplayPort Configuration is not selected (and the converter is driving its HPD output low), the converter shall track the current state of HPD, ready for appropriate indication when DisplayPort Configuration is subsequently selected.

Licensed to Jay Jin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

3.15.2.2 HPD Schematics

Figure 3-6 illustrates the schematic for an implementation of an HPD-to-PD converter in a UFP_D and a PD-to-HPD converter in a DFP_D and their connection using a USB Type-C to USB Type-C Cable assembly. As previously indicated, either one or both of the DFP_D and UFP_D may integrate the PD engine with the DisplayPort transmitter or receiver (as appropriate), rather than using an explicit HPD connection.

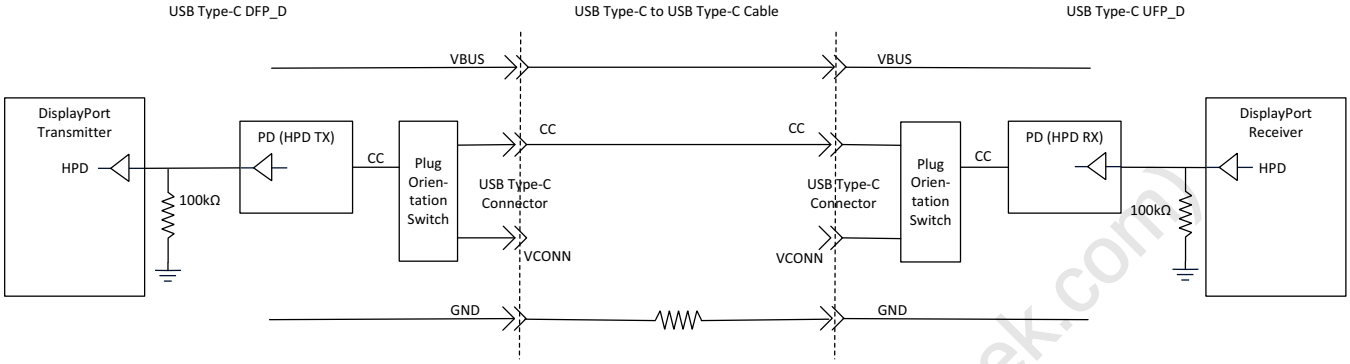


Figure 3-6: HPD Signaling Using USB Type-C to USB Type-C Cables

Figure 3-7 illustrates the schematic for an implementation of a PD-to-HPD converter in a DFP_D and an HPD-to-PD converter in a USB Type-C to DisplayPort Adaptor and its connection to a UFP_D on the DisplayPort connector. Because the USB Type-C to DisplayPort Adaptor is reversible, the HPD transmitter in the HPD-to-PD converter is disabled when signaling is transmitted in this direction.

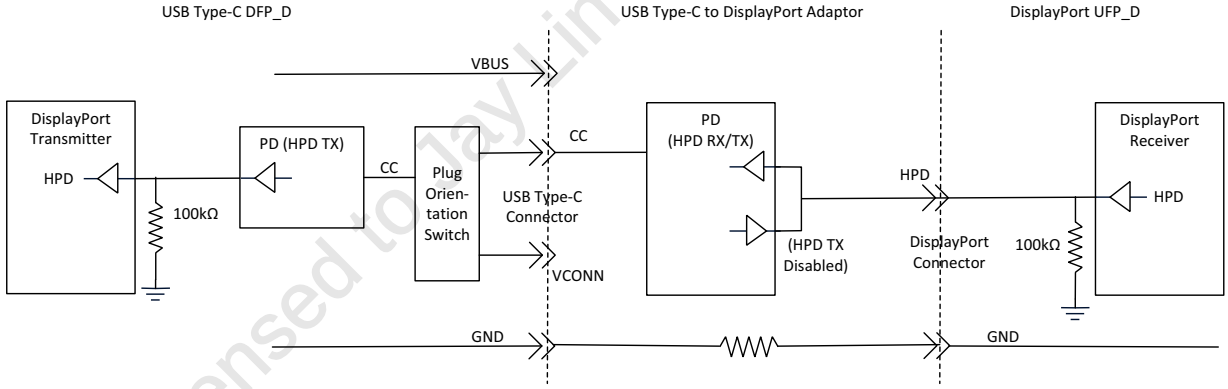


Figure 3-7: HPD Signaling Using USB Type-C to DisplayPort Adaptors

Licensed to Jay Lin (jinsc@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 3-8 illustrates the schematic for an implementation of an HPD-to-PD converter in a UFP_D and a PD-to-HPD converter in a DisplayPort to USB Type-C Adaptor and its connection to a DFP_D on the DisplayPort connector. Because the DisplayPort to USB Type-C Adaptor is reversible, the HPD transmitter in the HPD-to-PD converter is enabled when signaling is transmitted in this direction.

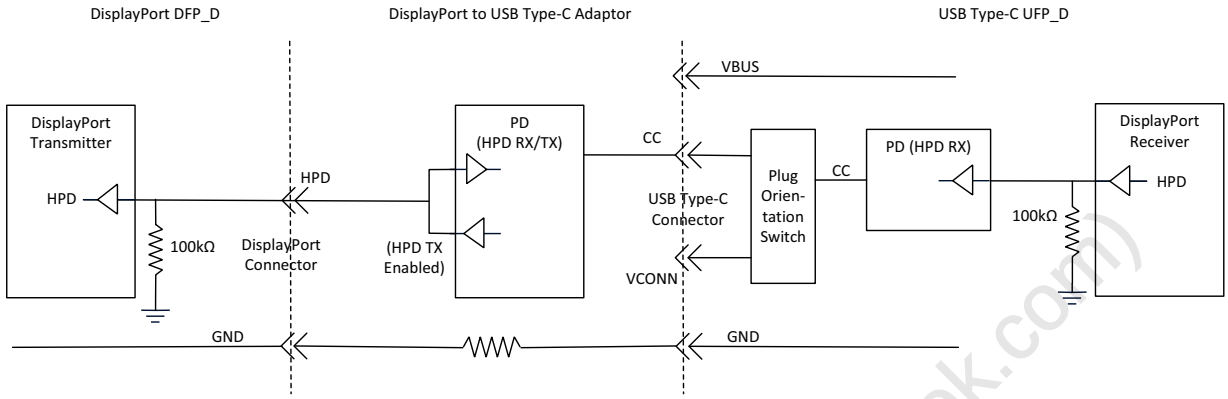


Figure 3-8: HPD Signaling Using DisplayPort to USB Type-C Adaptors

Figure 3-9 illustrates the schematic for an implementation of a PD-to-HPD converter in a DFP_D and an HPD-to-PD converter in a Protocol Converter. In this design, the Protocol Converter is used to convert DisplayPort protocol to a non-DisplayPort audio-video protocol.

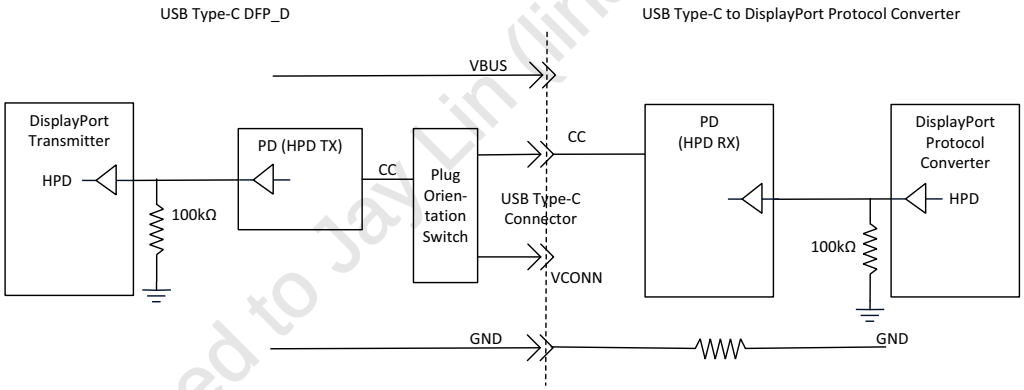


Figure 3-9: HPD Signaling Using USB Type-C to Protocol Converters

3.16 DisplayPort Connector CONFIG1 and CONFIG2 Pins

The DisplayPort plug's CONFIG1 and CONFIG2 pins in USB Type-C to DisplayPort Adaptors shall each be pulled low to GND with a 1MΩ pull-down resistor. Section 4.5 describes the CONFIG1 and CONFIG2 connection for a DisplayPort receptacle Adaptor to a Multi-function dock.

Note: *DisplayPort Dual-Mode Adaptors, as specified in DP++ v1.1, are not supported on the USB Type-C to DisplayPort Adaptor's DisplayPort connector.*

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4 USB Type-C Cables Used for DisplayPort

This section describes the Cable assemblies and Adaptors used for DisplayPort signaling through a USB Type-C connector.

All associations of USB Type-C functions with specific pins on the USB Type-C connector are informative. Reference should be made to *USB Type-C r1.1* for normative USB pin functionality. The mechanical and electrical properties of the USB Type-C Receptacle and Plug are specified in *USB Type-C r1.1*.

4.1 Scenario 1 USB Type-C Cable

Figure 4-1 illustrates an example of a USB Full-Featured Type-C Cable assembly (as defined in *USB Type-C r1.1*) used for DisplayPort. DisplayPort signaling speeds are as for USB Gen 2 (10.0Gbps) or as specified in *DP v1.3* (Passive cables only). Multi-function (simultaneous DisplayPort and USB Enhanced SuperSpeed) is supported.

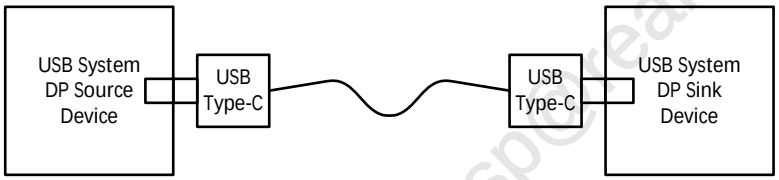


Figure 4-1: Scenario 1

Licensed to Jay Lin (linspc@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.1.1 Scenario 1 USB Type-C Cable Using GEN2_BR

This scenario uses DFP_D Pin Assignment A at the DisplayPort Source device and UFP_D Pin Assignment A at the DisplayPort Sink device. The figures illustrate DisplayPort use of the signal connections in the USB Full-Featured Type-C Cable assembly, as follows:

- Figure 4-2 illustrates when the plugs at both ends are inserted in normal orientation
- Figure 4-3 illustrates when the DFP_D plug is inverted and the UFP_D plug is inserted in normal orientation
- Figure 4-4 illustrates when the DFP_D plug is inserted in normal orientation and the UFP_D plug is inverted
- Figure 4-5 illustrates when the plugs at both ends are inverted

Note: *The figures omit the USB-PD Identification Circuitry that is incorporated in the Cable assembly.*

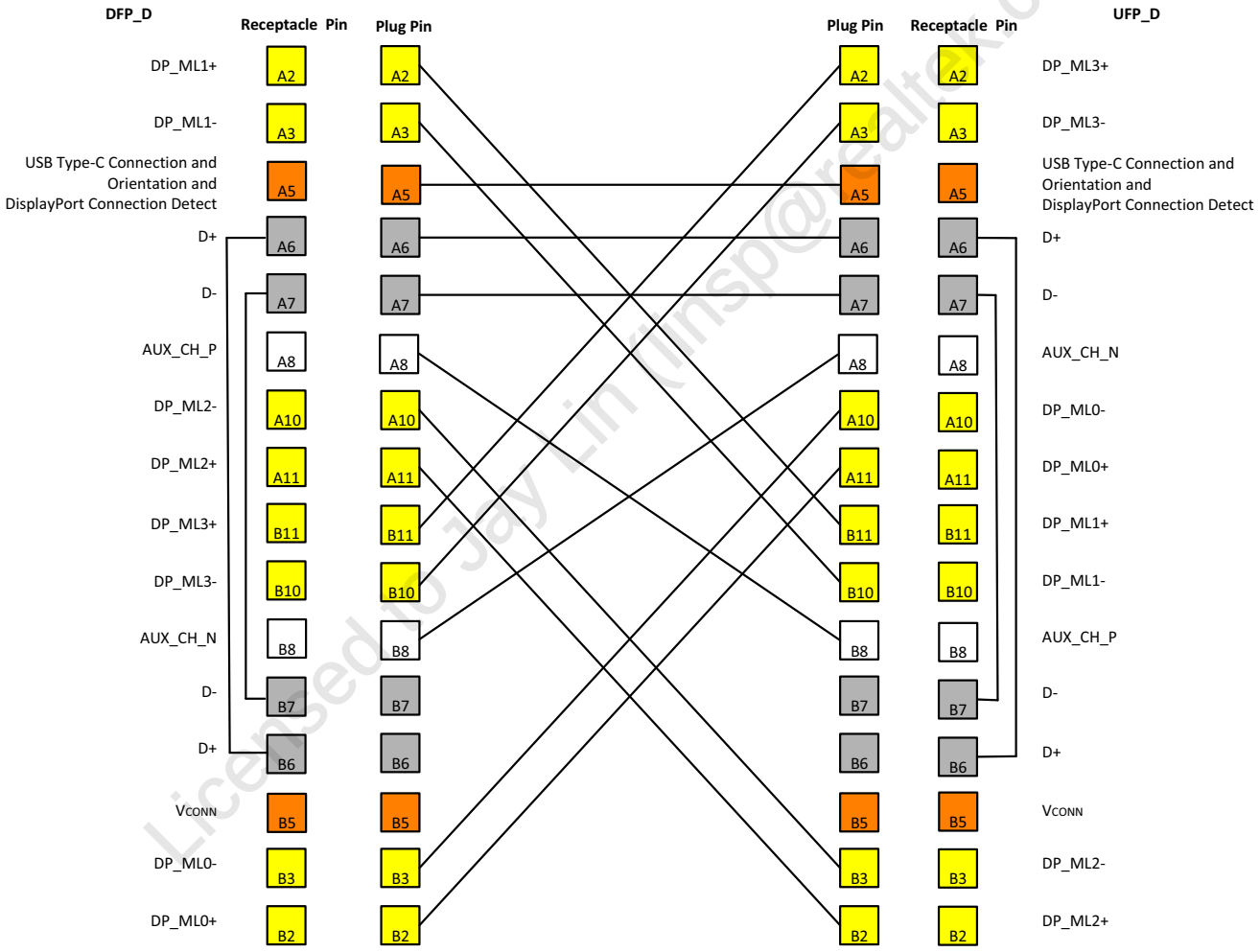


Figure 4-2: Scenario 1 GEN2_BR Cable Wiring – Normal Plug Orientation

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

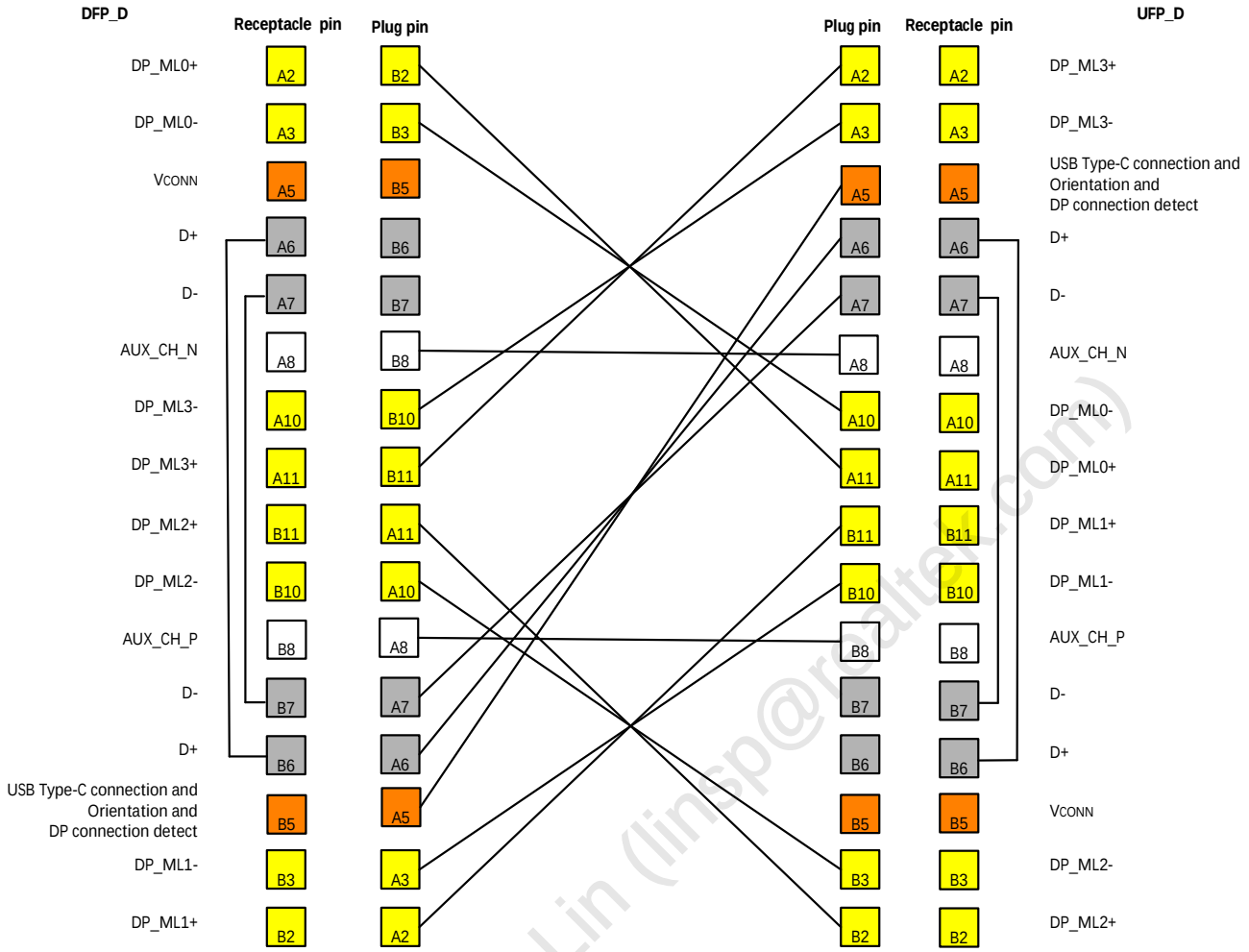


Figure 4-3: Scenario 1 GEN2_BR Cable Wiring – DFP_D Plug Inverted

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

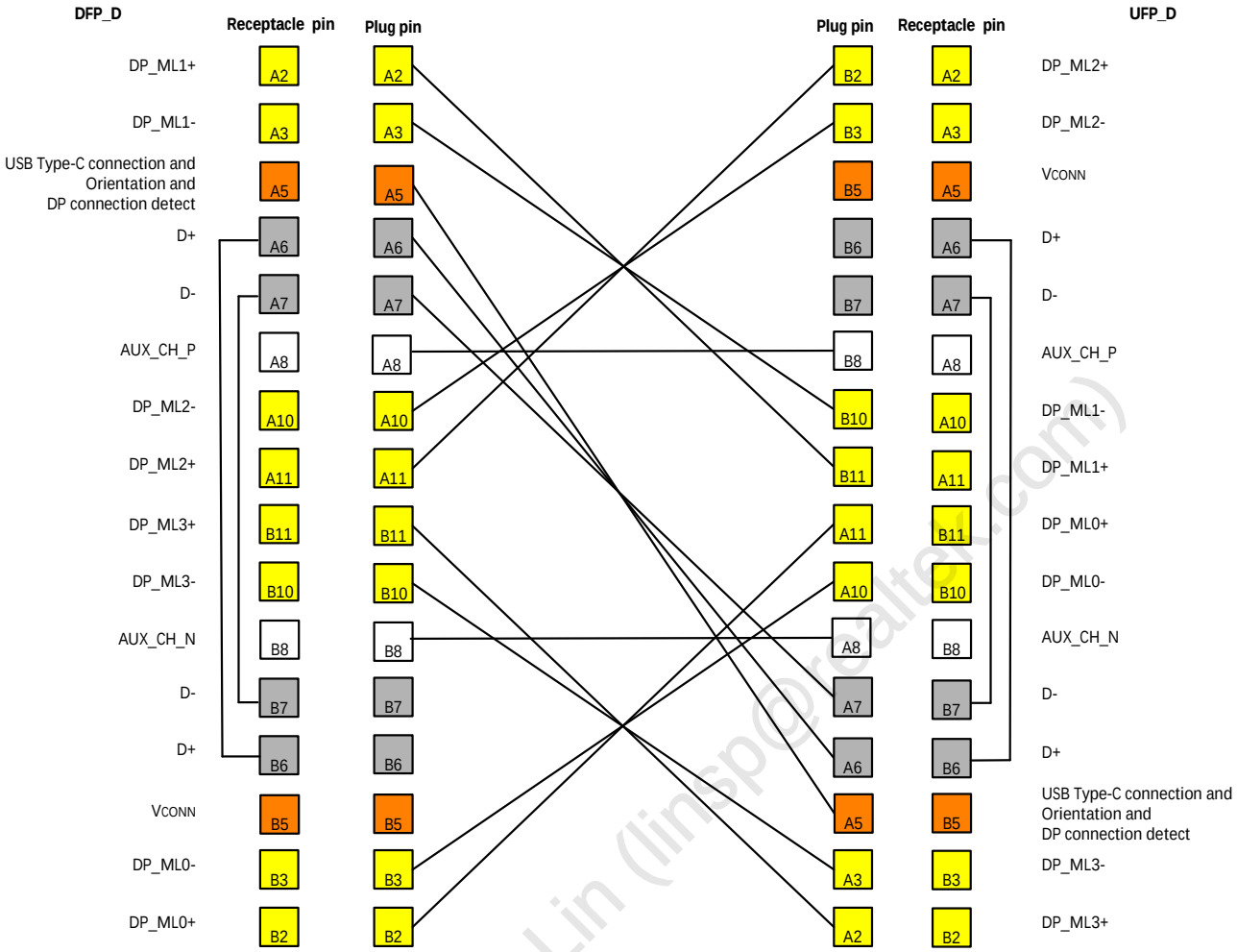


Figure 4-4: Scenario 1 GEN2_BR Cable Wiring – UFP_D Plug Inverted

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

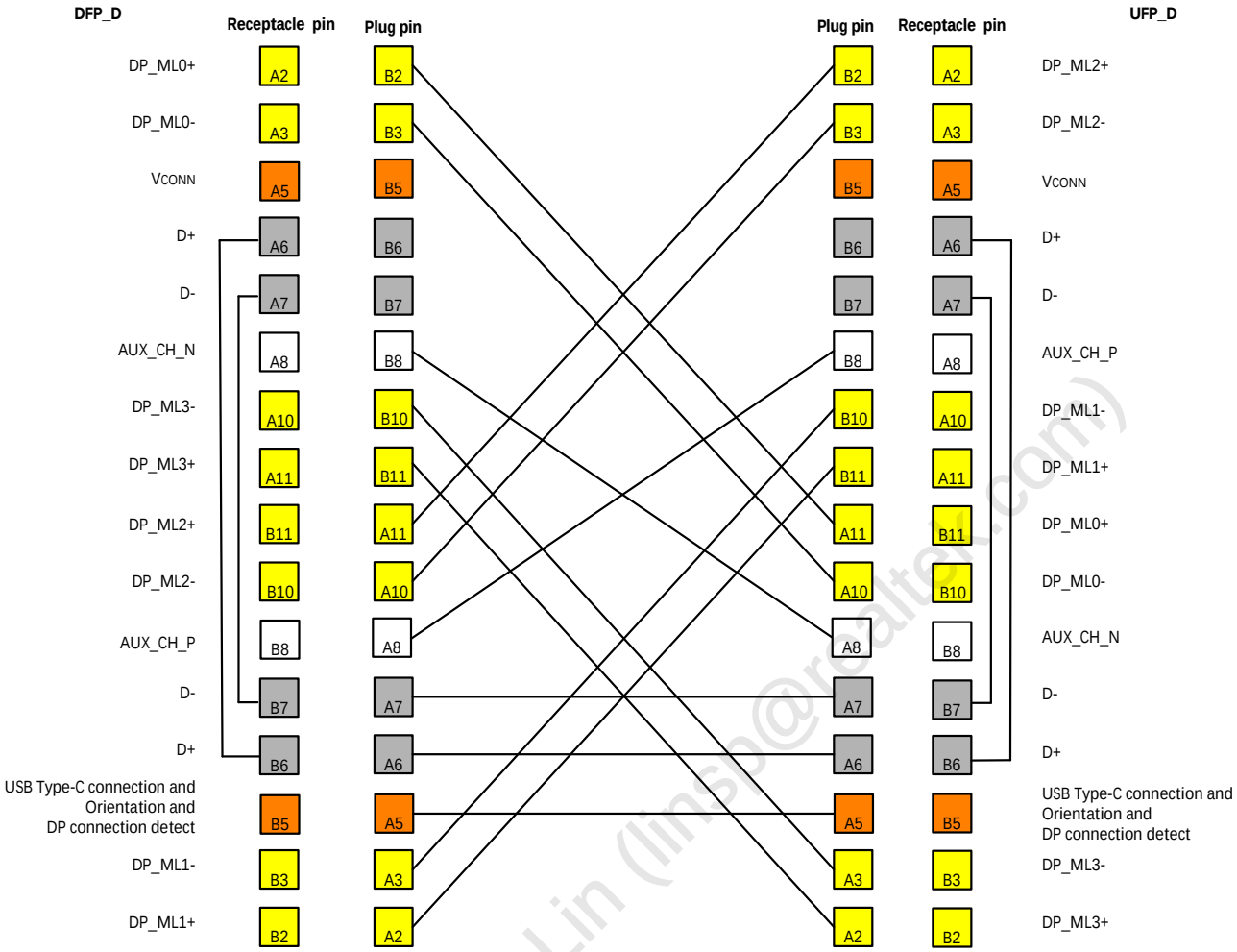


Figure 4-5: Scenario 1 GEN2_BR Cable Wiring – DFP_D and UFP_D Plug Inverted

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.1.2 Scenario 1 USB Type-C Cable using DP_BR

This scenario uses DFP_D Pin Assignment C at the DisplayPort Source device and UFP_D Pin Assignment C at the DisplayPort Sink device. The figures illustrate DisplayPort use of the signal connections in the USB Full-Featured Type-C Cable assembly, as follows:

- Figure 4-6 illustrates when the plugs at both ends are inserted in normal orientation
- Figure 4-7 illustrates when the DFP_D plug is inverted and the UFP_D plug is inserted in normal orientation
- Figure 4-8 illustrates when the DFP_D plug is inserted in normal orientation and the UFP_D plug is inverted
- Figure 4-9 illustrates when the plugs at both ends are inverted

Note: *The figures omit the USB-PD Identification Circuitry that is incorporated in the Cable assembly.*

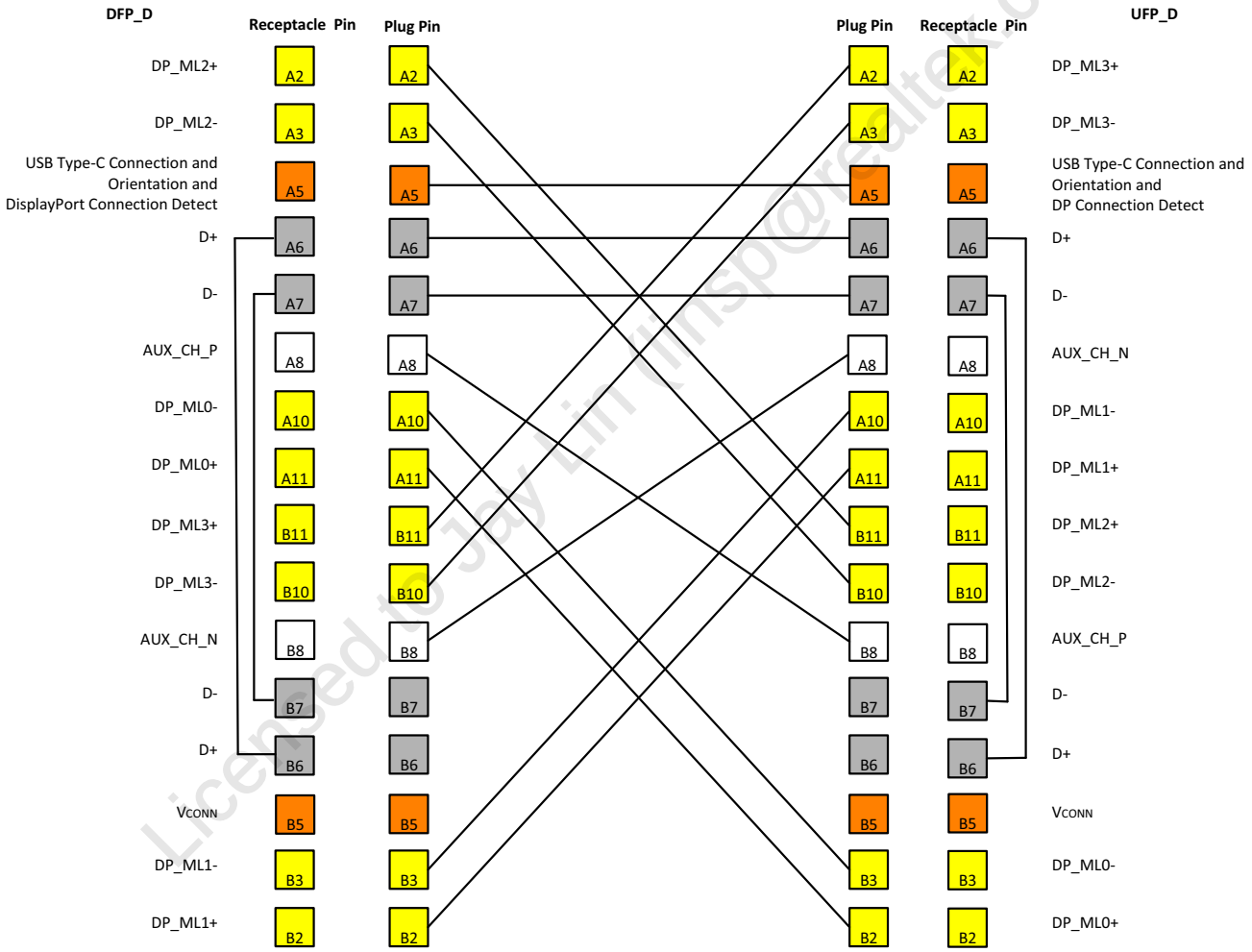


Figure 4-6: Scenario 1 DP_BR Cable Wiring – Normal Plug Orientation

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

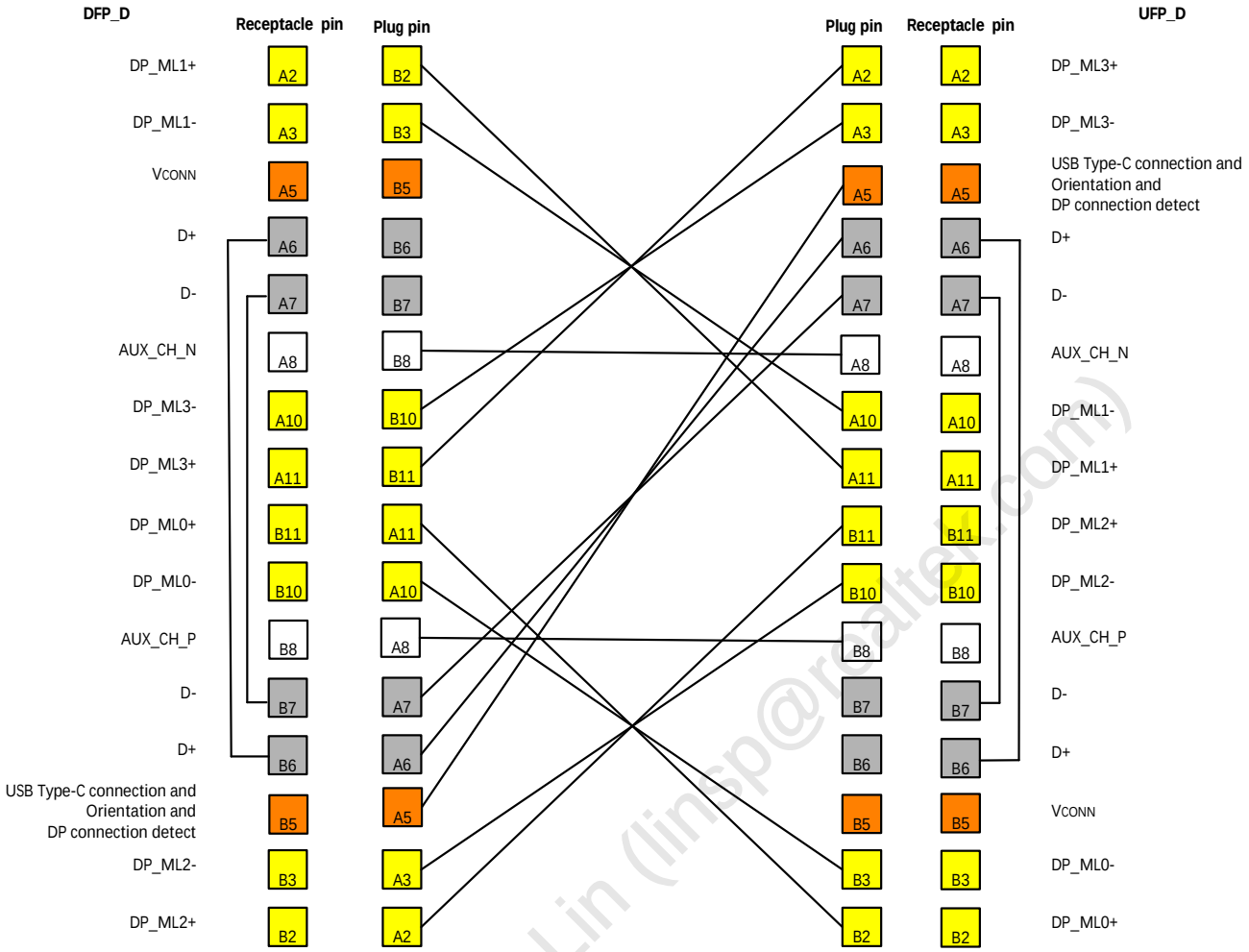


Figure 4-7: Scenario 1 DP_BR Cable Wiring – DFP_D Plug Inverted

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

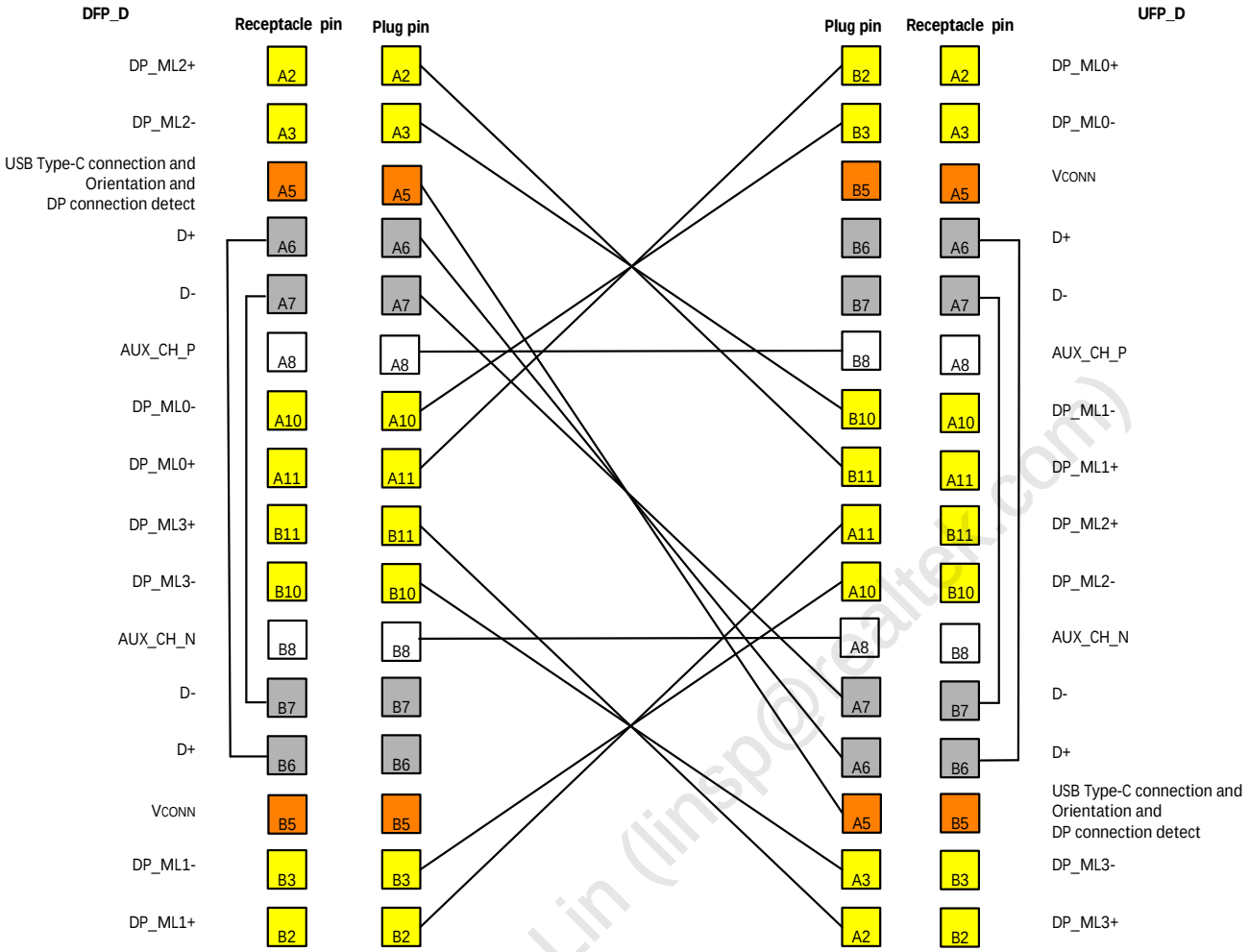


Figure 4-8: Scenario 1 DP_BR Cable Wiring – UFP_D Plug Inverted

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

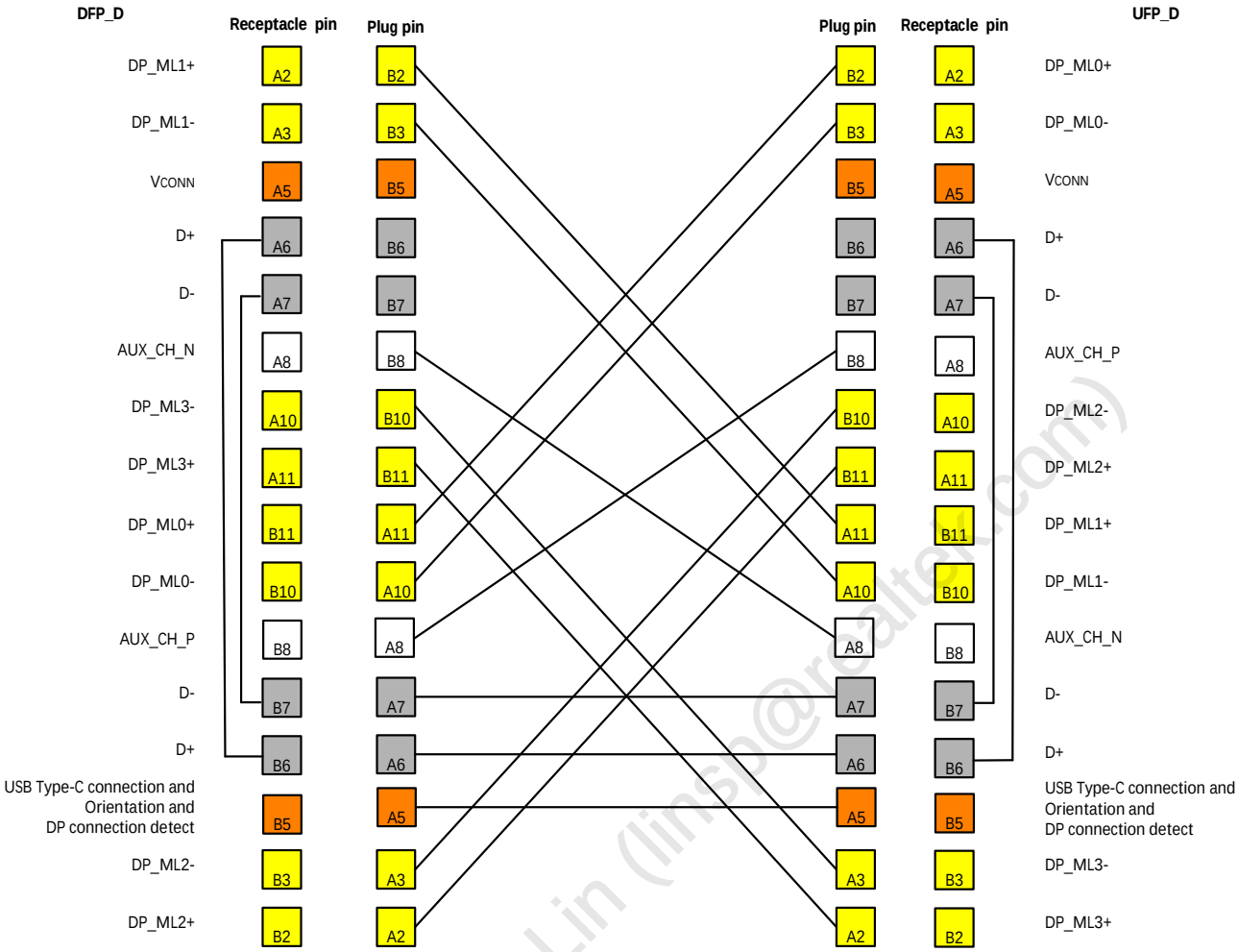


Figure 4-9: Scenario 1 DP_BR Cable Wiring – DFP_D and UFP_D Plug Inverted

Licensed to Jay Lin (linsb@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.2 Scenarios 2a and 2b USB Type-C to DisplayPort Cables

4.2.1 Scenario 2a USB Type-C DFP_D to DisplayPort UFP_D

Figure 4-10 illustrates a USB Type-C Plug to DisplayPort plug Cable assembly. Signaling speeds are as per *DP v1.3*.

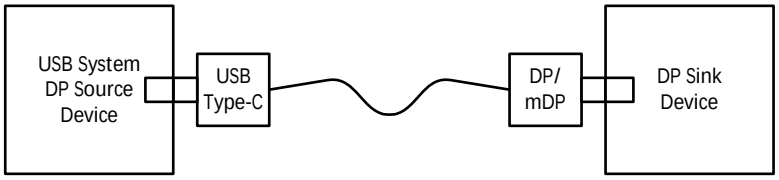


Figure 4-10: Scenario 2a

Table 4-1 lists DisplayPort use of the signal connections in the USB Type-C DFP_D to DisplayPort UFP_D Cable assembly. Figure 4-11 illustrates this use when the USB Type-C Plug is inserted in normal orientation. Figure 4-12 illustrates this use when the USB Type-C Plug is inserted in flipped orientation. This scenario uses DFP_D Pin Assignment E at the DisplayPort Source device. The pin numbering on the DisplayPort connector is that for the full-sized DisplayPort connector. See Table 4-1 for the mini DisplayPort connector pin numbering.

The Cable assembly shall contain logic to implement USB-PD and support the Discovery Process (see Section 5). The Cable assembly contains a *USB Billboard r1.0* device, which the PD logic shall connect to D+/D- in the event that switching to using DisplayPort as an Alternate Mode is unsuccessful. The Cable assembly's connections behind pins A10, A11, B10, and B11 on the USB Type-C Plug shall contain series AC capacitors as specified in *DP v1.3* (75 to 200nF), because these pins are required to be AC-coupled at the DFP_D by *DP v1.3*, but are DC-connected in the system behind the USB Type-C Receptacle. The series AC capacitors on these pins are also required by *USB Type-C r1.1*.

Note: *Figure 4-11 and Figure 4-12 omit the pull-down resistors on CONFIG1 and CONFIG2.*

Licensed to Jay Lin (jay@shinecable.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

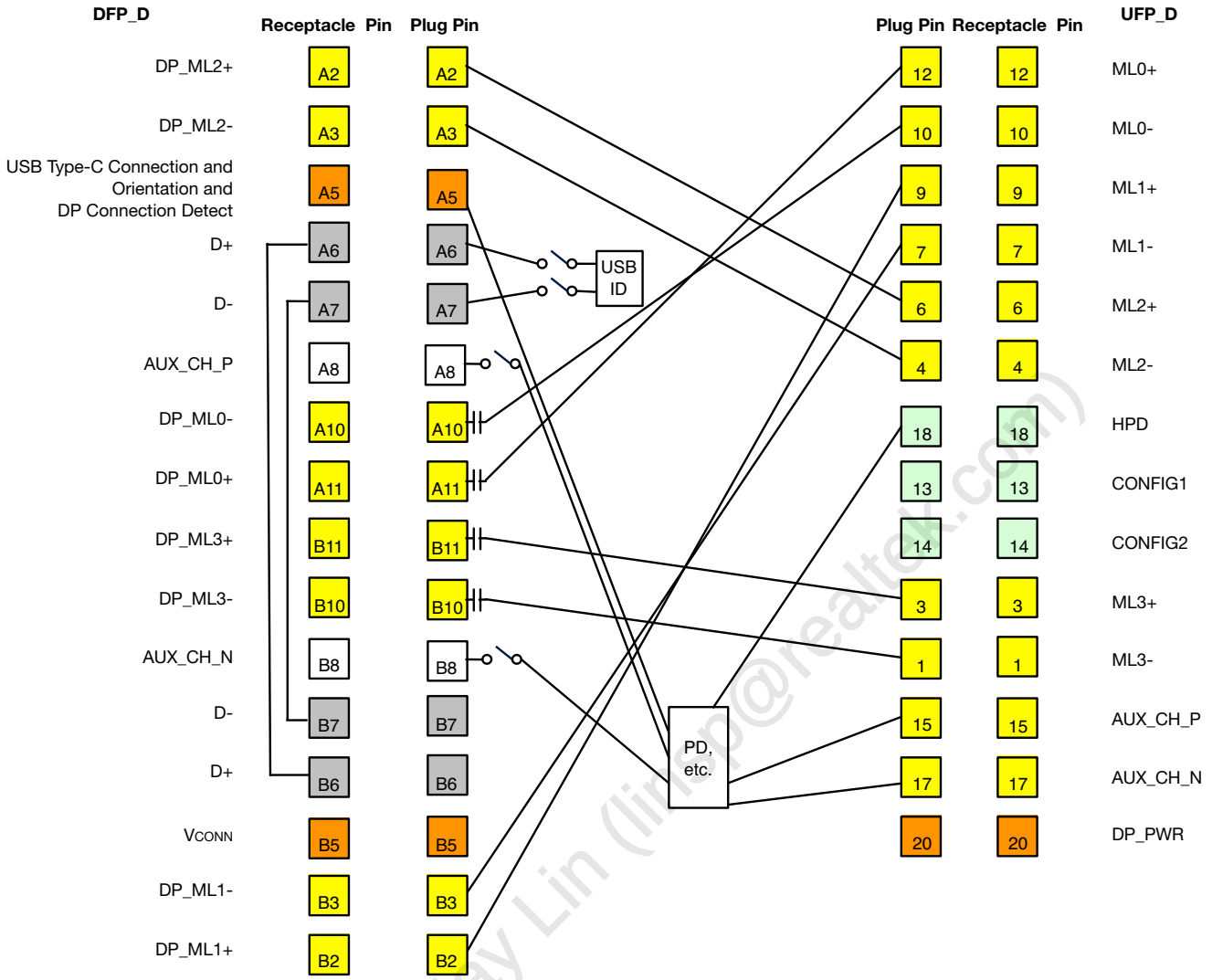


Figure 4-11: Scenario 2a Cable Wiring – Normal Plug Orientation

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

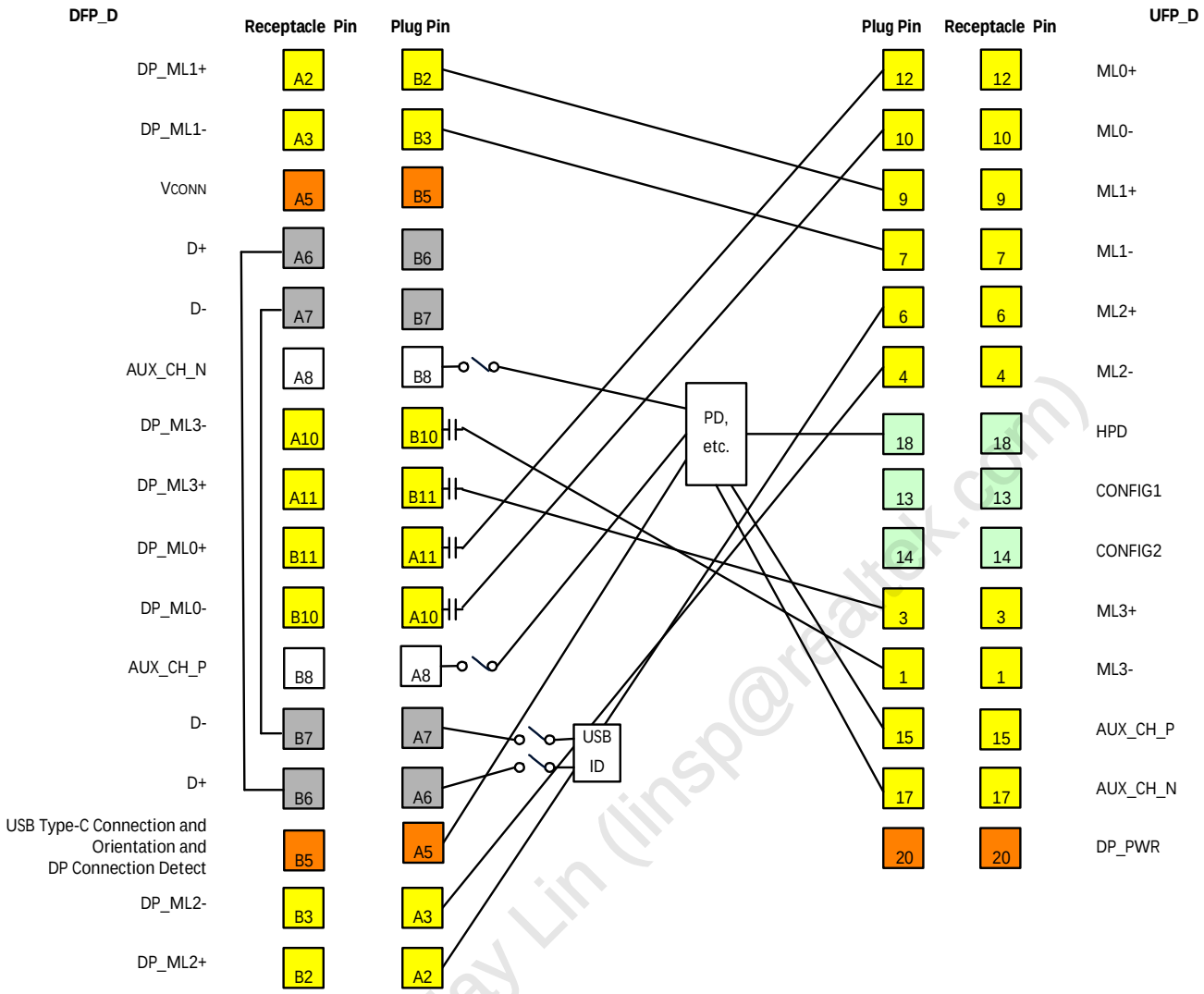


Figure 4-12: Scenario 2a Cable Wiring – DFP_D Plug Inverted

Licensed to Jay Lin (insp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 4-1: USB Type-C DFP_D to DisplayPort UFP_D Use

USB Type-C Plug #1 DFP_D		Wire		DisplayPort/mini DisplayPort Plug UFP_D		
Pin	**(DP) Signal Name**	**Wire Number**	**Signal Name**	**DP Pin**	**mDP Pin**	**(DP) Signal Name**
A1, B1, A12, B12	GND	1 [16]	GND_PWRrt1 [GND_PWRrt2]	2, 5, 8, 11, 16, 19	1, 7, 8, 13, 14, 19	GND
A4, B4, A9, B9	VBUS	2	Powers USB Billboard			
A5	CC	3	Cable logic (PD)			
A6	Dp1	4	Cable logic (USB Billboard)			
A7	Dn1	5	Cable logic (USB Billboard)			
			1M to GND	13	4	CONFIG1
			1M to GND	14	6	CONFIG2
A2	ML2+	6	SDPp1	6	11	ML2+
A3	ML2-	7	SDPn1	4	9	ML2-
B11	ML3+	8	SDPp2	3	5	ML3+
B10	ML3-	9	SDPn2	1	3	ML3-
B2	ML1+	10	SDPp3	9	17	ML1+
B3	ML1-	11	SDPn3	7	15	ML1-
A11	ML0+	12	SDPp4	12	12	ML0+
A10	ML0-	13	SDPn4	10	10	ML0-
A8	AUX_CH_P	14	Cable logic	15	16	AUX_CH_P
B8	AUX_CH_N	15	Cable logic	17	18	AUX_CH_N
			Cable logic	18	2	HPD
B5	VCONN	16	Powers Cable logic			
			No Connect	20	20	DP_PWR
Shell	Shield	Braid	Shield	Shell	Shell	Shield

Licensed to Jay Lin (msp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.2.2 Scenario 2b DisplayPort DFP_D to USB Type-C UFP_D

Figure 4-13 illustrates a DisplayPort plug to USB Type-C Plug Cable assembly. Signaling speeds are as per *DP v1.3*.

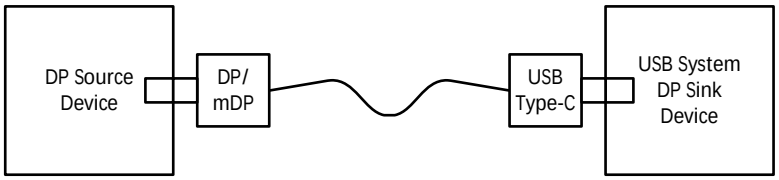


Figure 4-13: Scenario 2b

Table 4-2 lists DisplayPort use of the signal connections in the DisplayPort DFP_D to USB Type-C UFP_D Cable assembly. Figure 4-14 illustrates when the USB Type-C Plug is inserted in normal orientation. Figure 4-15 illustrates when the USB Type-C Plug is inserted in flipped orientation. This scenario uses UFP_D Pin Assignment E. The Cable assembly is the same Cable assembly that is used in Scenario 2a, but with DisplayPort signaling running in the opposite direction. The pin numbering on the DisplayPort connector is that for the full-sized DisplayPort connector. See Table 4-2 for the mini DisplayPort connector pin numbering.

Pins A10, A11, B10, and B11 are DC-coupled in the system behind the USB Type-C Receptacle; thus, the capacitors on these pins provide the optional AC-coupling permitted by *DP v1.3*.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

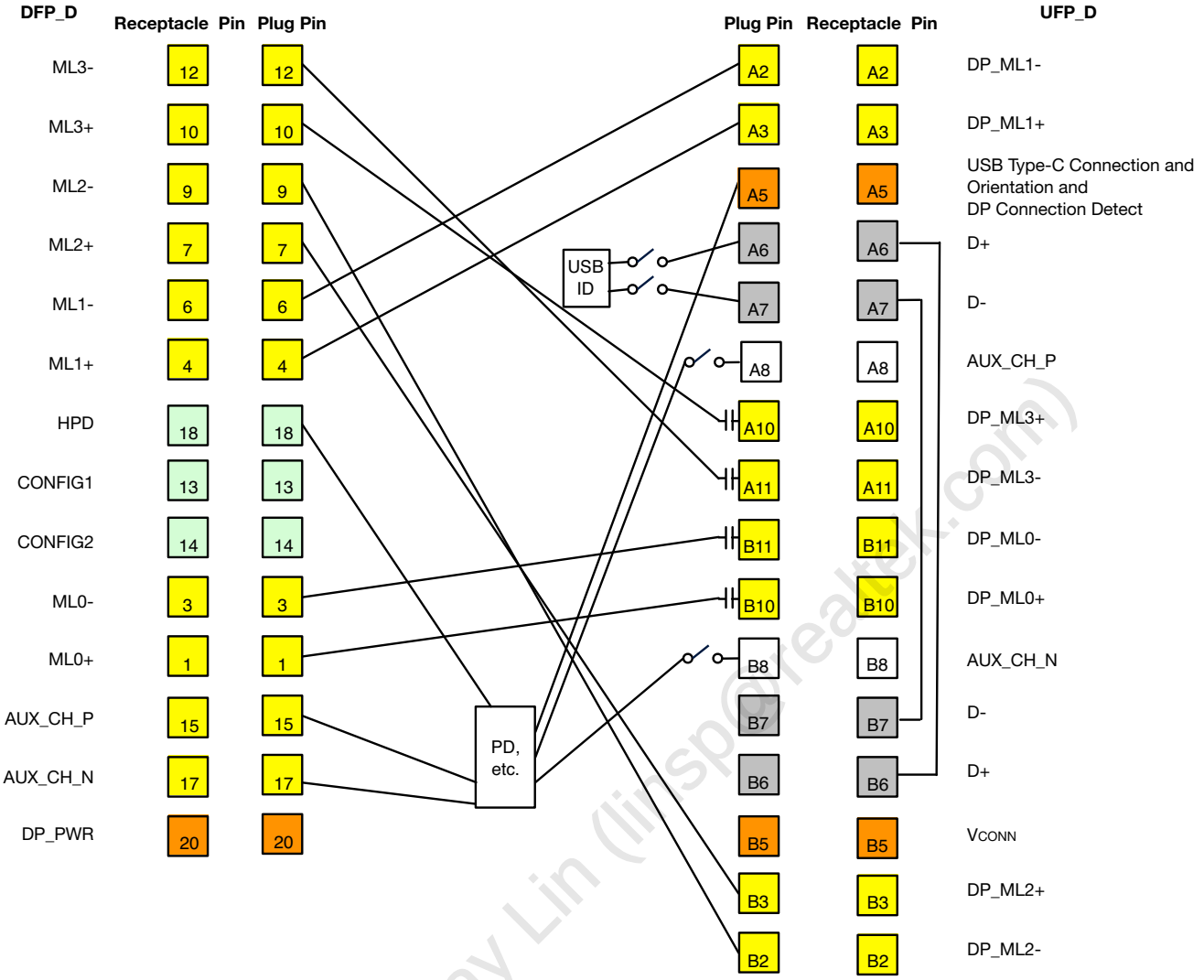


Figure 4-14: Scenario 2b Cable Wiring – Normal Plug Orientation

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

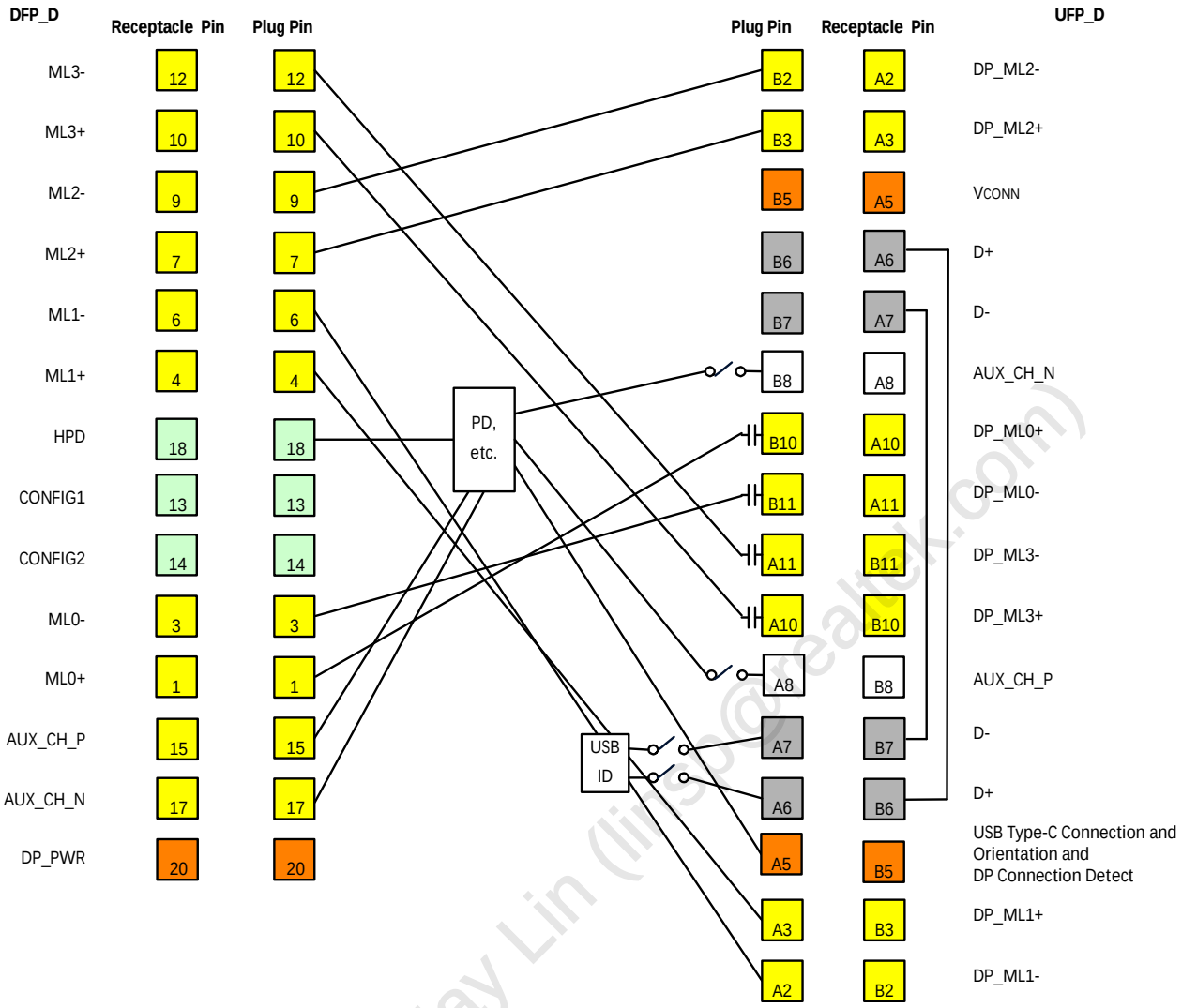


Figure 4-15: Scenario 2b Cable Wiring – UFP_D Plug Inverted

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 4-2: DisplayPort DFP_D to USB Type-C UFP_D Use

DisplayPort/mini DisplayPort Plug DFP_D			Wire		USB Type-C Plug UFP_D	
(DP) Signal Name	**DP Pin**	**mDP Pin**	**Signal Name**	**Wire Number**	**Pin**	**(DP) Signal Name**
GND	2, 5, 8, 11, 16, 19	1, 7, 8, 13, 14, 19	GND_PWRrt1 [GND_PWRrt2]	1 [16]	A1, B1, A12, B12	GND
			Powers USB Billboard	2	A4, B4, A9, B9	VBUS
			Cable logic (PD)	3	A5	CC
			Cable logic (USB Billboard)	4	A6	Dp1
			Cable logic (USB Billboard)	5	A7	Dn1
CONFIG1	13	4	1M to GND			
CONFIG2	14	6	1M to GND			
ML1-	6	11	SDPp1	6	A2	ML1-
ML1+	4	9	SDPn1	7	A3	ML1+
ML0-	3	5	SDPp2	8	B11	ML0-
ML0+	1	3	SDPn2	9	B10	ML0+
ML2-	9	17	SDPp3	10	B2	ML2-
ML2+	7	15	SDPn3	11	B3	ML2+
ML3-	12	12	SDPp4	12	A11	ML3-
ML3+	10	10	SDPn4	13	A10	ML3+
AUX_CH_P	15	16	Cable logic	14	A8	AUX_CH_P
AUX_CH_N	17	18	Cable logic	15	B8	AUX_CH_N
HPD	18	2	Cable logic			
			Powers Cable logic	16	B5	VCONN
DP_PWR	20	20	No Connect			
Shield	Shell	Shell	Shield	Braid	Shell	Shield

Licensed to Jay Lin (jlsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.2.3 USB Type-C to DisplayPort Cable Electrical Properties

The electrical specification of the Cable Assembly is based on the specification methodology for USB Type-C to USB Type-C Passive Cable Assemblies (see *USB Type-C r1.1*, Section 3.7.3). Insertion loss and Return loss are to be tested relative to an 85Ω differential environment at the USB Type-C connector and relative to a 100Ω differential environment at the DisplayPort connector. The appropriate parameters in the referenced methodology are to be adjusted accordingly. With respect to *fmax* and T_b in *USB Type-C r1.1*, Section 3.7.3.2.2, the values to be used shall be *fmax* = 12.15GHz and T_b = 123.4ps.

Note: *Using the methodology described in USB Type-C, the measurement results for the USB Type-C end of a Cable assembly do not include the USB Type-C Receptacle, but do include the DisplayPort receptacle at the DisplayPort end.*

4.2.3.1 Intra-pair Skew (Informative)

The intra-pair skew should be limited to 30ps. This is designed to support 2m cables at 15ps/m.

4.2.3.2 Insertion Loss Fit at Nyquist Frequencies

The Insertion Loss Fit at Nyquist Frequency (ILFitAtNq) shall meet the following requirements:

- $\geq$ -5.5dB at 1.35GHz (HBR)
- $\geq$ -8.5dB at 2.7GHz (HBR2)
- $\geq$ -11.0dB at 4.05GHz (HBR3)

1.35, 2.7, and 4.05GHz are the Nyquist frequencies for HBR, HBR2, and HBR3, respectively.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 4-16 through Figure 4-18 illustrate the Insertion Loss and the Insertion Loss Fit curves for an example Cable assembly. Each graph identifies the Insertion Loss and the Insertion Loss Fit for the Nyquist frequency corresponding to one of the three DisplayPort high-bit transmission rates, also providing the ILFitAtNq limit in each case.

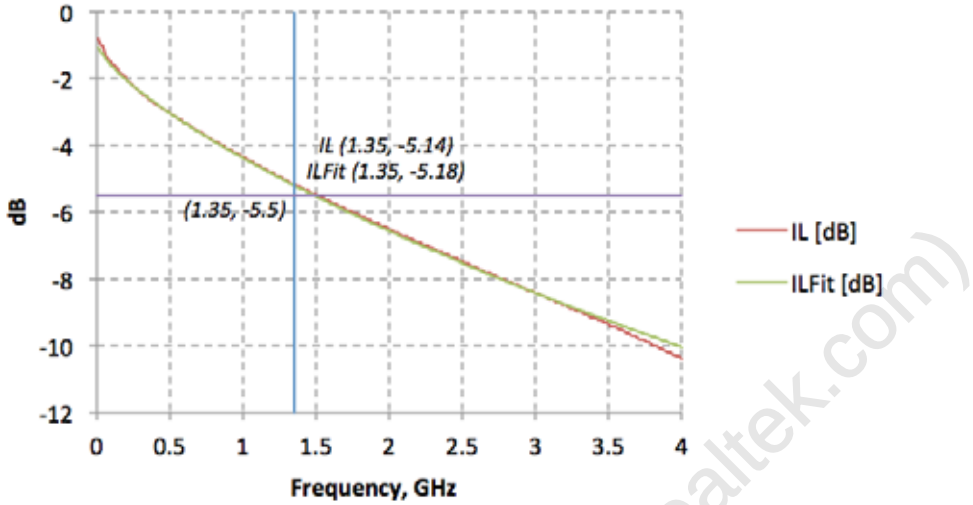


Figure 4-16: Insertion Loss Fit at Nyquist for HBR

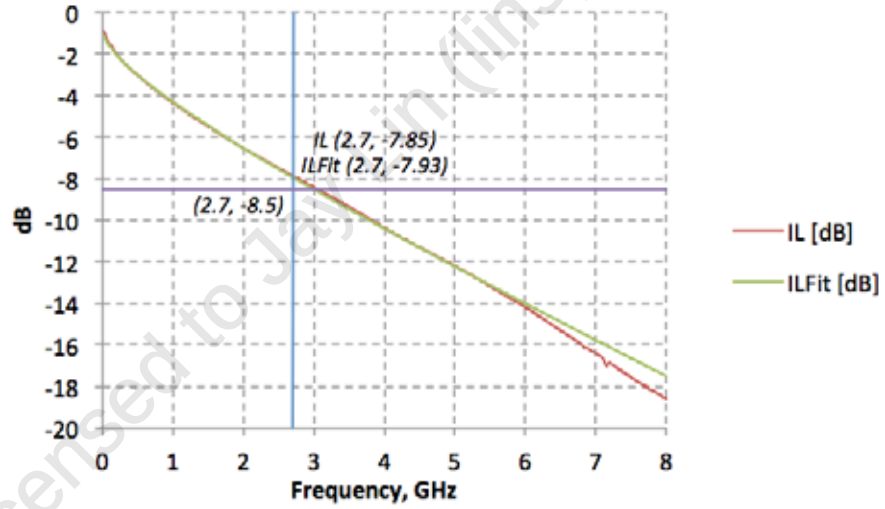


Figure 4-17: Insertion Loss Fit at Nyquist for HBR2

Licensed to Jay Jin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

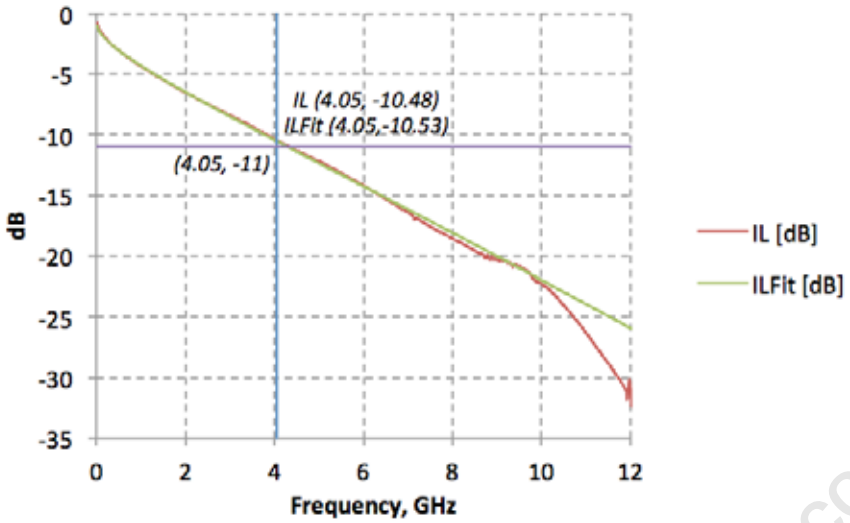


Figure 4-18: Insertion Loss Fit at Nyquist for HBR3

4.2.3.3 Integrated Multi-reflection

For the Input Pulse Spectrum (used in the calculation of integrated multi-reflection (IMR)), the DisplayPort HBR, HBR2, and HBR3 correspond to a UI of 370.4, 185.2, and 123.5ps, respectively. The rise time is 0.4UI.

IMR has dependency on ILFitAtNq. More IMR may be tolerated when ILFitAtNq decreases.

The IMR limit is specified as a function of ILFitAtNq:

$$IMR \leq 0.0579 * ILFitAtNq^2 + 1.64 * ILFitAtNq - 30.23$$

Figure 4-19 illustrates the compensation curve of IMR as a function of ILFitAtNq at HBR3.

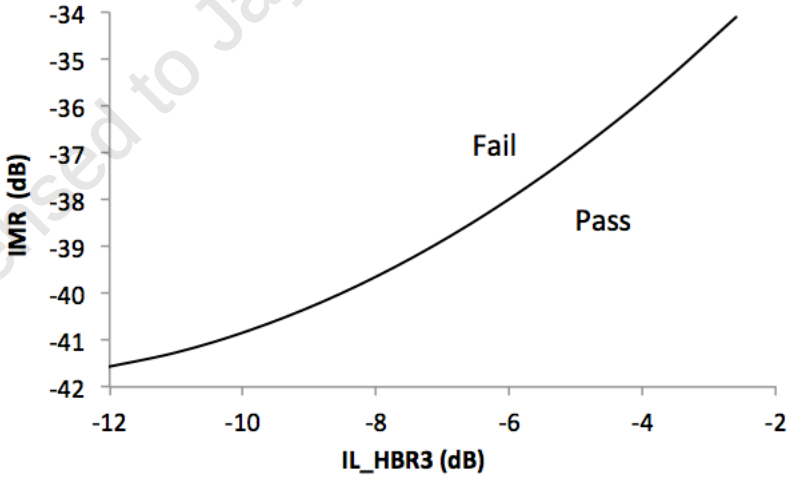


Figure 4-19: IMR as a Function of ILFitAtNq at HBR3

Licensed to Jay Lin (jlsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.2.3.4 Power Summed Integrated Crosstalk between DP Lanes

The power sum limits for integrated crosstalk between the DisplayPort lanes shall comply with the following limits:

- IFEXT $\leq$ -35dB (up to 12GHz)
- INEXT $\leq$ -35dB (up to 12GHz) when measured through the USB Type-C connector
- INEXT $\leq$ -33dB (up to 12GHz) when measured through the DisplayPort connector

4.2.3.5 Integrated Return Loss

IRL has a strong dependency on ILFitAtNq, and its limit is specified as a function of ILFitAtNq:

$$IRL \leq 0.0336 * ILFitAtNq^2 + 1.28 * ILFitAtNq - 10.87$$

Figure 4-20 illustrates the compensation curve of IRL as a function of ILFitAtNq at HBR3.

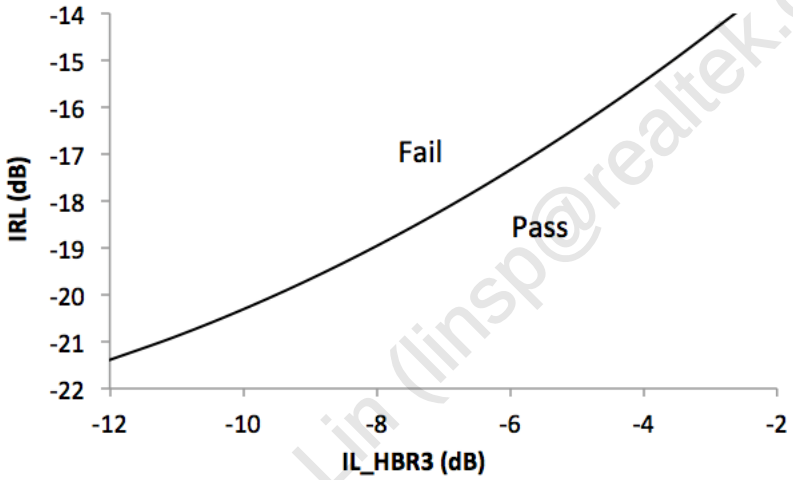


Figure 4-20: IRL as a Function of ILFitAtNq at HBR3

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.2.3.6 Differential to Common Mode Conversion

SCD12/SCD21 shall be ≤ -17dB from 100MHz to 8.1GHz.

4.2.4 USB Type-C to DisplayPort Cable Functionality

The Cable assembly includes logic to perform the following functions:

- Support of USB-PD
- DisplayPort DFP_D/UFP_D Detection

Both are described in the sub-sections that follow.

4.2.4.1 Support of USB-PD

The logic supports the USB-PD protocol in the role of a UFP_U, negotiates a power contract, and responds to PD Commands for the discovery of DisplayPort Mode capability, entry and exit by the connected DFP_U using structured Vendor-Defined Messages (VDMs) as specified in *USB PD r2.0 v1.1* and *USB Type-C r1.1*. The logic reports whether the DisplayPort connector is:

- Disconnected,
- Connected to a DisplayPort DFP_D, –or–
- Connected to a DisplayPort UFP_D

4.2.4.2 DisplayPort DFP_D/UFP_D Detection

The Cable assembly shall contain logic to detect the presence of a connected DisplayPort DFP_D or a connected DisplayPort UFP_D, and shall use messaging on USB-PD to report the DisplayPort connector status:

- Disconnected,
- Connected to a DisplayPort DFP_D, –or–
- Connected to a DisplayPort UFP_D

The Adaptor shall report a change of state between these three states, also using messaging on USB-PD.

The following description outlines a method by which connection detection can be performed. Any implementation with equivalent behavior may be used.

DP v1.3 defines pull-up and pull-down resistors for the AUX_CH_P/AUX_CH_N signal pair at both DisplayPort DFP_D and DisplayPort UFP_D, between the AC-coupling capacitors. This implementation allows a DisplayPort UFP_D to determine when a DisplayPort DFP_D has been connected. Table 4-3 lists the use of the pull-up and pull-down resistors for the AUX_CH_P/AUX_CH_N signal pair.

Table 4-3: DisplayPort Connector AUX States

Signal	DisplayPort DFP_D	DisplayPort UFP_D	Logic Value When Connected
AUX_CH_P	100kΩ to GND	1MΩ to 3.3V	0
AUX_CH_N	100kΩ to 3.3V	1MΩ to GND	1

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

A non-connected Adaptor creates a floating AUX bias between the AC-coupling capacitors (and therefore an unknown logic state). To avoid this, a very weak (4.7MΩ) pull-down resistor is added to AUX_CH_N. A weak (470kΩ) pull-down resistor is used on AUX_CH_P as part of the DisplayPort UFP_D detection mechanism.

After filtering to generate AUX logic states while ignoring actual AUX channel traffic and filtering HPD to avoid an IRQ-HPD being detected as a disconnect, the state of the DisplayPort connector may be determined.

Initially, the Adaptor places its HPD driver into a high-impedance state.

A powered DisplayPort DFP_D always has a 100kΩ pull-up resistor on AUX_CH_N. Thus, given that the Adaptor's SBU2 isolation switch is open, a logic 1 on AUX_CH_N indicates a connected DisplayPort DFP_D. If the Adaptor detects a connected DisplayPort DFP_D, it reports the presence of the connected DisplayPort DFP_D to the USB Type-C system over PD and awaits a DisplayPort Configure command from the USB Type-C system to switch into UFP_D Configuration. On receipt of this command, the Adaptor closes the SBU isolation switches, thus connecting the AUX signals between the UFP_D on the USB Type-C connector and DFP_D on the DisplayPort connector. The UFP_D on the USB Type-C connector will provide HPD status updates to the Adaptor using PD messages, and the Adaptor will control the HPD signal on the DisplayPort connector accordingly. Because the UFP_D on the USB Type-C connector has only a weak pull-up resistor on AUX_CH_P and a weak pull-down resistor on AUX_CH_N, the filtered states on these connections does not change. If the DFP_D is subsequently disconnected from the DisplayPort connector, the (filtered) state of AUX_CH_N will revert to a low state and the (filtered) state on AUX_CH_P will revert to a high state. The Adaptor will report the disconnected state to the UFP_D using USB PD messaging, and the UFP_D, in turn, will signal to isolate the SBU connections.

A powered DisplayPort UFP_D will provide the weak pull-up resistor on AUX_CH_P. Some displays do not provide this pull-up resistor, but assert HPD without DisplayPort DFP_D detection. Thus, given that the Adaptor's SBU1 isolation switch is open, either a logic 1 (using a threshold appropriate for the Adaptor's weak pull-down resistor on AUX_CH_P) on AUX_CH_P or HPD indicates a connected DisplayPort UFP_D. If the Adaptor detects the presence of a connected DisplayPort UFP_D, the Adaptor reports the presence of the connected DisplayPort UFP_D to the USB Type-C system over PD and awaits a DisplayPort Configure command from the USB Type-C system to switch into DisplayPort DFP_D Configuration. On receipt of this command, the Adaptor closes the SBU isolation switches, thus connecting the AUX signals on the DisplayPort connector toward the DisplayPort UFP_D. If the UFP_D is waiting to detect the presence of the DFP_D before asserting HPD, the UFP_D will detect this when the SBU isolation switches are closed. The Adaptor provides status updates for HPD to the DFP_D on the USB Type-C connector using PD messages. When the SBU isolation switches are closed, the 100kΩ pull-down resistor on AUX_CH_P in the DFP_D at the USB Type-C Receptacle will cause the (filtered) state of AUX_CH_P to transition to logic 0 and the 100kΩ pull-up resistor on AUX_CH_N in the DFP_D will cause the (filtered) state of AUX_CH_N to transition to logic 1. If the UFP_D is subsequently disconnected from the DisplayPort connector, the (filtered) state of HPD will transition to logic 0. The Adaptor will report the disconnected state to the DFP_D using USB PD messaging, and the DFP_D, in turn, will signal to isolate the SBU connections.

Unauthorized copy in circulation

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 4-4 enumerates the possible states of HPD, AUX_CH_P, and AUX_CH_N when there is no connection on the DisplayPort connector or the Adaptor is operating in DFP_D Configuration (i.e., connected to a UFP_D at the DisplayPort connector).

Table 4-4: DisplayPort Connector HPD and AUX States When Disconnected or in Adaptor DFP_D Configuration

HPD	AUX_CH_P	AUX_CH_N	Interpretation
0	0	0	Nothing is connected, or a powered-down device is connected.
0	0	1	An external DisplayPort DFP_D has been connected.
0	1	0	An external DisplayPort UFP_D has been connected and is waiting to detect the presence of the DisplayPort DFP_D (i.e., waiting for the 100kΩ resistors to be connected). Presence of DisplayPort UFP_D is reported.
0	1	1	Invalid.
1	0	0	An external DisplayPort UFP_D has been connected (one that does not implement the 1MΩ resistors, but directly asserts HPD).
1	0	1	DisplayPort UFP_D is connected. The Adaptor has already closed the SBU isolation switches, connecting the 100kΩ resistors from the DFP_D. This state is maintained until the DisplayPort UFP_D is disconnected.
1	1	0	Invalid (possible transitory state between 0 1 0 and 1 0 1 if the DisplayPort UFP_D detects a connection resulting from the Adaptor's weak pull-down resistor).
1	1	1	Invalid (this state may occur if the connected device is a DisplayPort DFP_D side DP++ Adaptor with DisplayPort Dual-Mode Adaptor with Display Data Channel (DDC) pull-up resistors, as specified in *DP++ v1.1*).

After the Adaptor is operating in DisplayPort UFP_D Configuration (i.e., connected to a DisplayPort DFP_D on the DisplayPort connector) and converting HPD from the USB-PD messages received on the USB Type-C connector and driving HPD to the DisplayPort connector, the table simplifies to that listed in Table 4-5.

Table 4-5: DisplayPort Connector HPD and AUX States in Adaptor UFP_D Configuration

HPD	AUX_CH_P	AUX_CH_N	Interpretation
1	0	0	A powered-down device is connected.
1	0	1	External DisplayPort DFP_D is connected and operational.
1	1	0	Transitional state when the previously connected DisplayPort DFP_D is disconnected, the Adaptor should switch out of DisplayPort UFP_D Configuration and report the new status to the USB Type-C UFP_D, which in turn should disable DisplayPort Configuration, taking HPD to a logic low state and isolating the SBU connection.
1	1	1	Invalid.

Licensed to Jay Lee (jlee@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.3 Scenarios 3a and 3c USB Type-C DFP_D to Protocol Converter

Figure 4-21 and Figure 4-22 illustrate a USB Type-C DFP_D to Protocol Converter. Signaling speeds are as per *DP v1.3*.

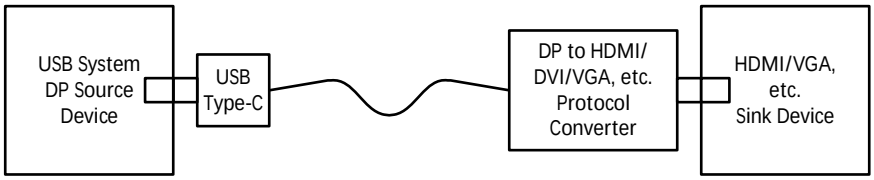


Figure 4-21: Scenario 3a

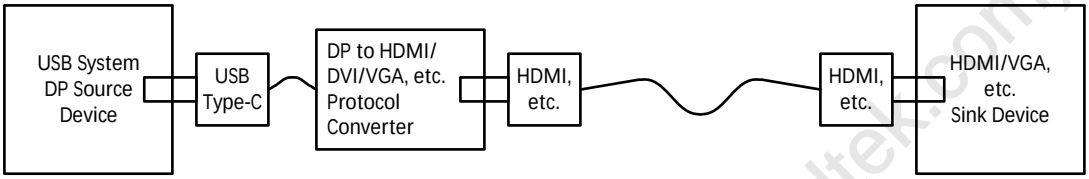


Figure 4-22: Scenario 3c

These scenarios use DFP_D Pin Assignment C. The implementation of AUX signaling and HPD in the Protocol Converter is similar to that used in a USB Type-C UFP_D (see Section 3.15).

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.3.1 HDMI Protocol Converters

HDMI Protocol Converters shall comply with the requirements for DP to HDMI Protocol Converter Adaptors specified in *DP v1.3*. HDMI Protocol Converters shall support four lanes of HBR2.

In addition, the DETAILED_CAP_INFO_AVAILABLE bit in the DOWN_STREAM_PORT_PRESENT register (DPCD Address 00005h, bit 4) shall have a value of 1 (i.e., the DFP_D Receiver Capability fields in registers starting at DPCD Address 00080h shall use the longer four-byte format).

The following features, optional in *DP v1.3*, shall be supported:

- 594M characters per second per channel
- 4K at 60Hz YCbCr 4:4:4, as specified in *HDMI v2.0*
- HDMI Sink maximum character per channel rate indication
- HDMI Sink Error status
- CEC, including low-power management and CEC-Tunneling-over-AUX with support for snooping and multiple-logical addresses
- Pass-through of YCbCr 4:2:0 color format
- Conversion from YCbCr 4:4:4 format or YCbCr 4:2:2 format to YCbCr 4:2:0
- Generation of an IRQ_HPD on HDMI status changes
- *HDCP v1.4* and *HDCP v2.2* on the USB Type-C connection
- *HDCP v1.4* and *HDCP v2.2* on the HDMI connection
- Conversion from DP Frame Sequential to HDMI Frame Pack 3D format
- *HDCP v2.2* repeater

4.3.1.1 Audio Return Channel

The Protocol Converter may include a *USB r2.0* bridge Device to convert the Audio Return Channel (ARC) to the USB Audio Device Class (audio input).

Support is optional for the Protocol Converter, and its support is exposed to the host as part of the Service Abstraction defined in *DP v1.3*. The Source device policy may or may not support CEC.

Notes: *CEC support is a pre-requisite for ARC for both the Protocol Converter and Source device.*

ARC is supported only on Source devices that are also USB hosts.

4.3.2 VGA Protocol Converters

VGA Protocol Converters shall support a minimum resolution of 1920x1200 at 60Hz, 24bpp reduced blanking, 1600x1200 at 60Hz, 24bpp full blanking. VGA Protocol Converters shall support at least one lane of HBR2.

4.3.3 DVI Protocol Converters

Single-link DVI Protocol Converters shall support a minimum resolution of 1920x1200 at 60Hz, 24bpp. DVI Protocol Converters shall support two lanes of HBR, but are not required to support bandwidths in excess of two lanes of HBR.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.4 Scenario 4a USB Type-C DFP_D to a Multi-function Dock

Figure 4-23 illustrates a USB Type-C DFP_D connected to a Multi-function dock.

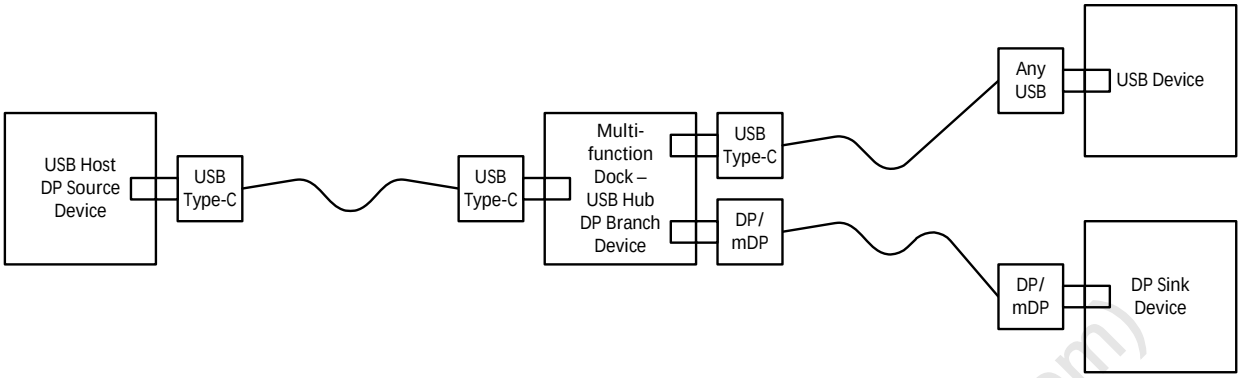


Figure 4-23: Scenario 4a

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.4.1 Scenario 4a USB Type-C Cable Using GEN2_BR Multi-function

This scenario uses DFP_D Pin Assignment B at the DisplayPort Source device and UFP_D Pin Assignment B at the DisplayPort Sink device. In the event that the Multi-function dock can be configured to apply all four DisplayPort lanes to the DisplayPort connector, Pin Assignment A is used instead. Note that the Multi-function dock must contain a DisplayPort Branch device that can adapt from GEN2_BR to DP_BR on the DisplayPort connector. The figures illustrate DisplayPort use of the signal connections in the USB Full-Featured Type-C Cable assembly, as follows:

- Figure 4-24 illustrates when the plugs at both ends are inserted in normal orientation
- Figure 4-25 illustrates when the DFP_D plug is inverted and the UFP_D plug is inserted in normal orientation
- Figure 4-26 illustrates when the DFP_D plug is inserted in normal orientation and the UFP_D plug is inverted
- Figure 4-27 illustrates when the plugs at both ends are inverted

Note: *The figures omit the USB-PD identification circuitry that is incorporated in the Cable assembly.*

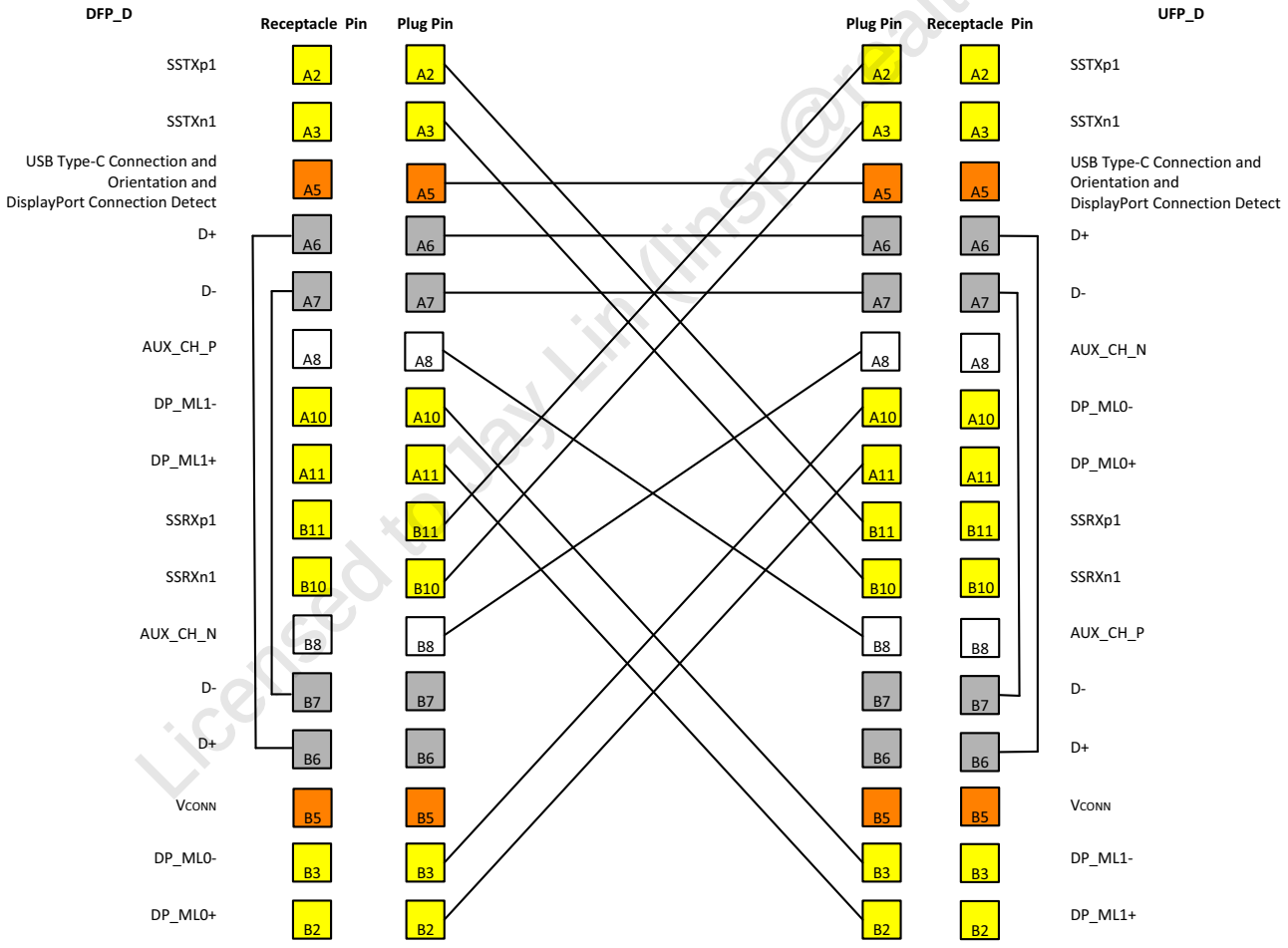


Figure 4-24: Scenario 4a GEN2_BR MF Cable Wiring – Normal Plug Orientation

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

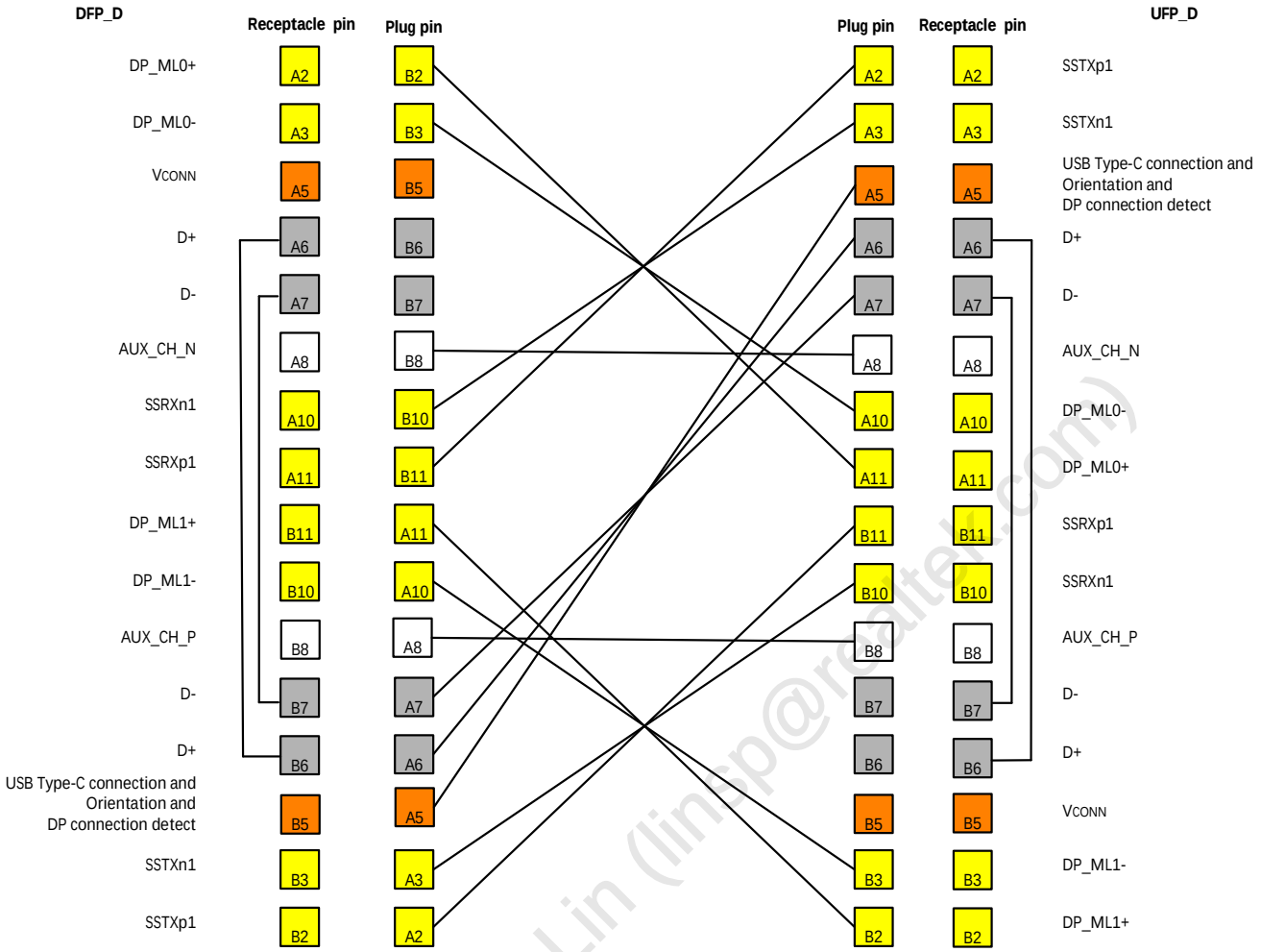


Figure 4-25: Scenario 4a GEN2_BR MF Cable Wiring – DFP_D Plug Inverted

Licensed to Jay Lin (insp@rdata.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

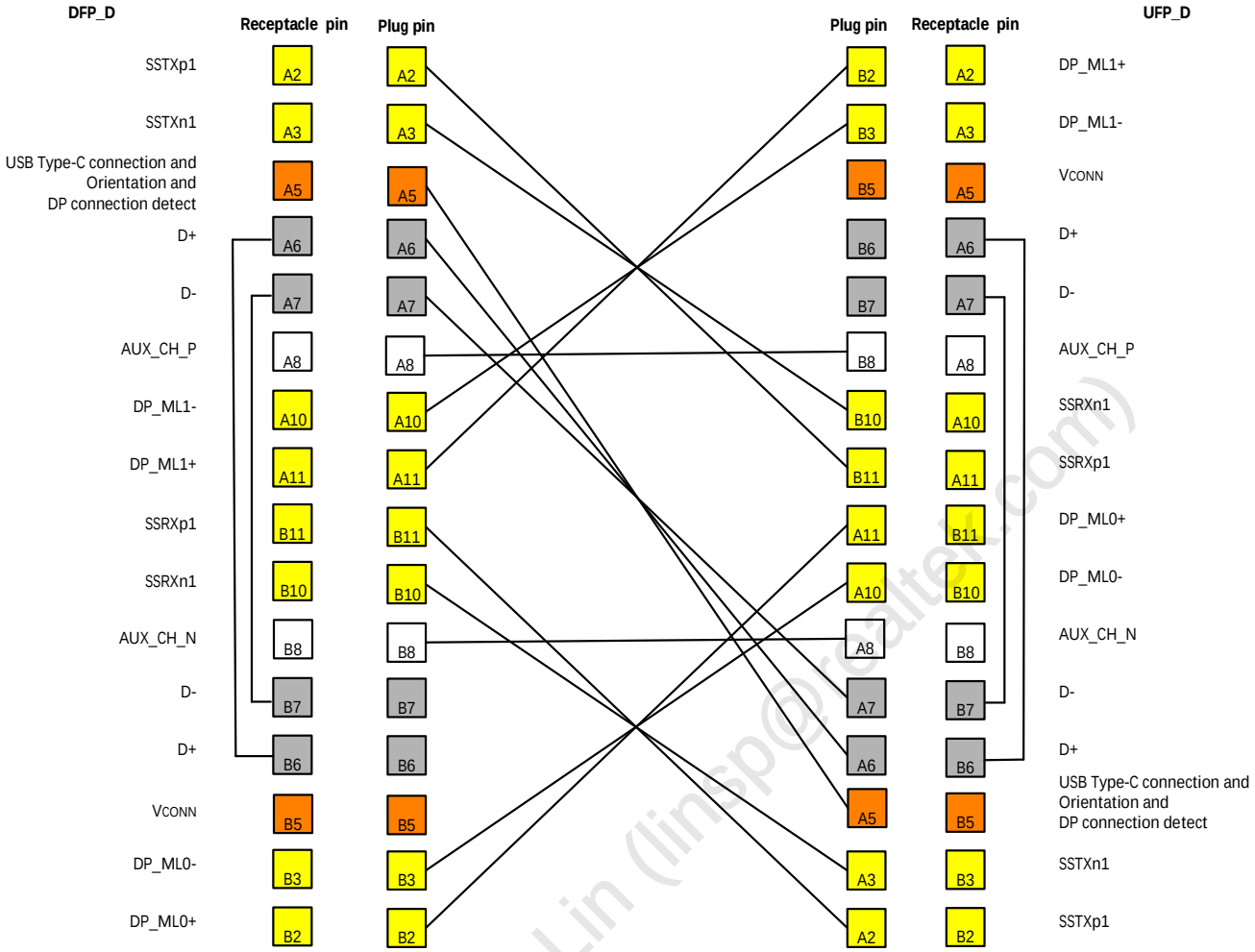


Figure 4-26: Scenario 4a GEN2_BR MF Cable Wiring – UFP_D Plug Inverted

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

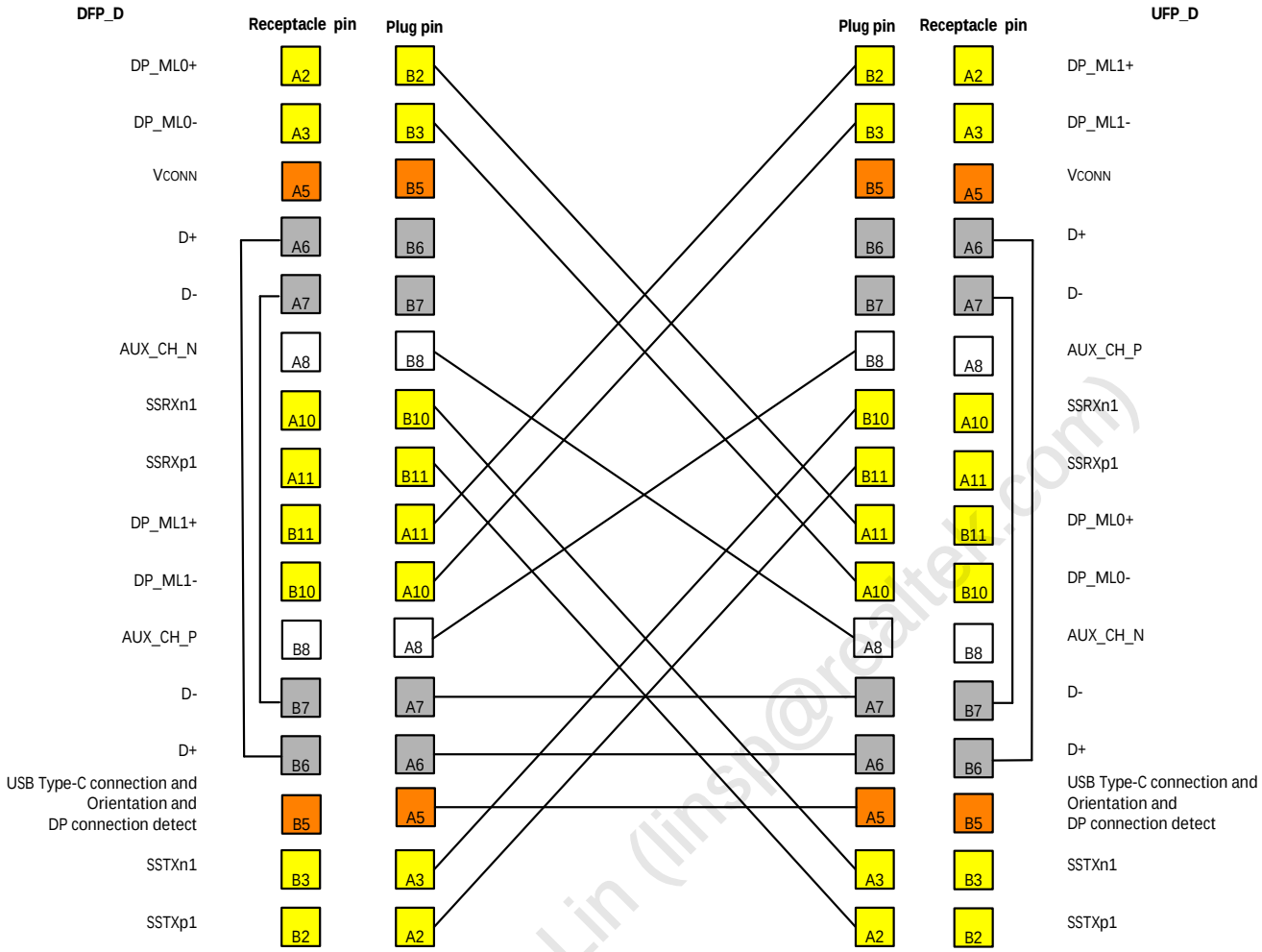


Figure 4-27: Scenario 4a GEN2_BR MF Cable Wiring – DFP_D and UFP_D Plug Inverted

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.4.2 Scenario 4a USB Type-C Cable Using DP_BR Multi-function

This scenario uses DFP_D Pin Assignment D at the DisplayPort Source device and UFP_D Pin Assignment D at the DisplayPort Sink device. In the event that the Multi-function dock can be configured to apply all four DisplayPort lanes to the DisplayPort connector, Pin Assignment C is used instead. The figures illustrate DisplayPort use of the signal connections in the USB Full-Featured Type-C Cable assembly, as follows:

- Figure 4-28 illustrates when the plugs at both ends are inserted in normal orientation
- Figure 4-29 illustrates when the DFP_D plug is inverted and the UFP_D plug is inserted in normal orientation
- Figure 4-30 illustrates when the DFP_D plug is inserted in normal orientation and the UFP_D plug is inverted
- Figure 4-31 illustrates when the plugs at both ends are inverted

Note: *The figures omit the USB-PD identification circuitry that is incorporated in the Cable assembly.*

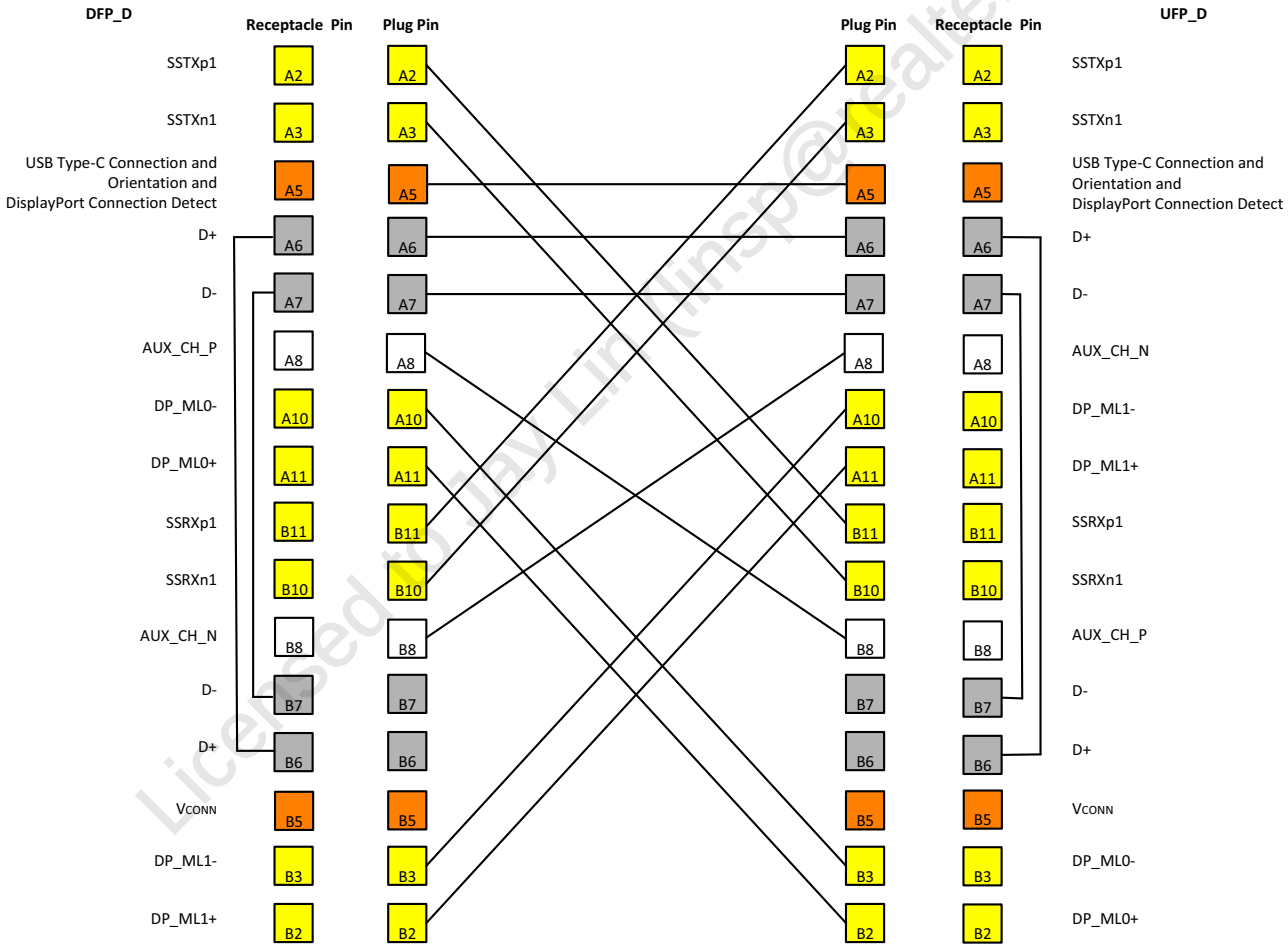


Figure 4-28: Scenario 4a DP_BR MF Cable Wiring – Normal Plug Orientation

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

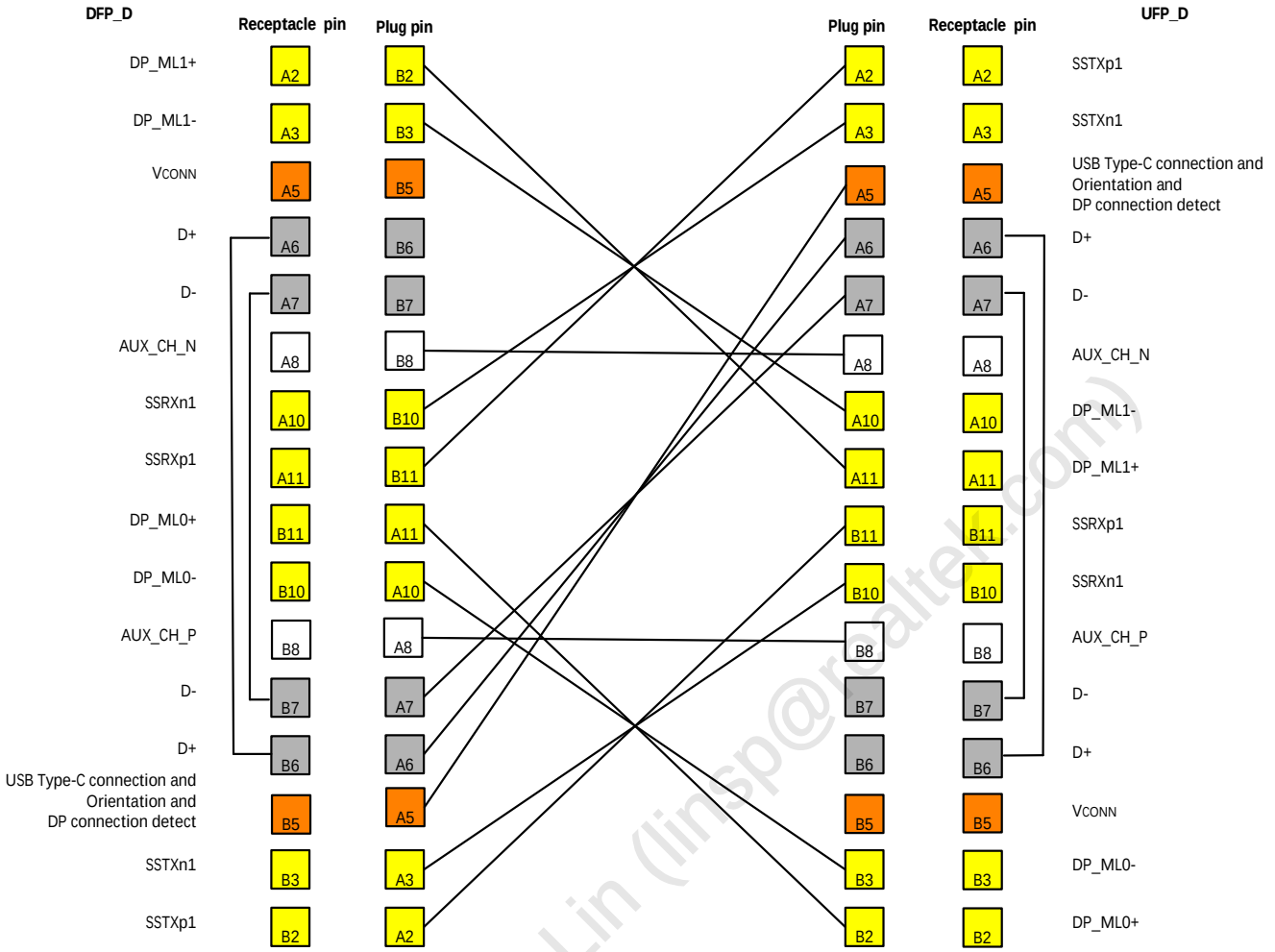


Figure 4-29: Scenario 4a DP_BR MF Cable Wiring – DFP_D Plug Inverted

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

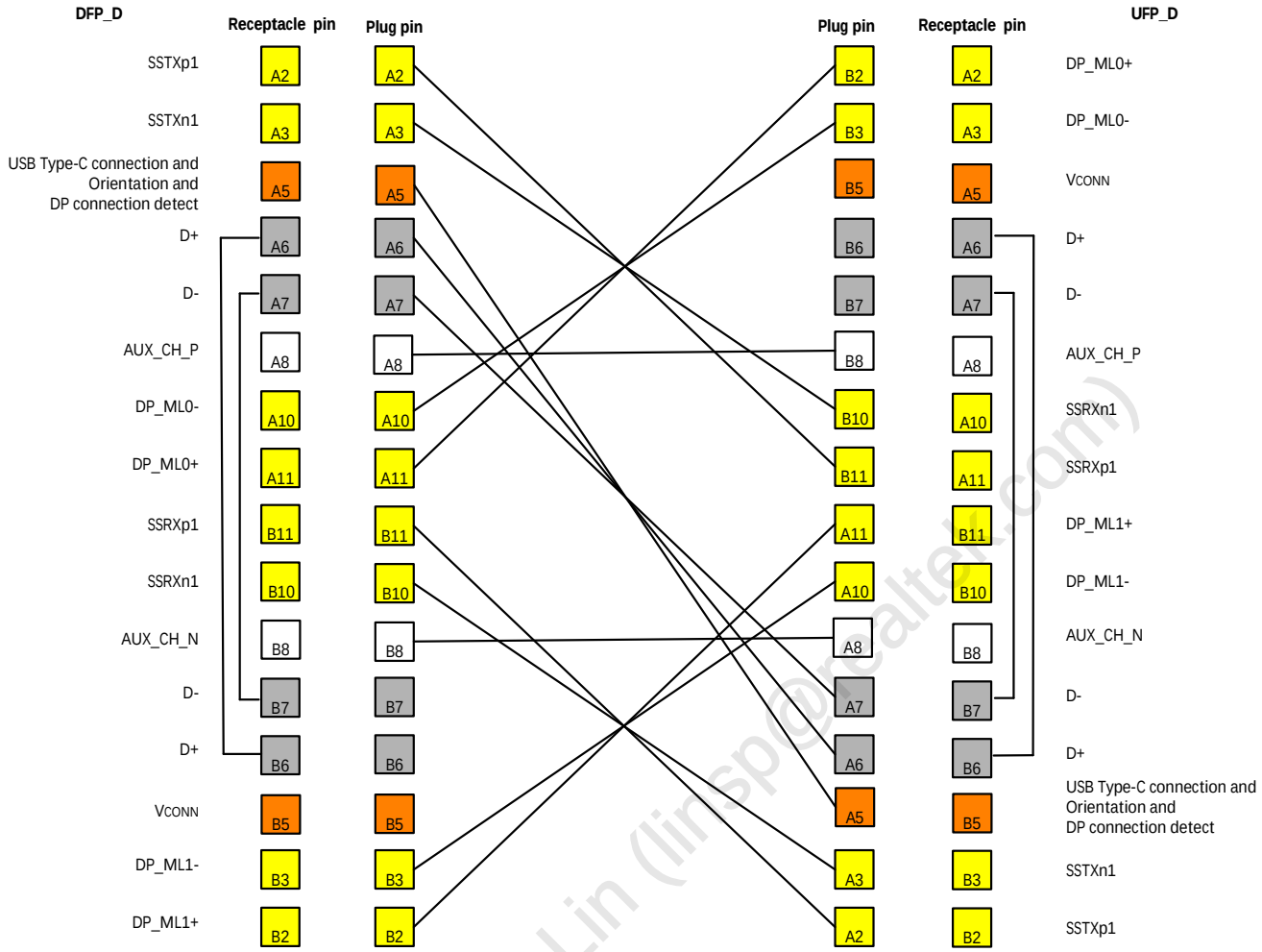


Figure 4-30: Scenario 4a DP_BR MF Cable Wiring – UFP_D Plug Inverted

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

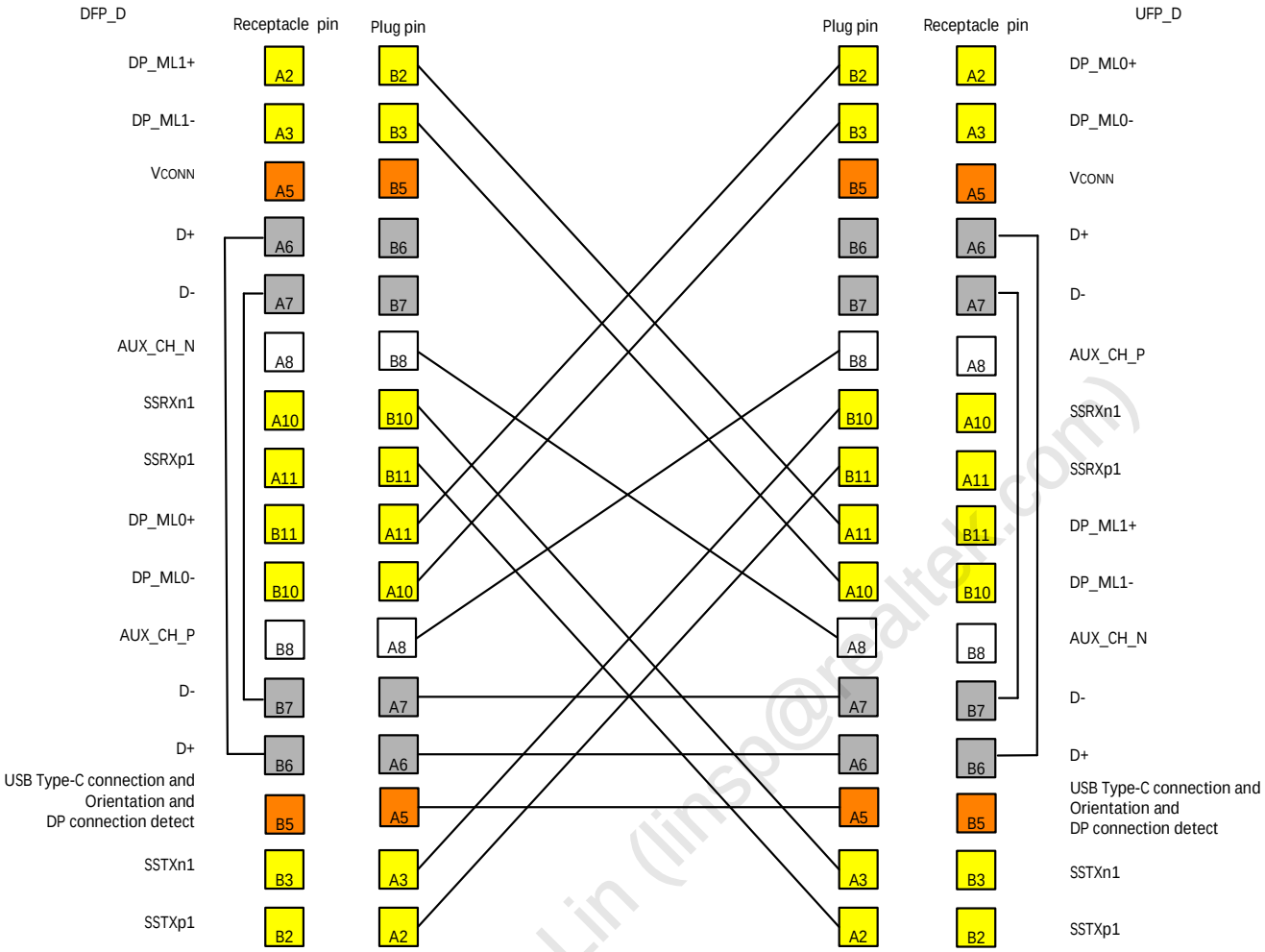


Figure 4-31: Scenario 4a DP_BR MF Cable Wiring – DFP_D and UFP_D Plug Inverted

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.5 Scenario 4b USB Type-C DFP_D to a *DockPort v1.0a* Adaptor

Figure 4-32 illustrates a USB Type-C DFP_D connected to a dock, as defined in *DockPort v1.0a*, using a *DockPort v1.0a* receptacle Adaptor. Table 4-6 lists DisplayPort use of the signal connections in the USB Type-C to *DockPort v1.0a* Adaptor. Figure 4-33 illustrates this use when the USB Type-C Plug is inserted in normal orientation. Figure 4-34 illustrates this use when the USB Type-C Plug is inserted in flipped orientation. The pin numbering on the DisplayPort connector is that for the full-sized DisplayPort connector. See Table 4-6 for mini DisplayPort connector pin numbering.

The Cable assembly shall contain logic to implement USB-PD and support the Discovery Process (see Section 5). The Cable assembly contains a *USB Billboard r1.0* device, which the PD logic shall connect to D+/D- in the event that switching to using DisplayPort as an Alternate Mode is unsuccessful. The connections in the Cable assembly behind pins A10, A11, B10, and B11 on the USB Type-C Plug shall contain series AC capacitors, as specified in *DP v1.3* (75 to 200nF), because these pins are required to be AC-coupled at the DFP_D by *DP v1.3* but are DC connected in the system behind the USB Type-C Receptacle. The series AC capacitors on these pins are also required by *USB Type-C r1.1*.

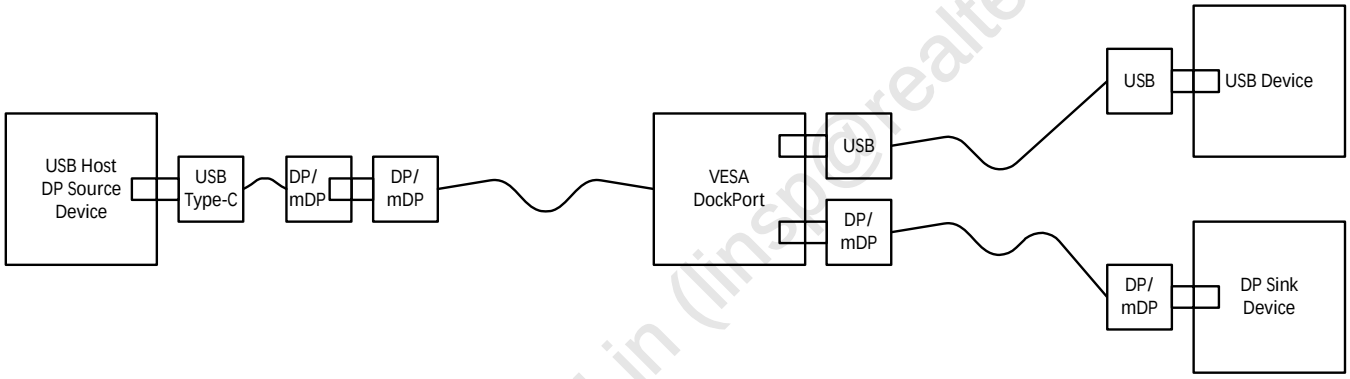


Figure 4-32: Scenario 4b

The Adaptor has additional functionality, as compared to the USB Type-C to DisplayPort Cable assemblies, to support *DockPort v1.0a* discovery and connect CONFIG1 and CONFIG2 to the *USB r2.0* signals from the host by way of a hub (shown as part of the PD logic block in Figure 4-33 and Figure 4-34). This scenario uses DFP_D Pin Assignment F at the DisplayPort Source device.

Note: *DFP_D Pin Assignment E is used in the case that the DockPort is in the mode that supports all four lanes of DisplayPort.*

Licensed to Jay Lin (jinsn@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

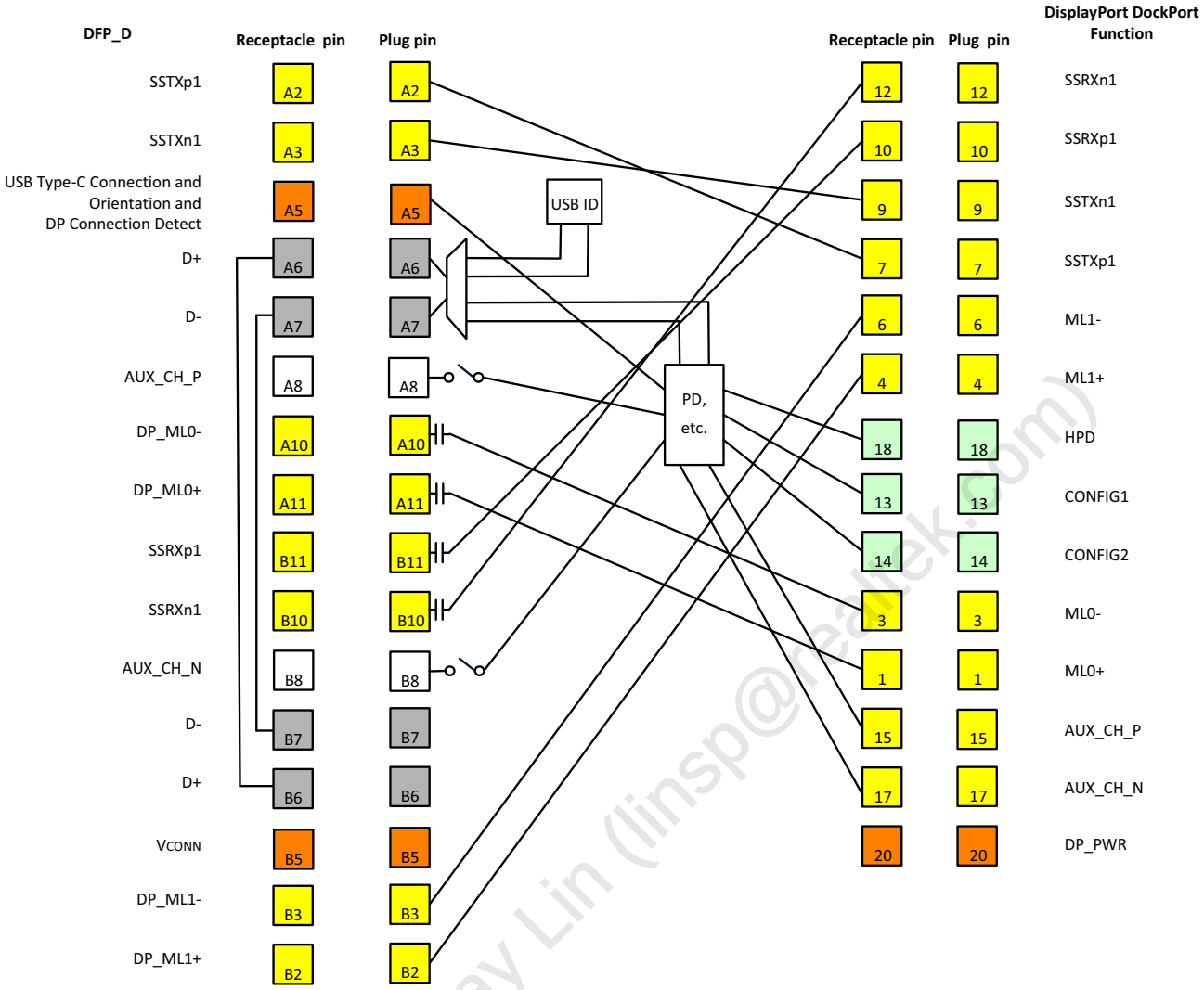


Figure 4-33: Scenario 4b *DockPort v1.0a* Receptacle Adaptor Wiring – Normal Plug Orientation

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

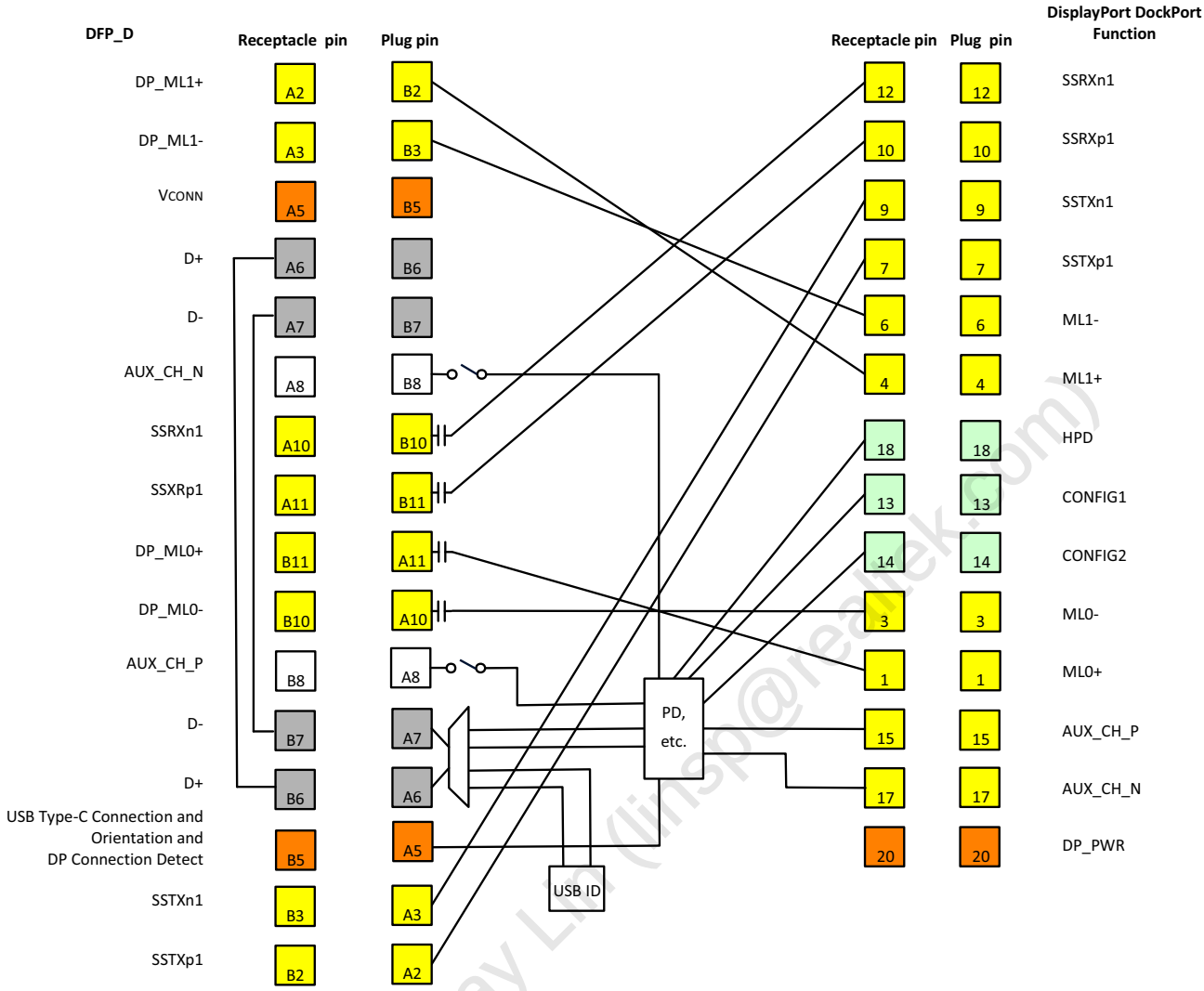


Figure 4-34: Scenario 4b *DockPort v1.0a* Receptacle Adaptor Wiring – Flipped Plug Orientation

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 4-6: USB Type-C DFP_D to *DockPort v1.0a* Receptacle

USB Type-C Plug DFP_D		Wire		DisplayPort/mini DisplayPort Receptacle DFP_D		
Pin	**(DP) Signal Name**	**Wire Number**	**Signal Name**	**DP Pin**	**mDP Pin**	**(DP) Signal Name**
A1, B1, A12, B12	GND	1 [16]	GND_PWRrt1 [GND_PWRrt2]	2, 5, 8, 11, 16, 19	1, 7, 8, 13, 14, 19	GND
A4, B4, A9, B9	VBUS	2	Powers USB Billboard			
A5	CC	3	Cable logic (PD)			
A6	Dp1	4	Cable logic (USB Billboard)/Dp1	13	4	CONFIG1
A7	Dn1	5	Cable logic (USB Billboard)/Dn1	14	6	CONFIG2
A2	SSTXp1	6	SDPp1	7	15	SSTXp1
A3	SSTXn1	7	SDPn1	9	17	SSTXn1
B11	SSRXp1	8	SDPp2	10	10	SSRXp1
B10	SSRXn1	9	SDPn2	12	12	SSRXn1
B2	ML1+	10	SDPp3	4	9	ML1+
B3	ML1-	11	SDPn3	6	11	ML1-
A11	ML0+	12	SDPp4	1	3	ML0+
A10	ML0-	13	SDPn4	3	5	ML0-
A8	AUX_CH_P	14	Cable logic	15	16	AUX_CH_P
B8	AUX_CH_N	15	Cable logic	17	18	AUX_CH_N
			Cable logic	18	2	HPD
B5	VCONN	16	Cable logic			
			No Connect	20	20	DP_PWR
Shell	Shield	Braid	Shield	Shell	Shell	Shield

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.5.1 USB Type-C DFP_D to a *DockPort v1.0a* Adaptor Electrical Properties

The electrical specification of the USB Type-C to *DockPort v1.0a* receptacle Adaptor is based on the specification methodology for the USB Full-Featured Type-C Plug to *USB r3.1*-A receptacle Adaptor assembly (see *USB Type-C r1.1*, Section 3.7.5.2). The Adaptor shall meet the requirements specified in *USB Type-C r1.1*, Table 3-30, including the value used for *fmax* and formulae used for *Vin(f)* and *Vdd(f)*.

Insertion loss and Return loss are to be tested relative to the following:

- 85Ω-differential environment at the USB Type-C connector
- 100Ω-differential environment at the DisplayPort receptacle connector

The appropriate parameters in the referenced methodology are to be adjusted accordingly.

Note: *Using the methodology described in USB Type-C r1.1, the measurement results for the USB Type-C end of a Cable assembly do not include the USB Type-C Receptacle; however, the results do include the DisplayPort plug at the DisplayPort end.*

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.6 Scenarios 4c and 4d USB Type-C DFP_D to Multi-function Cradle/Captive Cable Dock

Figure 4-35 illustrates a USB Type-C DFP_D connected to a Multi-function cradle dock.

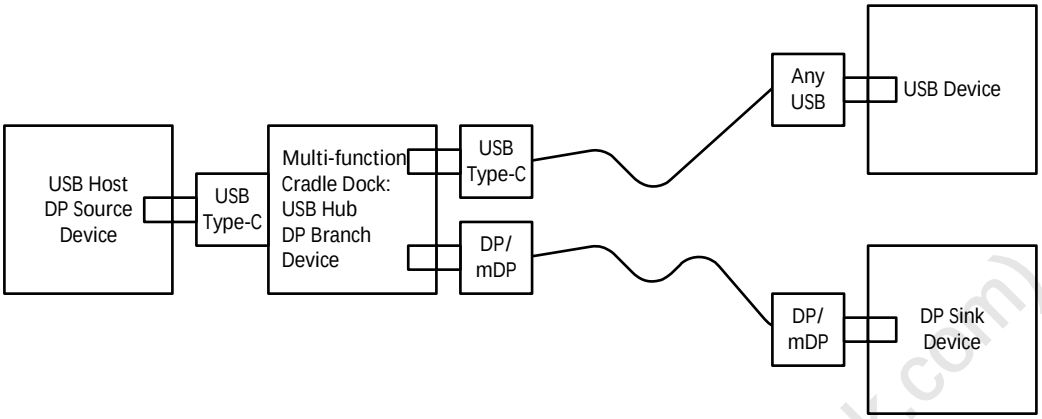


Figure 4-35: Scenario 4c

Figure 4-36 illustrates a USB Type-C DFP_D connected to a Multi-function Captive cable dock.

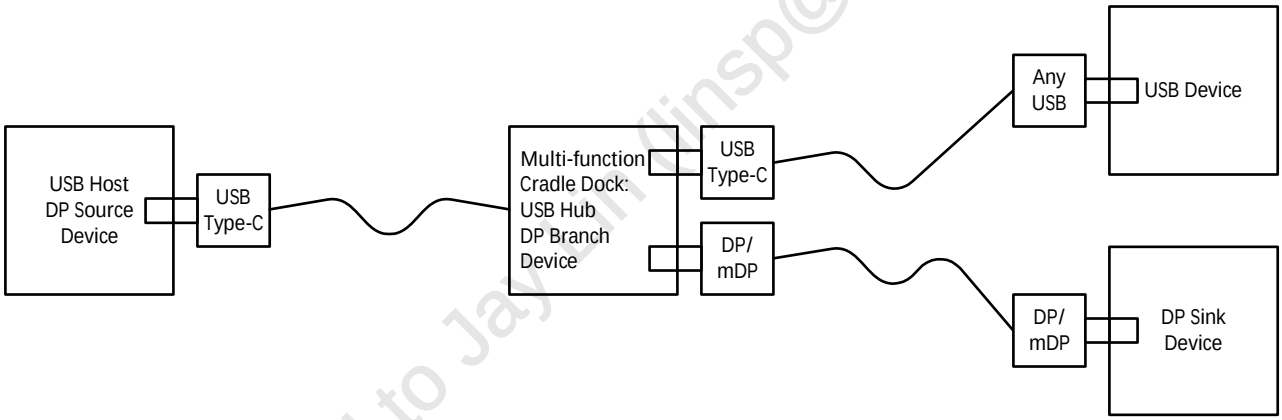


Figure 4-36: Scenario 4d

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.6.1 Scenarios 4c and 4d USB Type-C DFP_D to Multi-function Cradle/Captive Cable Dock Using GEN2_BR

These scenarios use DFP_D Pin Assignment B at the DisplayPort Source device. In the event that the Multi-function dock can be configured to apply all four DisplayPort lanes to the DisplayPort connector, Pin Assignment A is used instead. Note that the Multi-function dock must contain a DisplayPort Branch device that can adapt from GEN2_BR to DP_BR on the DisplayPort connector.

Table 4-7: USB Type-C Cable DisplayPort Use (Multi-function Cradle/Captive Cable Dock) GEN2_BR

USB Type-C Plug DFP_D	
Pin	**Signal Name**
A1, B1, A12, B12	GND
A4, B4, A9, B9	V_{BUS}
A5	CC
A6	Dp1
A7	Dn1
A2	SSTXp1
A3	SSTXn1
B11	SSRXp1
B10	SSRXn1
B2	ML0+
B3	ML0-
A11	ML1+
A10	ML1-
A8	AUX_CH_P
B8	AUX_CH_N
B5	V_{CONN}
Shell	Shield

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

4.6.2 Scenarios 4c and 4d USB Type-C DFP_D to Multi-function Cradle/Captive Cable Dock using DP_BR

These scenarios use DFP_D Pin Assignment D at the DisplayPort Source device. In the event that the Multi-function dock can be configured to apply all four DisplayPort lanes to the DisplayPort connector, Pin Assignment C is used instead.

Table 4-8: USB Type-C Cable DisplayPort Use (Multi-function Cradle/Captive Cable Dock) DP_BR

USB Type-C Plug DFP_D	
Pin	**Signal Name**
A1, B1, A12, B12	GND
A4, B4, A9, B9	V_{BUS}
A5	CC
A6	Dp1
A7	Dn1
A2	SSTXp1
A3	SSTXn1
B11	SSRXp1
B10	SSRXn1
B2	ML1+
B3	ML1-
A11	ML0+
A10	ML0-
A8	AUX_CH_P
B8	AUX_CH_N
B5	V_{CONN}
Shell	Shield

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

5 Discovery and USB-PD

This section describes the mechanisms for discovery of DisplayPort support on a USB Type-C connector and the negotiation procedures for entering and exiting DisplayPort Mode and establishing DisplayPort Configuration.

All descriptions of USB-PD signaling apart from DisplayPort Mode-specific commands are informative. Reference should be made to *USB PD r2.0 v1.1* for normative USB-PD functionality.

5.1 Connection Discovery

A system supporting a DFP_D on a USB Type-C Receptacle shall implement one of the following:

- DFP, as specified in the *USB Type-C r1.1* (with Rp pull-up resistor on CC1/CC2 for connection discovery)
- DRP, as specified in the *USB Type-C r1.1* (able to periodically use Rp to pull up CC1/CC2)
- UFP with accessory support, as described in the *USB Type-C r1.1*

After *USB Type-C r1.1* discovery, all these ports are described in this Standard as DFP_U, assuming that the DRP resolves as a DFP. If the DRP resolves into a UFP, it is described in this Standard as a UFP_U.

A system supporting a UFP_D on a USB Type-C Receptacle shall implement one of the following:

- DRP, as specified in the *USB Type-C r1.1* (able to periodically use Rp to pull up CC1/CC2)
- UFP with accessory support, as described in the *USB Type-C r1.1*

After *USB Type-C r1.1* discovery, both of these ports are described in this Standard as UFP_U, assuming that the DRP resolves as a UFP. If the DRP resolves into a DFP, it is described in this Standard as a DFP_U.

Note: *A UFP with accessory support, as described in the USB Type-C r1.1, is not required to operate as a USB Host when it uses Rp to pull up CC1/CC2 and discovers a DisplayPort-capable UFP.*

If one or both of the connected systems are DRPs, as specified in the *USB Type-C r1.1*, the DRP(s) resolve into one being DFP, as specified in the *USB Type-C r1.1*, and one being UFP on first connection. Note that they may exchange roles using a DR_Swap USB-PD message. The roles are static (DFP_U or UFP_U) during the time that DisplayPort discovery and DisplayPort Mode is used.

If a DR_Swap message is issued during DisplayPort discovery, the following occurs:

- Current DFP, as specified in the *USB Type-C r1.1*, abandons the Discovery Process, and
- New DFP, as specified in the *USB Type-C r1.1*, initiates a new Discovery Process

In this Standard, the new DFP is described as DFP_U and the new UFP is described as UFP_U.

Licensed to Ivy Lin (linsn@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

A system supporting DisplayPort on a USB Type-C Plug shall implement a UFP, as specified in the *USB Type-C r1.1*, with an Rd pull-down resistor on CC. The system may be either of the following:

- Self-powered, in which case the system shall present zOPEN on VCONN, –or–
- VCONN-powered accessory, in which case the system shall present Ra on VCONN

If the system is a VCONN-powered accessory, it is strongly recommended for the accessory to reduce the power consumption through Ra after the system detects that VCONN is provided. The system shall support BMC PD signaling on CC.

The system supporting DisplayPort on a USB Type-C Receptacle that is also the DFP_U (including a UFP with accessory support, as described in the *USB Type-C r1.1*, that has detected a powered accessory) provides VCONN if the system detects Ra on the CC1/2 pin not being used for PD communications.

5.2 DisplayPort Mode Discovery, Entry, and Exit

Discovery of support for DisplayPort and entry into/exit from DisplayPort Mode shall be performed using USB-PD messages as described in the Discovery Process defined in *USB PD r2.0 v1.1* and *USB Type-C r1.1*. These shall be augmented for the specific case of DisplayPort as described in this section.

The 16-bit USB Standard ID assigned by the USB-IF to DisplayPort (FF01h) is denoted in this section by DP_SID.

Only a receptacle-based DFP_U supporting DisplayPort may behave as an Initiator for the USB-PD messages related to DisplayPort Mode, apart from the Attention command, which may be issued only by the UFP_U (which may be receptacle- or plug-based).

Note: *This Standard does not support plug-based DFP_Us.*

Licensed to Jay Lin (jinsh@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

5.2.1 Structured VDM Header

The Structured Vendor Defined Message (VDM) Header (as specified in *USB PD r2.0 v1.1*) shall be used for the DisplayPort-specific commands (i.e., DisplayPort Status Update and DisplayPort Configure, in conjunction with DP_SID). Table 5-1 is reproduced from *USB PD r2.0 v1.1*, Table 6-19, and updated to list the additional DisplayPort-specific commands.

Table 5-1: VDM Header

Bit(s)	Description	Values
4:0	Command	0h = RESERVED, shall not be used. 1h = Discover Identity. 2h = Discover SVIDs. 3h = Discover Modes. 4h = Enter Mode. 5h = Exit Mode. 6h = Attention. 7h – Fh = RESERVED, shall not be used. 10h = DisplayPort Status Update. 11h = DisplayPort Configure. 12h – 1Fh = RESERVED for DP_SID use.
5	RESERVED	RESERVED (always 0).
7:6	Command Type Initiator shall clear to 0.	00 = Initiator. 01 = Responder ACK. 10 = Responder NAK. 11 = Responder BUSY.
10:8	Object Position	000 = RESERVED. 001 – 110 = Index into the list of Vendor Defined Objects (VDOs) to identify the desired Mode VDO. 111 = Exit all Modes (equivalent of a power-on reset). Shall not be used with the Enter Mode command.
12:11	RESERVED	RESERVED (always 0).
14:13	Structured VDM Version	00 = Version 1.0. All other values are RESERVED.
15	VDM Type	1 = Structured VDM.
31:16	Standard or Vendor ID	Base SID (for Discover SVIDs command) or DP_SID, a 16-bit unsigned integer, assigned by the USB-IF.

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

The UFP_U may support SVIDs other than the DisplayPort SVID, in which case the UFP_U may present them in an arbitrary order in its response. The DFP_U shall search all the SVIDs returned so as to discover the DP_SID.

The UFP_U may support a DFP_D, a UFP_D, or both a DFP_D and UFP_D. An example of supporting both is a USB Type-C to DisplayPort Adaptor cable, which is required to be reversible. The UFP_U, when sending the Responder Discover Modes VDM, shall indicate its DisplayPort capabilities in the VDO, as specified in Table 5-2.

Future versions of this Standard may describe other modes associated with DP_SID. Such modes will be identified by having a non-zero value in bits 31:24 of the VDO. The DFP_U shall examine the list of modes returned until it finds 0s in bits 31:24 of the VDO and a non-zero value in bits 23:0 of the VDO (i.e., DisplayPort capabilities). The DFP_U and UFP_U shall use the corresponding offset (indexed from 1) as the Object Position in the Enter Mode, DisplayPort Configure, DisplayPort Status Update, Attention, and Exit Mode commands.

Note: *Any value other than 0 or 7 indicates that the command is to be interpreted in the context of some Mode other than DisplayPort Mode. Value 7 is valid only for Enter Mode and Exit Mode. If the value in the Object Position is not valid for DisplayPort Mode or some other Mode, USB PD r2.0 v1.1 requires that it be NAK'd.*

Table 5-2: DisplayPort Capabilities (VDO in Responder Discover Modes VDM)

Bit(s)	Description	Values
1:0	Port Capability	00 = RESERVED. 01 = UFP_D-capable (including Branch device). 10 = DFP_D-capable (including Branch device). 11 = Both DFP_D and UFP_D-capable.
5:2	Signaling for Transport of DisplayPort Protocol	xxx1 = Supports *DP v1.3* signaling rates and electrical specification (shall always be set apart from diagnostic purposes). xx1x = Supports USB Gen 2 signaling rate and electrical specification. x1xx = RESERVED. 1xxx = RESERVED.
6	Receptacle Indication	0 = DisplayPort interface is presented on a USB Type-C Plug. 1 = DisplayPort interface is presented on a USB Type-C Receptacle.
7	*USB r2.0* Signaling Not Used	0 = *USB r2.0* signaling may be required on A6 – A7 or B6 – B7 while in DisplayPort Configuration. 1 = *USB r2.0* signaling is not required on A6 – A7 or B6 – B7 while in DisplayPort Configuration.

Licensed to Jay Lin (linjsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 5-2: DisplayPort Capabilities (VDO in Responder Discover Modes VDM) (Continued)

Bit(s)	Description	Values
15:8	DFP_D Pin Assignments Supported (reported by a DFP_D receptacle or UFP_D (direct attach) plug)	00000000 = DFP_D pin assignments are not supported. xxxxxxx1 = Pin Assignment A is supported. xxxxxx1x = Pin Assignment B is supported.[a] xxxxx1xx = Pin Assignment C is supported. xxxx1xxx = Pin Assignment D is supported.[a] xxx1xxxx = Pin Assignment E is supported. xx1xxxxx = Pin Assignment F is supported.[a] x1xxxxxx = RESERVED. 1xxxxxxx = RESERVED.
23:16	UFP_D Pin Assignments Supported (reported by a UFP_D receptacle or DFP_D (direct attach) plug)	00000000 = UFP_D pin assignments are not supported. xxxxxxx1 = Pin Assignment A is supported. xxxxxx1x = Pin Assignment B is supported.[a] xxxxx1xx = Pin Assignment C is supported. xxxx1xxx = Pin Assignment D is supported.[a] xxx1xxxx = Pin Assignment E is supported. xx1xxxxx = RESERVED. x1xxxxxx = RESERVED. 1xxxxxxx = RESERVED.
31:24	RESERVED	RESERVED (always 0).

a. A system supporting DisplayPort on a USB Type-C connector that supports Multi-function operation (i.e., supports the simultaneous transmission of USB SuperSpeed signaling and DisplayPort signaling) shall offer both the Multi-function pin assignment and associated DisplayPort-only pin assignment. The associated DisplayPort-only pin assignment is identified by the previous letter (e.g., Pin Assignment D for Multi-function and Pin Assignment C for DisplayPort only). The system shall offer both pin assignments, even if the system supports only one or two lanes of DisplayPort signaling.

The DFP_U shall ensure that the UFP_U's capabilities are consistent with its own capabilities, taking care to note that a UFP_U that uses a USB Type-C Plug describes the pin assignments supported for the corresponding receptacle (i.e., a UFP_D will describe the DFP_D pin assignments to which it connects), whereas a UFP_U that uses a USB Type-C Receptacle describes its pin assignments directly (i.e., a UFP_D will describe its own UFP_D pin assignments). In general, the DFP_U will select a pin assignment for its USB Type-C Receptacle (one of A through F) and configure the port partner with the same pin assignment (same identifying letter).

An Adaptor may require more than the USB Type-C level of power on VCONN to fully power its DisplayPort functionality. The DFP_U shall ensure that the VCONN power specified in this Standard is available prior to issuing an Enter Mode command addressed to DP_SID. The Adaptor shall not exceed the USB Type-C-specified level of power consumption on VCONN prior to receiving an Enter Mode command addressed to DP_SID.

Licensed to Jay Li (lingeareatek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

5.2.2 DisplayPort Enter Mode

The Enter Mode command/Exit Mode command pair addressed to the DisplayPort SVID behave as brackets for the various other DisplayPort Mode-specific commands, which may only be issued during the time that DisplayPort Mode is established.

On receiving the Responder ACK in a response to the Enter Mode command, the DFP_U proceeds to the next step towards configuring the port by sending a DisplayPort Status Update command to the UFP_U. If the UFP_U responds to the Enter Mode VDM with a Responder NAK, the mode is not entered and the UFP_U shall expose a *USB Billboard r1.0* device. The UFP_U is not allowed to return a Responder BUSY.

After DisplayPort Mode is entered, the DFP_U and UFP_U shall remain in that mode until one of the three following events occur:

- Exit Mode command is successful
- Hard Reset occurs on PD
- Device is disconnected

An Adaptor may consume power from VCONN up to the limit specified in Section 6.1 after receiving an Enter Mode command addressed to the DisplayPort SVID and selecting the DisplayPort Mode.

A UFP_U may transmit a USB-PD Attention command at any time once in DisplayPort Mode, and shall do so whenever its status changes.

5.2.3 DisplayPort Status Update

The DFP_U may send a Status Update command at any time, and generally will do so immediately after receiving the Responder ACK in response to a DisplayPort Enter Mode command. The DFP_U provides its latest DisplayPort Status in the VDO. The UFP_U replies by in turn transmitting its DisplayPort Status in the VDO. The UFP_U is not allowed to return a Responder NAK. There is no requirement for the UFP_U to carry out any specific behavior based on the information that it receives concerning the DFP_U's status, and any use that it makes of this information is implementation specific. However, this information may be useful for diagnostic purposes.

A DFP_U or UFP_U that integrates a DFP_D or a UFP_D shall set the appropriate connected status flag permanently to 1. A DFP_U or UFP_U that integrates both a DFP_D and a UFP_D shall set both of the appropriate connected status flags permanently to 1.

To resolve the direction of DisplayPort signaling when two systems are both capable of behaving as both a DFP_D and a UFP_D are connected together, a vendor-dependent method of disabling one or the other function shall be provided. After one of the two functions is disabled, the respective system shall issue an updated DisplayPort Status Update command with only the DFP_D Connected or UFP_D Connected bit set, as appropriate.

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 5-3 lists the DisplayPort Status as transmitted in the VDO.

Table 5-3: DisplayPort Status

Bit(s)	Description	Values
1:0	DFP_D/UFP_D Connected	00 = Neither DFP_D nor UFP_D is connected, or Adaptor is disabled. 01 = DFP_D is connected. 10 = UFP_D is connected.[a] 11 = Both DFP_D and UFP_D are connected.
2[b]	Power Low	0 = Adaptor is functioning normally or is disabled. 1 = Adaptor has detected low power and disabled DisplayPort support.
3[b]	Enabled	0 = Adaptor DisplayPort functionality is disabled. 1 = Adaptor DisplayPort functionality is enabled and operational.
4[b]	Multi-function Preferred	0 = No preference for Multi-function. 1 = Multi-function preferred.
5[b]	USB Configuration Request	0 = Maintain current configuration. 1 = Request switch to USB Configuration (if in DisplayPort Configuration).
6[b]	Exit DisplayPort Mode Request	0 = Maintain current mode. 1 = Request exit from DisplayPort Mode (if in DisplayPort Mode).
7[c]	HPD State	0 = HPD_Low. 1 = HPD_High.[d]
8[c]	IRQ_HPD	0 = No IRQ_HPD since last status message. 1 = IRQ_HPD.[e]
31:9	RESERVED	RESERVED (always 0).

a. An Adaptor that contains a DisplayPort Branch device with its DisplayPort receiver connected to the USB Type-C Plug shall report UFP_D connected, regardless of whether the Branch device has DFP_Ds connected.

b. These status bits apply only to DisplayPort Status sent by a UFP_U to a DFP_U. When sending DisplayPort Status, the DFP_U shall clear these bits. When receiving DisplayPort Status, the UFP_U shall ignore these bits.

c. These status bits apply only to DisplayPort Status sent by a UFP_D to a DFP_D. When sending DisplayPort Status, the DFP_D shall clear these bits. When receiving DisplayPort Status, the UFP_D shall ignore these bits.

d. HPD state is reported after glitch filtering, IRQ_HPD filtering (HPD state is maintained high when reporting an IRQ_HPD) and de-bouncing, when applied.

e. IRQ_HPD may be generated only when the HPD state is HPD_High.

Licensed to ... (...@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

5.2.4 DisplayPort Configure

The switch to use DisplayPort signaling is controlled using the DisplayPort Configure command with the VDO containing the details of the requested configuration. A DFP_U may transmit a DisplayPort Configure command at any time while in DisplayPort Mode. Prior to issuing the command, the DFP_U shall place the USB Type-C pins that are to be re-configured to DisplayPort Configuration into the Safe state (as specified in *USB Type-C r1.1* (i.e., ensure that there is no USB signaling on these pins)). The UFP_U may enable DisplayPort signaling immediately upon receipt of this command. It shall respond to this command with a Responder ACK once it has ensured that the re-configured pins are connected to DisplayPort.

Note: *The UFP_U may directly connect the re-configured pins to the appropriate DisplayPort functionality (e.g., to pins on a DisplayPort connector, or to a DisplayPort Protocol Converter). In this case, there is no re-connection required and the Responder ACK may be immediately issued. Upon receipt of the Responder ACK to this command, the DFP_U shall switch the re-configured pins on its USB Type-C Receptacle from the Safe state to DisplayPort, connect the SBU pins, and then, if a DFP_D, indicate the HPD state to the DisplayPort transmitter, or if a UFP_D, transmit the HPD state within a DisplayPort Status Update command.*

The DFP_U may change the number of DisplayPort lanes that are connected as a result of re-configuration. In this case, as well as placing any additional USB Type-C pins into the Safe state before sending the DisplayPort Configure command, the DFP_U shall ensure that the HPD state is taken low and then isolate the SBU pins. The DFP_U shall not re-connect the SBU pins until after receiving the Responder ACK and then re-connecting the re-configured High-Speed pins. In addition, the DFP_U, when a DFP_D, shall ensure that HPD_Low is maintained for a minimum of 3ms to ensure that its DisplayPort transmitter (e.g., GPU) sees an HPD disconnect event (and not an IRQ_HPD).

The DFP_U may switch to USB Configuration using the DisplayPort Configure command with the Select Configuration field programmed to USB. Prior to issuing the command, the DFP_U shall ensure that the HPD state is low, and then isolate the USB Type-C pins that were previously re-configured to DisplayPort Configuration, including the SBU pins. The DFP_U, when a DFP_D, shall ensure that HPD_Low is maintained for a minimum of 3ms to ensure that its DisplayPort transmitter (e.g., GPU) sees an HPD disconnect event (and not an IRQ_HPD). The UFP_U may start to use USB signaling immediately upon receipt of this command. It shall respond to this command with a Responder ACK once it has connected the re-configured pins to USB, where USB functionality on these pins is supported. Upon receipt of the Responder ACK to this command, the DFP_U shall switch the previously re-configured pins on the USB Type-C Receptacle to USB.

The DFP_U shall take into account the UFP_U's capabilities together with the limitations of the cable connecting the two systems (in the case of a USB Type-C to USB Type-C Cable) when determining the configuration of DisplayPort on the interface and select a configuration that is mutually supported. If multiple configurations are available, the configuration to be selected is a matter of DFP_U policy.

Licensed to Jay Lee (linshicai@rtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Table 5-4 lists the DisplayPort Configurations. The DFP_U shall select one of the signaling rates and configurations and one pin assignment, unless it is switching back to USB Configuration, in which case the pin assignment shall be cleared to 0. The Configuration shall indicate the DisplayPort role of the UFP_U (either DFP_D or UFP_D), independently of whether the UFP_U is implemented with a receptacle or plug. The UFP shall select the corresponding UFP_D pin assignment if it is a UFP_D implemented on a receptacle or DFP_D implemented on a plug, and shall select the corresponding DFP_D pin assignment if it is a DFP_D implemented on a receptacle or UFP_D implemented on a plug.

Notes: *Setting of appropriate multiplexers to support DisplayPort is associated with the DisplayPort Configure command.*

It is not necessary for the DFP_U to exit DisplayPort Mode to change the DisplayPort Configuration settings on the interface.

Table 5-4: DisplayPort Configurations

Bit(s)	Description	Values
1:0	Select Configuration	00 = Set configuration for USB.[a] 01 = Set configuration for UFP_U as DFP_D.[a] 10 = Set configuration for UFP_U as UFP_D.[a] 11 = RESERVED.
5:2	Signaling for Transport of DisplayPort Protocol	0000 = Signaling unspecified (used only when Select Configuration field is set for USB Configuration). 0001 = Select *DP v1.3* signaling rates and electrical settings. 0010 = Select Gen 2 signaling rates and electrical specifications. All other values are RESERVED.
7:6	RESERVED	RESERVED (always 0).
15:8	Configure UFP_U Pin Assignment	00000000 = De-select pin assignment. 00000001 = Select Pin Assignment A. 00000010 = Select Pin Assignment B. 00000100 = Select Pin Assignment C. 00001000 = Select Pin Assignment D. 00010000 = Select Pin Assignment E. 00100000 = Select Pin Assignment F. All other values are RESERVED.
31:16	RESERVED	RESERVED (always 0).

a. Selecting configuration for USB results in selecting USB Configuration. Either selecting UFP_U as DFP_D or selecting UFP_U as UFP_D results in selecting DisplayPort Configuration.

Licensed to Jay Lin (jinsc@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

A UFP_U supporting DisplayPort on a USB Type-C connector that supports Multi-function operation (i.e., supports the simultaneous transmission of USB SuperSpeed signaling and DisplayPort signaling) may be instructed by the DFP_U to configure the related DisplayPort signaling-only pin assignment (i.e., Pin Assignment A, C, or E). If the UFP_U supports only one or two lanes of DisplayPort, the UFP_U shall ensure, when so instructed, that the channels normally associated with USB SuperSpeed signaling that are not being used in this pin assignment are in the Safe state (as specified in *USB Type-C r1.1*).

The UFP_U shall respond with a Responder NAK in the event that it is unable to switch to the requested Configuration.

Note: *This Responder NAK indicates an error in the DFP_U.*

5.2.5 Attention

The Attention VDM is used by a UFP_U to signal to the DFP_U that there has been some change in the UFP_U's status since last reported. The UFP_U shall report its updated status in the VDO following the header, using the format specified in Table 5-3. See Section 5.2.3 for full details of the use of status reporting.

5.2.6 DisplayPort Exit Mode

The Exit Mode command is used by a DFP_U to instruct the UFP_U to exit DisplayPort Mode. The DFP_U shall issue an Exit Mode command only when the port is configured to be in USB Configuration. If the UFP_U receives an Exit Mode command when in DisplayPort Configuration, it shall switch to USB Configuration and then continue with the other actions that it performs on receiving an Exit Mode command.

Note: *Receipt of an Exit Mode command while not configured in USB Configuration indicates an error in the DFP_U.*

The UFP_U that receives the Exit Mode VDM shall return to V_{CONN} default power consumption, and exit the node, returning to normal USB operation. It shall then send the Responder ACK response. The device is not allowed to return either a Responder NAK or Responder BUSY.

After the DFP_U receives the Responder ACK, the DFP_U exits into the state that it enters on a new connection.

Note: *The Exit Mode command may be addressed directly to DP_SID plus the Object Position that identifies the mode, or may be addressed to DP_SID plus Object Position 111b (Exit All Modes).*

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

5.3 VDM Flows

Figure 5-1 illustrates the initial actions that occur when a new connection is detected. These include:

- Starting the tAMETimeout timer (see *USB Type-C r1.1*, Section 5.1.3) at the UFP_U, on expiration of which *USB Billboard r1.0* is connected if VBUS is being received;
- Determining the cable or Adaptor properties by using Discover Identity addressed to SOP';
- Negotiating a PD contract – Mandatory, but for simple devices a contract not to supply/ consume power is sufficient; and
- Establishing the port's data role – Optional, but if used must be performed before VDMs related to Alternate Modes are sent, or after successfully exiting DisplayPort Mode. A DFP_U/UFP_U role swap, if performed, must be completed before entering DisplayPort Mode.

The result is to move to either the DFP_U or the UFP_U message flows, depending on the port role.

Note: *Implementers should take care to ensure that sufficient time is allowed to discover the connected display during system boot and resume from sleep.*

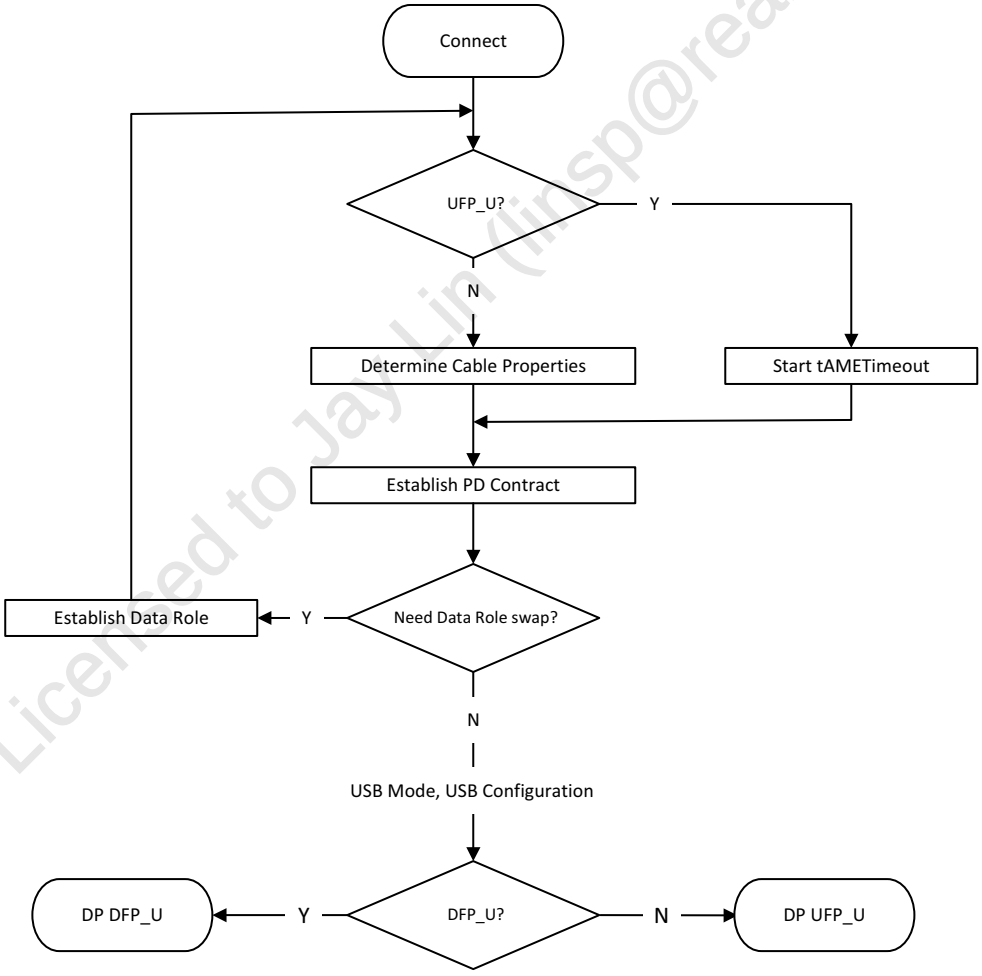


Figure 5-1: Connect Initial Actions

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 5-2 illustrates the following, up to the point where the port is ready to be configured to use DisplayPort Configuration:

- Message flow at the DFP_U for DisplayPort Mode discovery and entry
- Generation of Status Update commands
- Receipt of Attention messages

While waiting to be configured to use DisplayPort, the port is in a DisplayPort Mode/USB Configuration Wait state. If, while waiting in this state, the DFP_U is a UFP_D, it may detect a change in HPD state or an IRQ_HPD. In this case, the UFP_D shall transmit a Display Status Update Command and await a DisplayPort Status Update Response (not shown in Figure 5-2).

When the DFP_U and UFP_U are attached by way of a USB Type-C to USB Type-C Cable, the DFP_U shall ensure that the capabilities of the DFP_U, cable (where present), and UFP_U are compatible with one another (i.e., that the connection is not DFP_D-to-DFP_D or UFP_D-to-UFP_D, and that there is a signaling configuration supported by all three). If the DFP_U, cable, and UFP_U are incompatible, the DFP_U shall remain in USB Mode.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

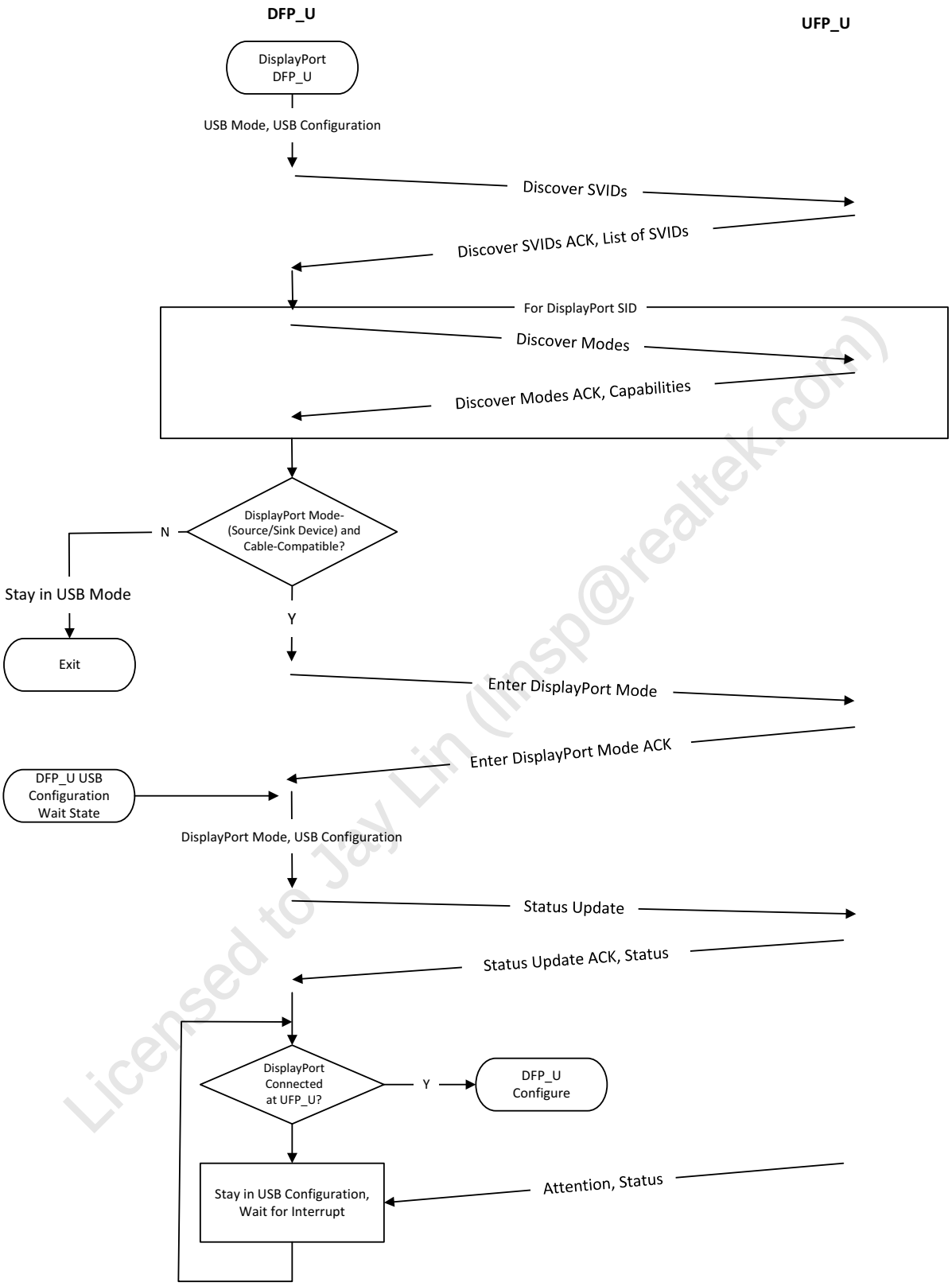


Figure 5-2: DFP_U DisplayPort Mode Discovery and Entry

Licensed to Jay Lin (insp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 5-3 illustrates the message flow at the DFP_U for configuration, up to the point that the SuperSpeed and SBU connections start to use DisplayPort Configuration.

Note: *If the UFP_U requests Multi-function Mode and the DFP_U supports Multi-function Mode, it is a matter of DFP_U implementation policy whether to select Multi-function Mode or remain in DisplayPort Mode.*

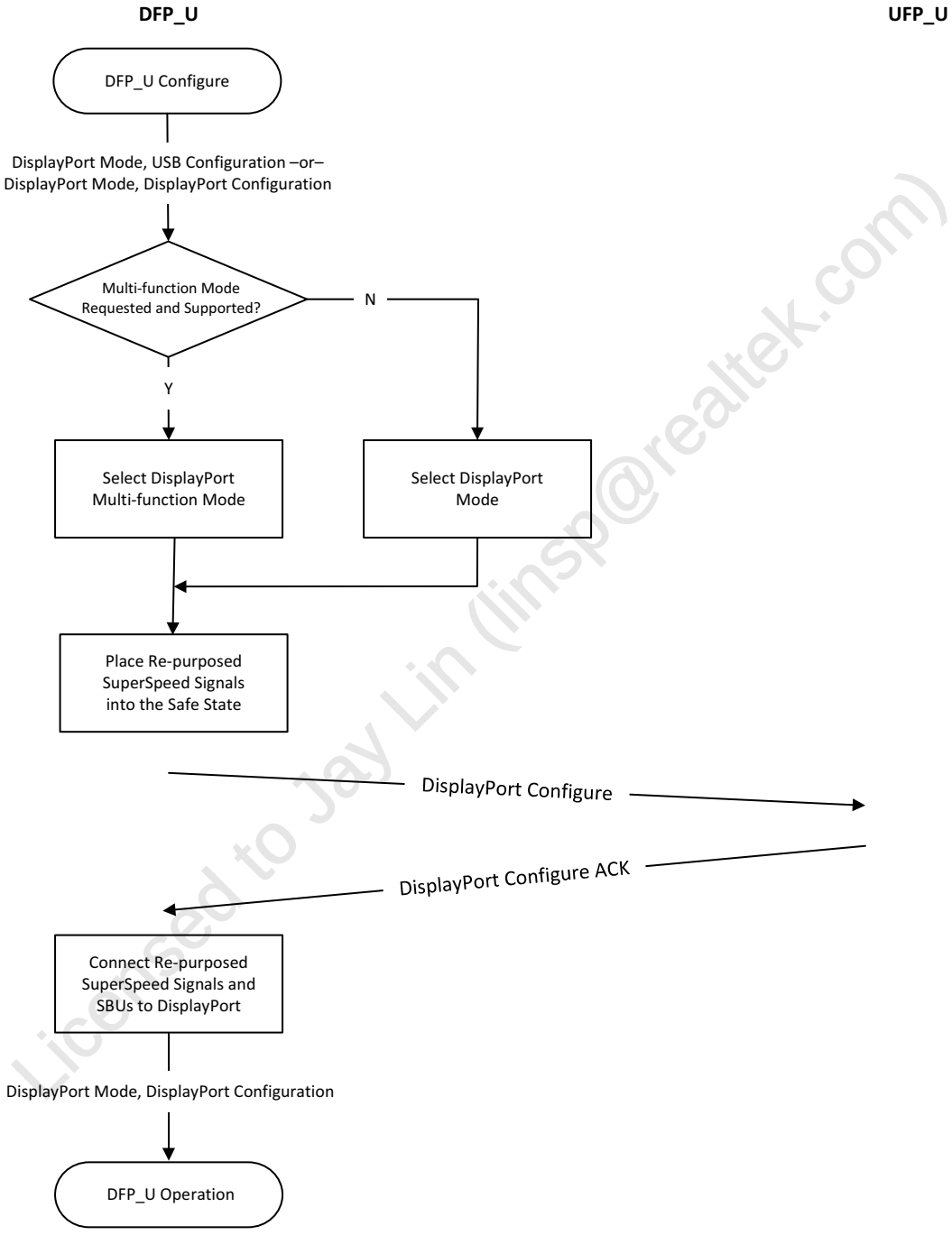


Figure 5-3: DFP_U DisplayPort Configuration

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 5-4 illustrates the message flow when the DFP_U is operating in DisplayPort Configuration and receives a request for re-configuration by a DisplayPort Status transmitted as the VDO in an Attention command received from the UFP_U. The UFP_U may send a DisplayPort Status that requests either of the following:

- Different configuration
- Switch to USB Configuration instead of DisplayPort Configuration

If the DisplayPort Status indicates that the UFP_U no longer has a connected display, the DFP_U shall initiate a transition to USB Configuration. In addition, the DFP_U may make a policy change (e.g., in preparation to exiting DisplayPort Mode prior to using PD for a data role swap).

If the DFP_U is a UFP_D, it may detect a change in HPD state or an IRQ_HPD. In this case, the UFP_D shall transmit a Display Status Update Command and await a DisplayPort Status Update Response (not shown in Figure 5-4).

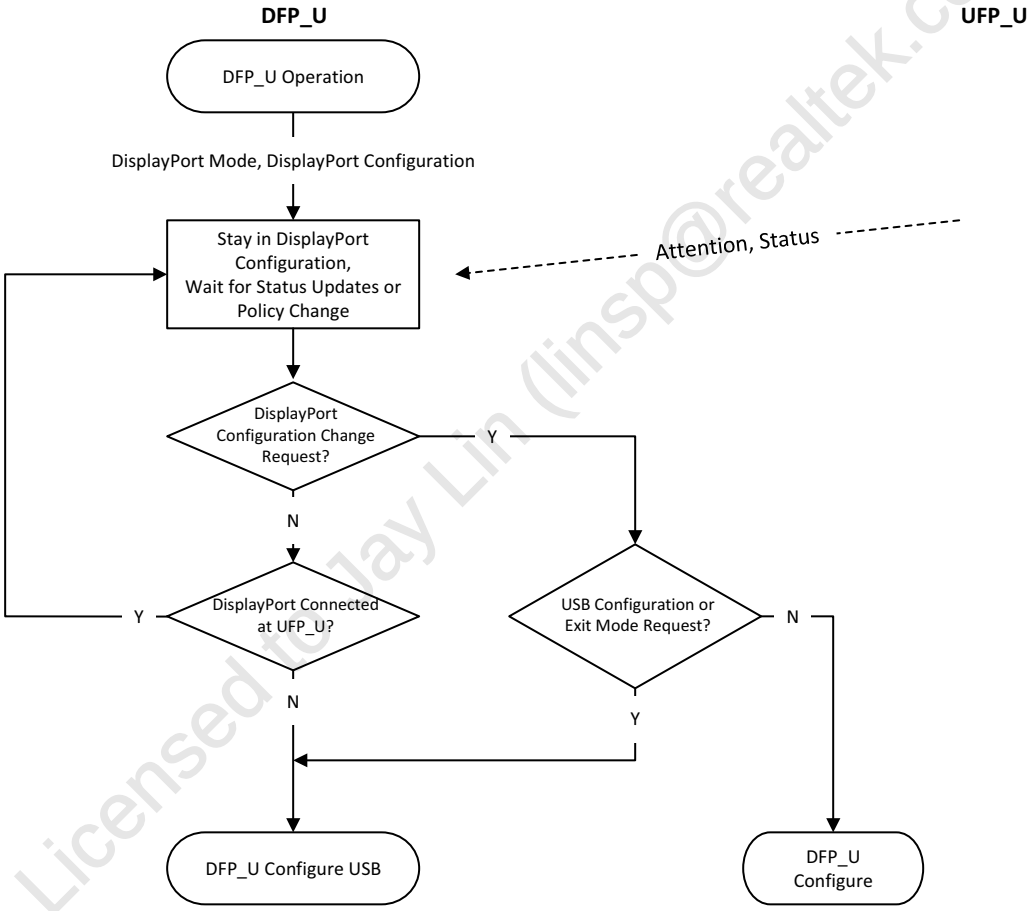


Figure 5-4: DFP_U DisplayPort Operation and Re-configuration

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 5-5 illustrates the sequence at the DFP_U for re-configuring the port to USB operation. This may be performed as a result of one of the following:

- Configuration request to re-configure from DisplayPort Configuration
- Result of an Exit Mode request, which can be completed only when in USB Configuration

Prior to transmitting the DisplayPort Configure command specifying USB Configuration, the DFP_U places the previously re-configured pins into the USB Safe state and isolates the SBU pins, as described in Section 5.2.6.

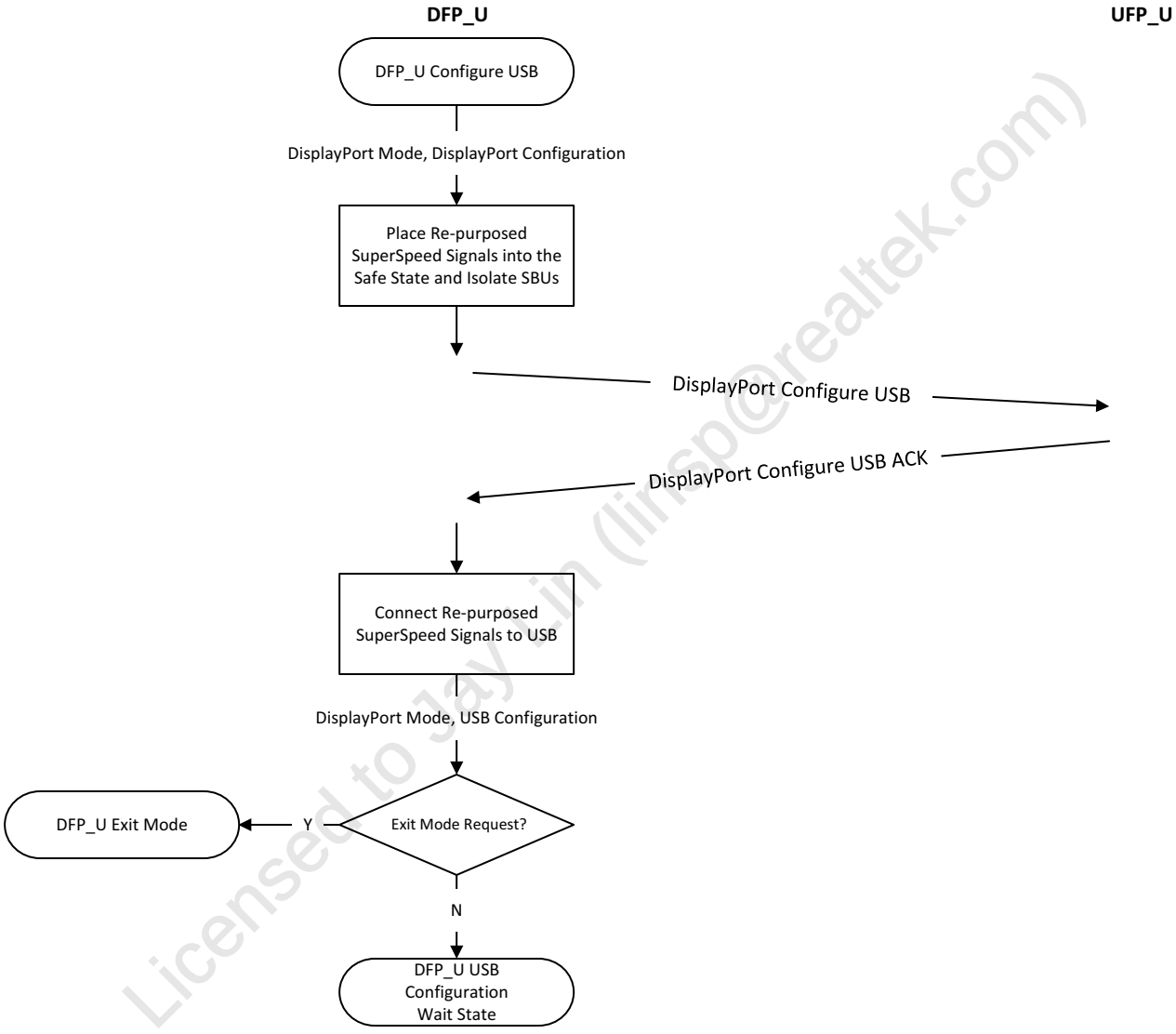


Figure 5-5: DFP_U Re-configure to USB Configuration

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 5-6 illustrates the Exit Mode sequence at the DFP_U. This sequence is entered only when the port configuration has been set or reset to USB Configuration.

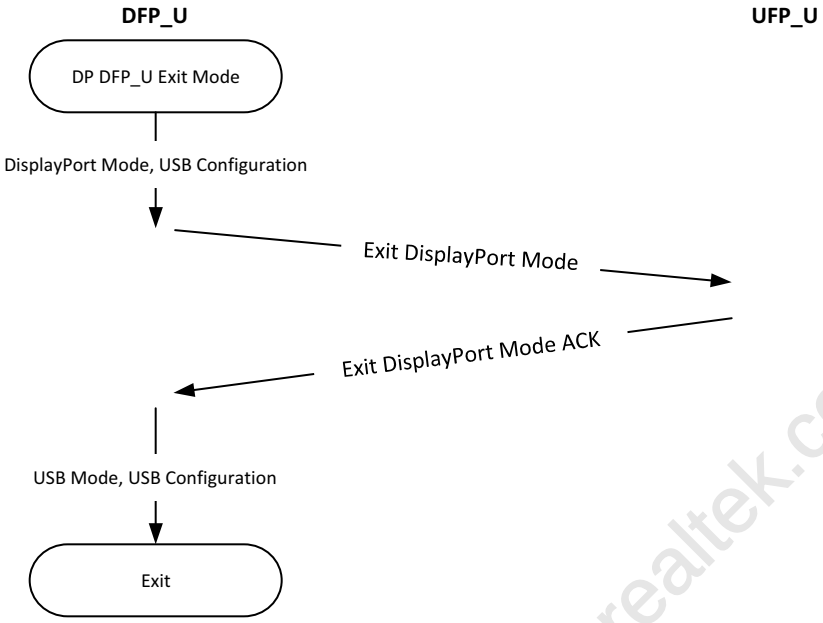


Figure 5-6: DFP_U Exit Mode

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 5-7 illustrates the message flow for the UFP_U DisplayPort Mode discovery and entry, up to the point where the port enters a Wait state ready to be configured to use DisplayPort Configuration.

Note that *USB Type-C r1.1* requires that the UFP_U present a *USB Billboard r1.0* device class if the UFP_U fails to receive and act upon an Enter Mode request within tAMETimeout (see *USB Type-C r1.1*, Section 5.1.3). In this case, the UFP_U shall await further commands from the DFP_U (e.g., it may receive a DisplayPort Enter Mode command even though it has already presented USB Billboard).

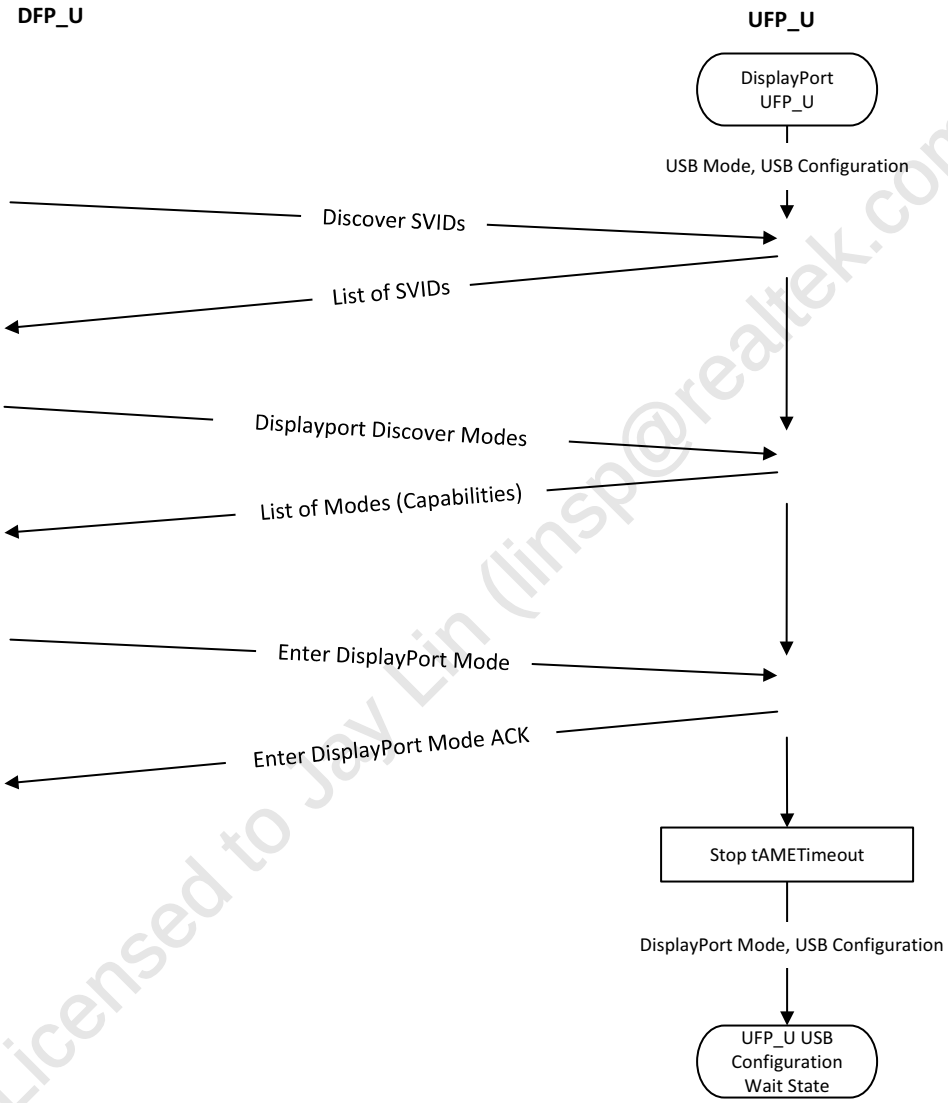


Figure 5-7: UFP_U DisplayPort Mode Discovery and Entry

Licensed to Joy Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 5-8 illustrates the message flow for the UFP_U while in the USB Wait state. In this state, the UFP_U may receive a command to do any one of the following:

- Report status
- Configure the UFP_U into DisplayPort Configuration
- Exit DisplayPort Mode

During this state, the UFP_U may detect a change in connection status on its DisplayPort side (e.g., a DFP_D or UFP_D is attached, or a change in HPD state or an IRQ_HPD) and consequently generate an Attention message toward the DFP_U. On entry to the USB Wait state after receiving a DisplayPort Enter Mode command, the UFP_U shall wait to receive a DisplayPort Status command from the DFP_U, and then provide its status in its response. The UFP_U shall not generate an Attention Message until a subsequent state change.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

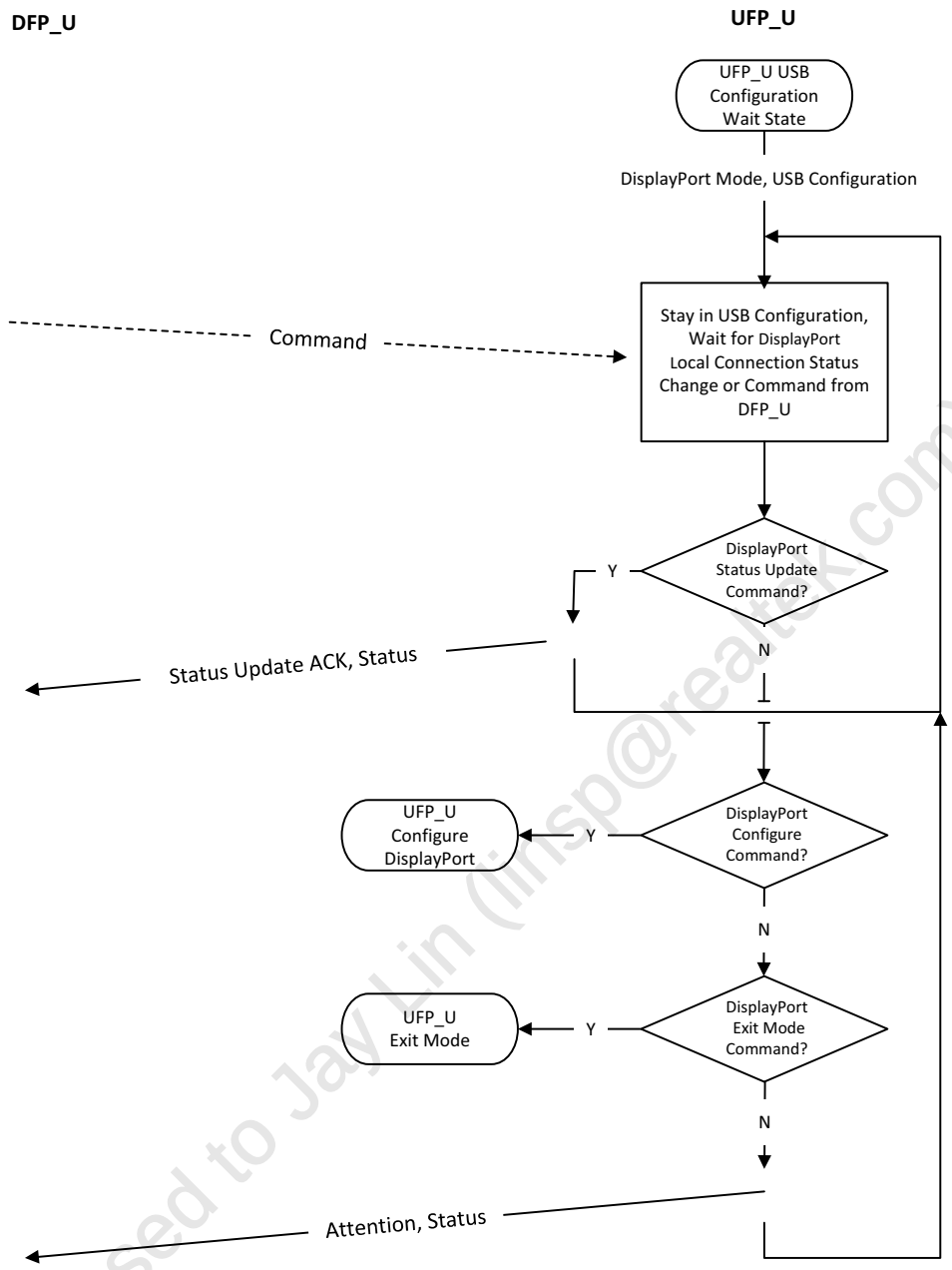


Figure 5-8: UFP_U USB Configuration Wait State

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 5-9 illustrates the message flow at the UFP_U for configuration, up to the point that the SuperSpeed and SBU connections start to use DisplayPort Configuration.

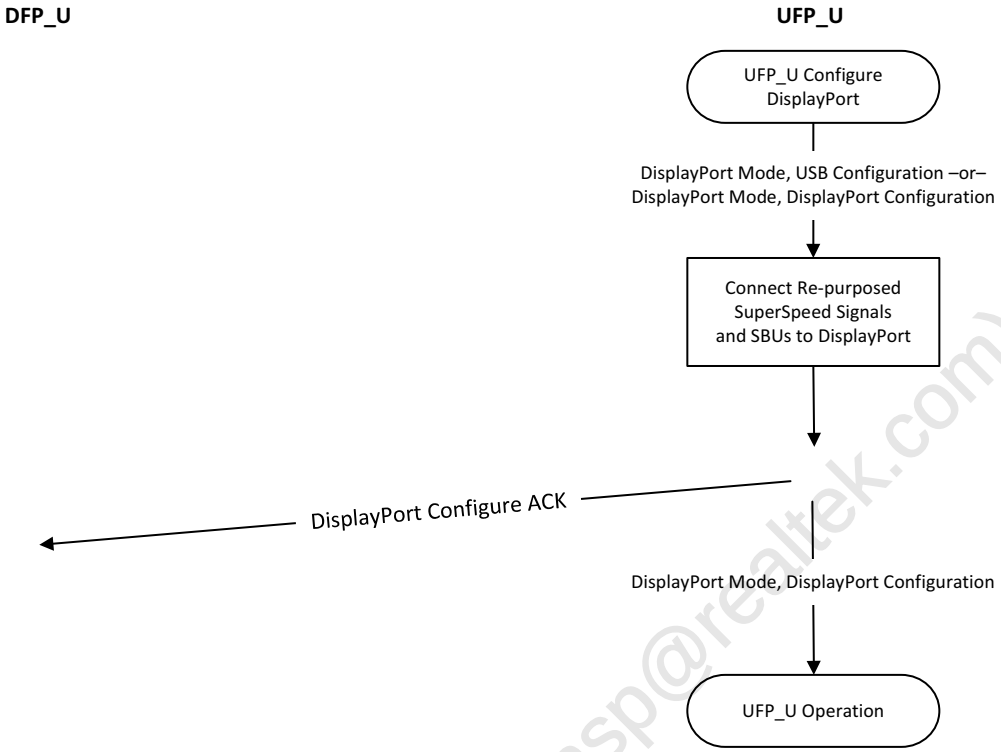


Figure 5-9: UFP_U DisplayPort Configuration

Licensed to Jay Lin (insp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 5-10 illustrates the message flow for UFP_U operation while in DisplayPort Configuration. If the UFP_U detects a change in the connection status on its DisplayPort side (e.g., a DFP_D or UFP_D is disconnected, or a change in HPD state or an IRQ_HPD), the UFP_U sends an Attention message to the DFP_U. Simultaneously, the UFP_U waits for a command from the DFP_U. The UFP_U may receive one of the following:

- Status Update command
- Configuration command for a different configuration
- Command to switch to using USB Configuration instead of DisplayPort Configuration

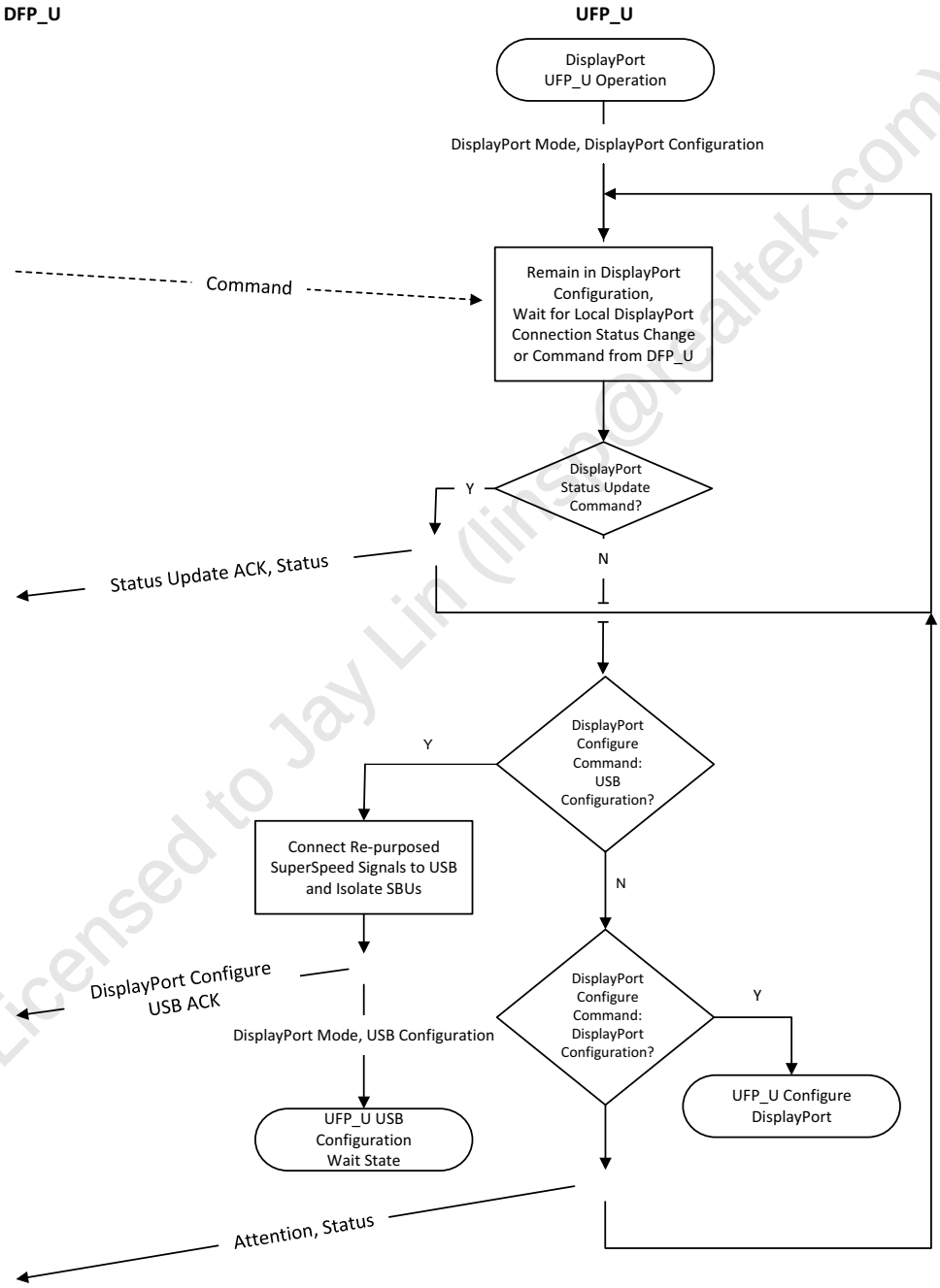


Figure 5-10: UFP_U DisplayPort Operation

Licensed to Jay Lin (jinsc@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

Figure 5-11 illustrates the Exit Mode sequence at the UFP_U. The UFP_U shall acknowledge the command, but take no further action. The UFP_U shall not send Attention messages after acknowledging the Exit Mode command. The UFP_U shall not re-start its tAMETimeout (USB Billboard connect) timer (i.e., *USB Billboard r1.0* is never displayed after the connection enters DisplayPort Mode).

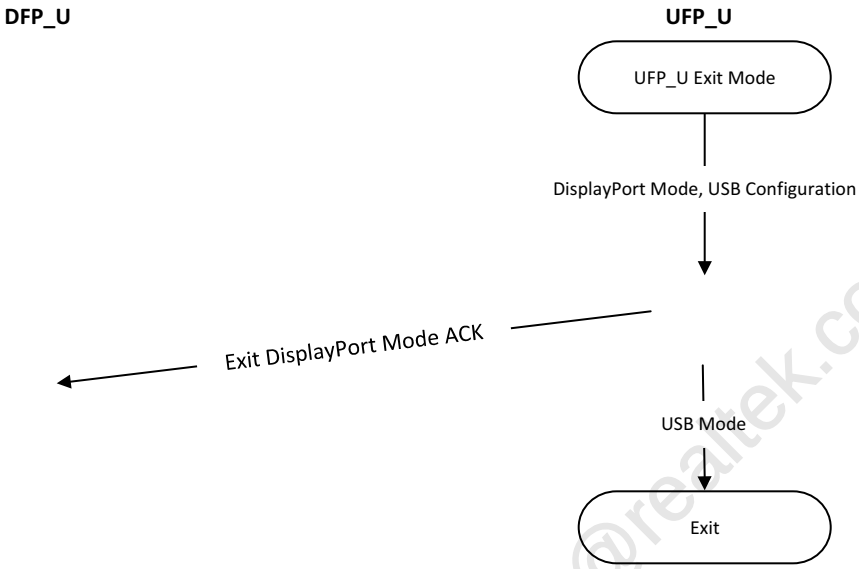


Figure 5-11: UFP_U Exit Mode

A port may be disconnected or subject to a Hard Reset at any time. The port resets the DisplayPort Mode. Figure 5-12 illustrates the sequence of actions on a Hard Reset or disconnect at both the DFP_U and UFP_U. The UFP_U shall reduce its power consumption to no more than the default VCONN. No Alternate Mode messaging occurs. After a Hard Reset, the port behaves as if a new connection was detected.

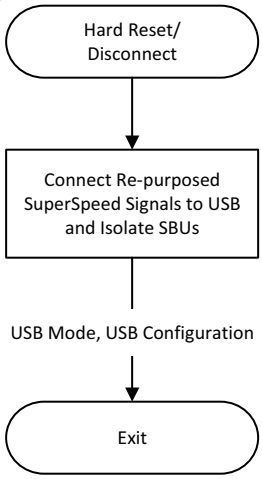


Figure 5-12: Hard Reset and Disconnect

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

6 Miscellaneous

This section describes power management, low-power operation, and equivalent USB functionality.

6.1 VCONN and VBUS

As specified in *USB Type-C r1.1*, VCONN is provided by the DFP_U at a voltage between 2.7 and 5.5V. Because DisplayPort Adaptors may require more power than that specified in *USB Type-C r1.1*, a DFP_U supporting DisplayPort on a USB Type-C Receptacle shall support 1.5W at any voltage within this range when operating in DisplayPort Mode.

The Adaptor shall operate at any voltage between 2.7 and 5.5V (as presented at the Adaptor's USB Type-C Plug, there may be an IR drop on a VCONN wire within the Adaptor's cable), and the Adaptor shall not consume in excess of 1.5W. PD communications may be maintained if the voltage falls below 2.7V; however, the power consumption shall not exceed 70mW and DisplayPort Adaptor functionality may be disabled. The Adaptor shall report whether it continues to operate but disables DisplayPort functionality. (See Table 5-3.)

During the Discovery Process and at any time that DisplayPort Mode is not entered, the Adaptor shall comply with *USB Type-C r1.1* VCONN power consumption requirements.

A DFP_U supporting DisplayPort Mode may either be a USB Power Provider, or UFP with accessory support that has detected a Powered Accessory (as defined in *USB Type-C r1.1*), and in this respect is operating as a DFP_U but not providing VBUS. Correspondingly, no assumptions can be made about the presence or direction of power on VBUS.

An Adaptor supporting DisplayPort on a USB Type-C Plug shall operate as a VCONN-powered accessory. It is not required to provide power on VBUS, and does not depend on the presence of power on VBUS (except to power the presentation of *USB Billboard r1.0* class, when required). A UFP with accessory Support supporting DisplayPort on a USB Type-C Receptacle shall not depend on power being available on VBUS. An Adaptor shall power the *USB Billboard r1.0* class from VBUS, not VCONN, so that Billboard is presented should the Adaptor be plugged into a DFP_U that does not support Alternate Modes and does not support VCONN.

6.2 Low-power Operation

6.2.1 Display Sleep

If the DFP_U is also a DFP_D, the DFP_U is in control of both the display power state and the DisplayPort Mode setting. When, as a DFP_D, it puts the display to sleep (i.e., by writing to DisplayPort DPCD Address 00600h within the UFP_D), the DFP_D shall not exit DisplayPort Mode (i.e., issue an Exit Mode command), and both the DFP_U and UFP_U shall maintain PD operation.

If the DFP_U is also a UFP_D (e.g., a display), the DFP_D is in control of the display power state; however, the DFP_U on the connected port is in control of the DisplayPort Mode setting. When the DFP_D (UFP_U) puts the display to sleep (i.e., by writing to DisplayPort DPCD Address 00600h within the UFP_D), the DFP_D shall not request an exit from DisplayPort Mode (i.e., request the DFP_U to issue an Exit Mode command), and both the DFP_U and UFP_U shall maintain PD operation.

Licensed to Jay Huang, huangsuc@realtek.com.tw

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

During DisplayPort low power (i.e., after a write of 010b or 101b to DisplayPort DPCD Address 00600h within the UFP_D), power consumption from VConn shall be limited to 50 mA unless the Adaptor includes a Protocol Converter to DVI or HDMI, in which case the limitation is 160 mA. The transition to the lower level of power consumption shall be completed within 250 ms of the write to DisplayPort DPCD Address 00600h within the UFP_D.

6.2.2 USB Suspend General Properties (Informative)

The system (DFP_U) may place the USB port into the USB Suspend state at any time. Typically, the system will do this when it is placed into a low-power state. The DFP_U maintains USB connectivity information on the port during the USB Suspend state (i.e., detects physical disconnect and connect events).

If the "USB Suspend Supported" bit is set in PD contract, the UFP_U is required to reduce its consumption of VBUS power to no more than 25mW (intended to be sufficient to maintain PD if powered from VBUS) and *USB Type-C r1.1* requires VCONN to support only 7.5mA (i.e., maximum of 35mW).

If the "USB Suspend Supported" bit is not set in PD contract, VBUS power and VCONN (if needed to power an Adaptor or for Active cable) are maintained.

6.2.3 USB Type-C DFP_U

DisplayPort functionality shall be maintained during the USB Suspend state. The DFP_U shall ensure that the PD contract has the "USB Suspend Supported" bit cleared. The DFP_U shall not exit Alternate Mode simply because USB has been placed into its Suspend state. Note that DisplayPort connectivity information is maintained during the USB Suspend state.

During system low-power, the DisplayPort sub-system may put the display to sleep; however, the DisplayPort sub-system may also maintain DisplayPort connectivity information (e.g., so that a new DP connection can be treated as a wake event):

- If DFP_U = DFP_D (DisplayPort Source device, HPD receiver)

  If the DFP_D (DisplayPort Source device) needs to maintain HPD connectivity information then, prior to placing the system into low power, the DFP_D shall follow *DP v1.3* low-power requirements and the DFP_U (=DFP_D) shall maintain Alternate Mode. The DFP_U also must maintain support for the full level of VCONN power if the cable is an Active cable or is being used to power the Adaptor; thus, the DFP_U shall ensure that the PD contract has the "USB Suspend Supported" bit cleared.

  If the DFP_D does not need to maintain HPD connectivity information then, prior to placing the system into low power, the DFP_U (=DFP_D) shall exit Alternate Mode. The DFP_U shall then place the USB connection into the Suspend state and reduce its support of VCONN to 7.5mA. In this case, exit from low power behaves like a new DisplayPort connection discovery.

- If DFP_U = UFP_D (DisplayPort Sink device, HPD transmitter)

  DisplayPort does not support the notion of the display autonomously going into low power, thus entry into system low power behaves like a DisplayPort disconnect. The DFP_U shall exit Alternate Mode and shall then reduce its support of VCONN to 7.5mA.

Licensed to Jay Lin (jlin@spd.realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

6.2.4 USB Type-C UFP_U that Supports a VCONN-Powered Adaptor

The UFP_U does not provide VBUS, and there is no USB data connectivity. There is therefore no concept of USB Suspend.

If the UFP_U goes into a system low-power state:

- If UFP_U = DFP_D (DisplayPort Source device, HPD receiver)

  If the DFP_D (DisplayPort Source device) needs to maintain HPD connectivity information then, prior to placing the system into low power, the DFP_D shall follow *DP v1.3* low-power requirements and the UFP_U (=DFP_D) shall maintain Alternate Mode. The UFP_U also must maintain support for the full level of VCONN power to power the Adaptor; thus, the UFP_U shall ensure that the PD contract has the "USB Suspend Supported" bit cleared.

  If the DFP_D does not need to maintain HPD connectivity information then, prior to placing the system into low power, the DFP_U shall exit Alternate Mode. The UFP_U shall then place the system into low power and reduce its support of VCONN to 7.5mA. In this case, exit from low power behaves like a new DisplayPort connection discovery.

- If UFP_U = UFP_D (DisplayPort Sink device, HPD transmitter)

  DisplayPort does not support the notion of the display autonomously going into low power, thus entry into system low power behaves like a disconnect. The UFP_U shall exit Alternate Mode and shall then reduce its support of VCONN to 7.5mA.

6.3 Equivalent USB Functionality

USB Type-C r1.1 strongly encourages devices that support Alternate Modes to provide equivalent USB functionality, where such exists, for best user experience (see *USB Type-C r1.1*, Section 5.1). An example of equivalent USB functionality for DisplayPort is the support of *USB AV r1.0* for the uncompressed transport of video with a resolution of 1080p60 24bpp RGB.

6.4 USB Billboard URL

A device that is required to present a USB Billboard device compliant to *USB Billboard r1.0* shall include within the string indexed by **iAlternateModeString[n]** for DisplayPort Alternate Mode either a manufacturer-specific permanently available URL or the following VESA URL:

www.vesa.org/dp-usb-type-c

An example string is as follows:

DisplayPort to VGA Adaptor. For further assistance, see www.vesa.org/dp-usb-type-c.

Licensed to Jay Lin (jaylin@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

A DisplayPort Mode Examples

This appendix provides two examples of using DisplayPort Mode.

A.1 Purpose

Examples are offered to illustrate how Alternate Modes, negotiated using USB-PD, are used to provide DisplayPort functionality beyond the basic functionality defined in *USB Type-C r1.1*. The examples are not intended to be complete designs.

A.1.1 DisplayPort Mode Examples

There are many use scenarios for including DisplayPort on a USB Type-C connector. In all cases, the system with a USB Type-C connector may be a USB host or device, and, independently, may behave as a DFP_D or UFP_D.

Some possible examples include:

- Transmission of DisplayPort over a USB Type-C to USB Type-C Cable;
- Transmission of DisplayPort over a USB Type-C to DisplayPort plug or receptacle cable (either direction);
- Transmission of DisplayPort over a USB Type-C to other video format connector cable (e.g., HDMI or VGA). In these cases, the Adaptor would also contain a DisplayPort-to-other video format Protocol Converter;
- Simultaneous transmission of DisplayPort and *USB r3.1* to a dock (DisplayPort is limited to two lanes in this example, because two of the USB SuperSpeed data signal pairs are used for USB Enhanced SuperSpeed signaling).

These particular examples are intended only to illustrate the overall Discovery Process and Alternate Mode entry and exit flow using PD VDMs. Note that additional messaging is used to support DisplayPort Mode-specific Configuration, Status, and Control.

A.2 USB Type-C USB Host to DisplayPort Video Adaptor Example

This example describes a USB host with a USB Type-C Receptacle that can also behave as a DFP_D, and connects to an Adaptor that has a USB Type-C Plug and DisplayPort plug to adapt to a DisplayPort monitor.

Note that a USB device can support the same DisplayPort functionality.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

A.2.1 DisplayPort Video Adaptor Assumptions

The DisplayPort Video Adaptor is a cable with a USB Type-C Plug on one end and a DisplayPort connector on the other. It is an active Adaptor that contains circuitry (e.g., in the plug overmold, lump in the cable, etc.) that communicates with the USB host to expose its capabilities and enable its functionality.

The DisplayPort Video Adaptor does not provide power to charge the USB host because DisplayPort does not support the provision of power over the cable.

The Host in this example has the capability to source native DisplayPort video (i.e., behave as a DisplayPort Source device).

The host and DisplayPort Video Adaptor both use the specific USB SID associated with DisplayPort (DP_SID) to identify themselves. DP_SID is used firstly to indicate support for DisplayPort as an Alternate Mode, and secondly to distinguish commands that are specific to the support of DisplayPort as an Alternate Mode.

The host is permitted to use the DisplayPort Video Adaptor only if it identifies the Adaptor as a DisplayPort Video Adaptor. Failure to meet this results in a connection to a *USB Billboard r1.0* device that will provide error information.

The DisplayPort Video Adaptor is powered by V_{CONN} for the switch, CC logic, PD engine, and DisplayPort functions; however, *USB r2.0*-related functions must be powered by V_{BUS}.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

A.2.2 Block Diagrams

Figure A-1 illustrates a block diagram of the Host system with its multiplexer. Figure A-2 illustrates a block diagram of the DisplayPort Video Adaptor Cable assembly.

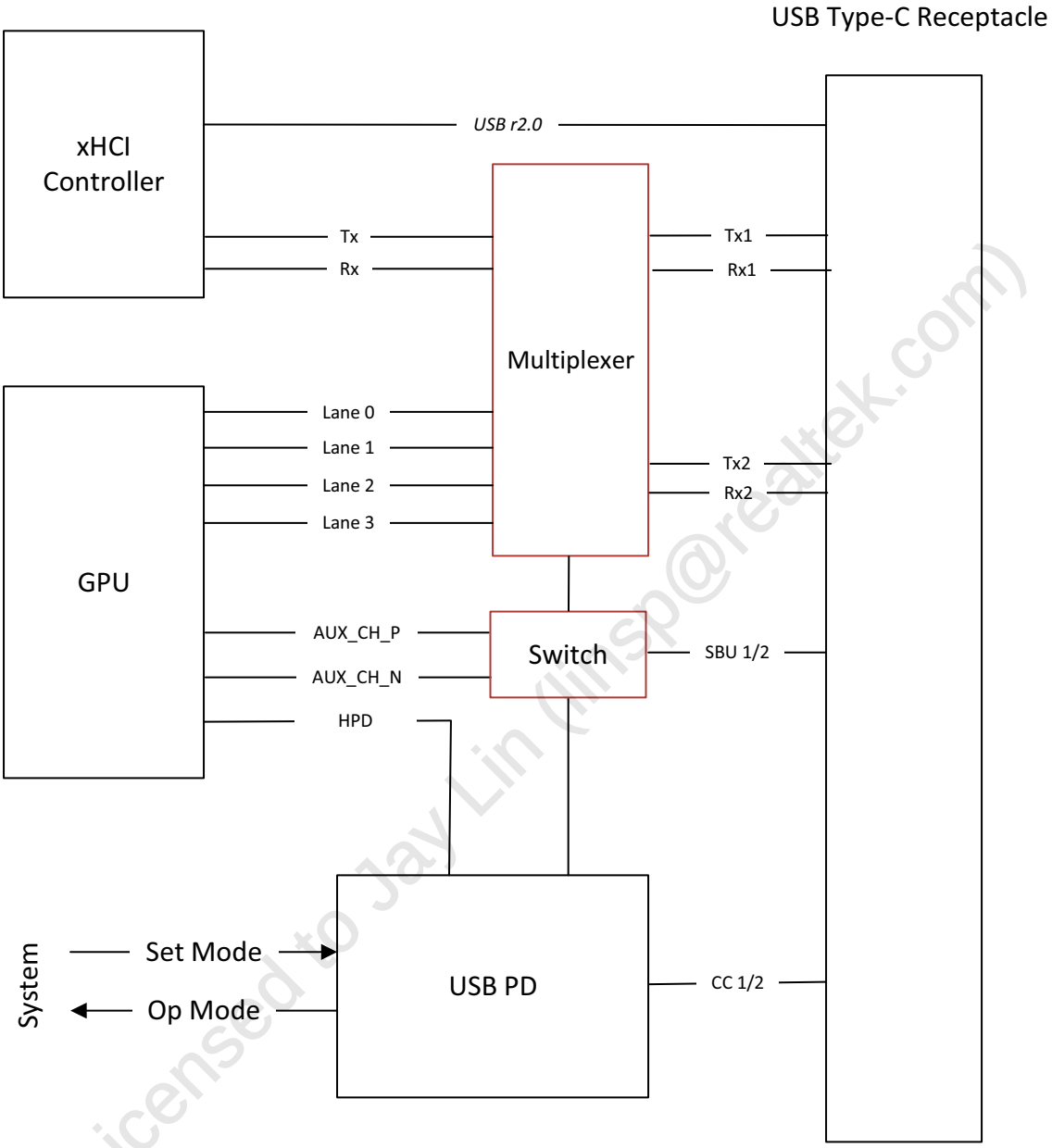


Figure A-1: Host System Block Diagram

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

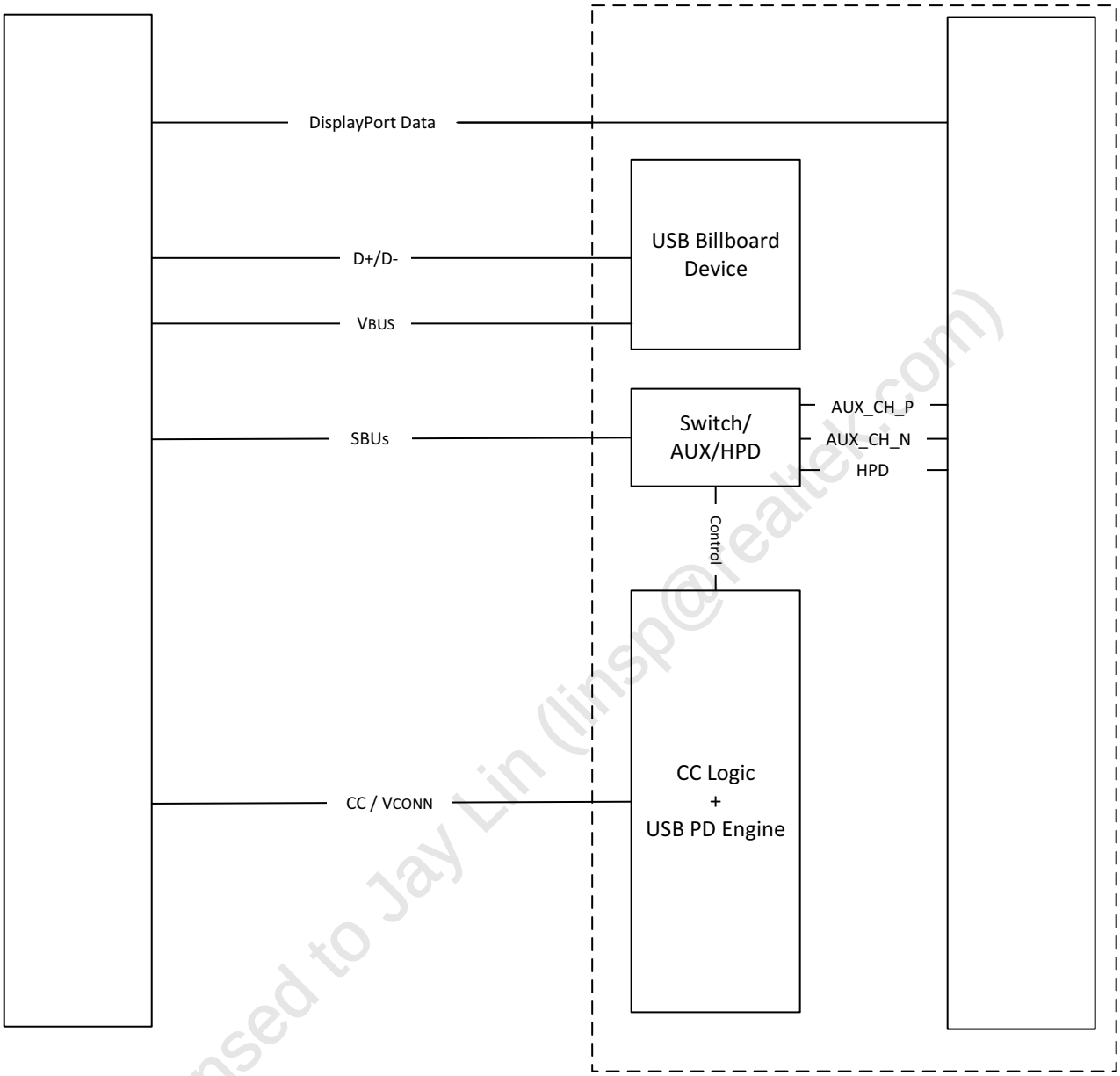


Figure A-2: DisplayPort Video Adaptor Cable Block Diagram

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

The host has a multiplexer that can steer one of the following:

- USB SuperSpeed signals
- All four lanes of the DisplayPort signals
- Two lanes of the DisplayPort signals together with the USB SuperSpeed signals to the USB Type-C Receptacle

The multiplexer or a combination of the multiplexer and DisplayPort port output manages the USB Type-C Plug flipping. When the USB-PD block is instructed by the Host system to re-configure the pins on the port to output DisplayPort signaling, the multiplexer is activated when the attached system identifies itself as a DisplayPort Video Adaptor cable in the Discovery Process and responds to the Enter Mode request with an ACK. If the DisplayPort Video Adaptor responds at any point during the Discovery Process with a NAK, the Host shall remain in the default *USB r2.0*/*USB r3.1* Configuration. The Host sees that the operational mode has changed based on the OpMode signal.

The DisplayPort Video Adaptor Cable assembly provides a USB-PD interface to communicate with the host. By default, the Cable assembly also provides a *USB r2.0* device interface that is used by hosts that do not support DisplayPort Video Adaptors to detect that a DisplayPort Video Adaptor is connected and to provide information for the user about the Adaptor and the host system's inability to use it. The DisplayPort Video Adaptor also contains circuitry to isolate the SBU connections until DisplayPort Mode is established. The DisplayPort Video Adaptor is powered by V_{CONN} for the switch, CC logic, PD engine, and DisplayPort functions; however the *USB r2.0*-related functions must be powered by V_{BUS}. The same Video Adaptor can be plugged into a USB host or device (with dual-role discovery; i.e., either may behave as a DisplayPort Source device). In the latter case, V_{BUS} is not available; therefore, V_{BUS} is not used by the Adaptor.

A.2.3 Solution

The host system and the DisplayPort Video Adaptor use USB-PD DisplayPort-defined Messages to communicate and take advantage of advanced features.

The flow of messages from the point after the DisplayPort Video Adaptor is plugged into the USB Type-C Receptacle, plug orientation has been determined, and the host/device relationship established is as follows:

- Host system does not support PD
  - After tAMETimeout, the host will see the *USB r2.0* device interface with *USB Billboard r1.0* and will report to the user that an unsupported device has been connected, identifying the type of device from the USB descriptor information presented in the USB Billboard.
- Host system supports PD and advanced functions, but does not support DisplayPort Video Adaptor(s)
  - Host may look for other advanced functions, but does support the DisplayPort Video Adaptor even though its presence is discovered, because the video Adaptor responds with DP_SID that the host does not recognize as identifying something that it supports.
  - Because DisplayPort Video Adaptors are not supported, after tAMETimeout the host will see the *USB r2.0* device interface with *USB Billboard r1.0* and will report to the user that an unsupported device has been connected, identifying the type of device from the USB descriptor information presented in the USB Billboard.

Licensed to Jay Lin (jinsplendor@gmail.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

- Host system supports PD and the DisplayPort Video Adaptor
  - Host looks for advanced functions and discovers a DisplayPort Video Adaptor by way of responses to USB-PD messages, using DP_SID.
  - Because a usable DisplayPort Video Adaptor is discovered, the host enters DisplayPort Mode and then re-configures its USB Type-C Receptacle to output DisplayPort signals on the connector. The video Adaptor cancels its tAMETimeout on entry to DisplayPort Mode.

This example uses the basic extensions when connected that do the following:

Base Set VDMs used

- Discover SVIDs
- Discover Modes (DP_SID)
- Enter Mode (DP_SID, Alternate Mode offset)
- Exit Mode (DP_SID, Alternate Mode offset)

Process Outline

The Host functions in its role as a DFP_U.

1 DFP_U detects presence of a device (CC pins) and connector orientation.

2 DFP_U applies 5V to VBUS and VCONN.

3 DFP_U uses PD to make a power contract with the UFP_U.

4 DFP_U runs Discover SVID process.

  a Sends Discover SVIDs message.

  b Receives Response message with a list of SVIDs for which the UFP_U supports modes.

5 DFP_U runs Discover Modes process.

  a Sends Discover Mode VDM for each of the SVIDs previously discovered that it supports, including DP_SID.

  b Receives Response message with a list of modes associated with DP_SID.

  c If a DisplayPort Video Adaptor is not discovered – stays in USB Mode, discovers the *USB Billboard r1.0* after tAMETimeout, and informs user – done.

  d Else.

6 DFP_U runs Enter Mode process.

  a Sends Enter Mode VDM with DP_SID and offset to mode.

  b Receives response:

    1 Responder ACK – DFP_U is now attached to the DisplayPort Video Adaptor, running in DisplayPort Mode and USB Configuration.

    2 Responder NAK – DFP_U returns to default *USB r3.1* Configuration.

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

7 DFP_U discovers Adaptor status, then waits until the Adaptor reports that it has a connected display.

8 DFP_U runs the DisplayPort Configure process.

   a Places signals to be re-configured into the Safe state.

   b Sends a DisplayPort Configure command to the Adaptor.

   c Receives Responder ACK response.

   d Configures signals to use DisplayPort.

9 If the DFP_U receives a Status update from the device that requires a change in configuration, the DFP_U re-runs the Configure process (possibly re-configuring the interface back into USB Configuration).

10 DFP_U continues to run mode until:

   a Explicitly exited by an Exit Mode VDM.

   b Hard Reset signaling on PD.

   c Power is removed from the host.

   d Adaptor is disconnected.

The host and the DisplayPort Video Adaptor will use the base set VDMs without change, together with the DP_SID-specific VDMs.

Note that the DisplayPort Video Adaptor is required to follow and/or not interfere with **all** rules for normal PD operation.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

A.3 USB Dock with DisplayPort Example

In this example, a device uses its USB Type-C connector as a cradle dock for a host.

Note: *There are additional mechanical requirements that are not covered in this example.*

A.3.1 USB Dock with DisplayPort Assumptions

- Cradle dock provides mechanical alignment and attachment in addition to those provided by the USB Type-C connector
- Dock uses USB-PD to charge/power the host
- Dock sources two lanes of native DisplayPort video on a DisplayPort receptacle
- Dock contains a USB hub that exposes three USB ports and attached internal USB devices:
  - USB Ethernet device (RJ-45 connector), and
  - USB audio device (a 3.5mm phone jack)

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

A.3.2 Block Diagram

Figure A-3 illustrates a block diagram of a USB Dock that also supports a DisplayPort DFP_D.

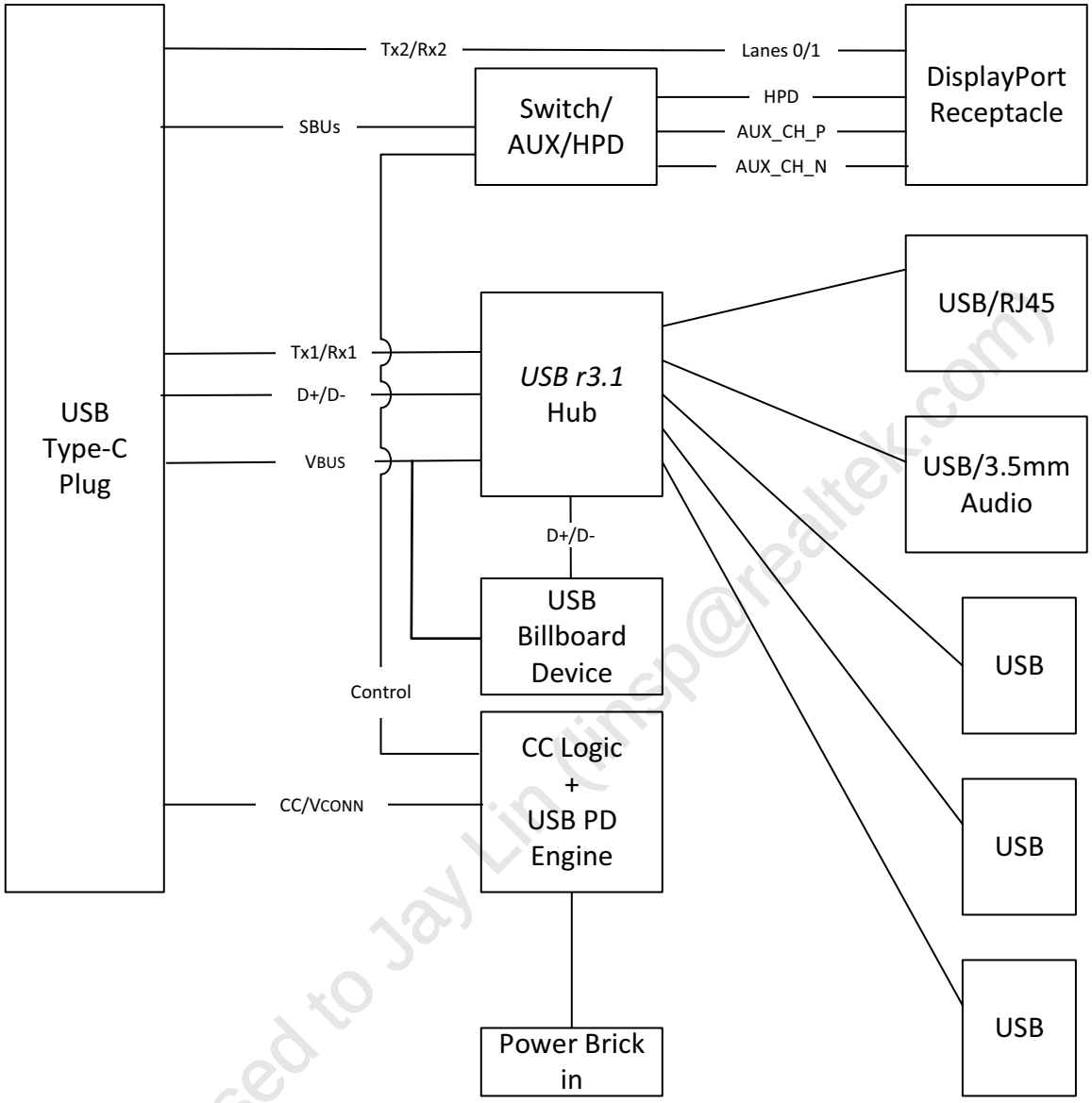


Figure A-3: Example USB Dock with DisplayPort Block Diagram

The dock communicates to the host that it can behave as a two-lane DisplayPort Source device and participates in the negotiation process to place the host into the Alternate Mode that supports the dock. Because USB-PD is used, it may also be used to negotiate power for the dock and host. In this example, the USB SuperSpeed signals allow the dock to work as a USB hub with a host that does not support PD or DisplayPort.

A more-complex dock can be designed that multiplexes SSTx1/SSRx1 to be either the USB SuperSpeed channel or to support an additional two lanes of DisplayPort. Such a dock is outside the scope of this example, but is not precluded by this Standard's support for Alternate Modes.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

A.3.3 Solution

The host system and USB dock with DisplayPort use USB-PD VDMs to communicate and take advantage of advanced features.

The flow of messages from the point where the USB dock with DisplayPort is plugged into the USB Type-C Receptacle, plug orientation has been determined, and host/device relationship established is as follows:

- Host system does not support PD
  - Host will discover the USB hub and operate as it would when connected to a USB hub.
  - After tAMETimeout, the Host will see the *USB r2.0* device interface with *USB Billboard r1.0* and report to the user that an unsupported device has been connected, identifying the type of device from the USB descriptor information presented in USB Billboard.
- Host system supports PD and advanced functions, but does not support the USB dock with DisplayPort
  - Host will discover the USB hub and operate as it would for a USB hub.
  - Host may look for other advanced functions, but does support the USB dock with DisplayPort.
  - Because the USB dock with DisplayPort is not supported, after tAMETimeout the host will see the *USB r2.0* device interface with *USB Billboard r1.0* and will report to the user that an unsupported device has been connected, identifying the type of device from the USB descriptor information presented in USB Billboard.
- Host system supports the USB dock with DisplayPort
  - Host looks for advanced functions by discovering whether the device reports SVIDs for advanced functions. The host then discovers a USB dock with DisplayPort by way of responses to USB-PD messages, using DP_SID.
  - Because a usable USB dock with DisplayPort is discovered, the host enters DisplayPort Mode, signals the dock to enter DisplayPort Mode, then re-configures the USB Type-C Receptacle to output DisplayPort signals on SSTx2/SSRx2 on the connector. The USB dock with DisplayPort cancels its tAMETimeout on entry to DisplayPort Mode. The host sends USB SuperSpeed signals on SSTx1/SSRx1 and operates USB through the hub in the dock as normal.

Licensed to Jay Lin (lin_sf@eaton.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

This example uses the basic extensions when docked that do the following:

Base Set VDMs used

- Discover SVIDs
- Discover Modes (DP_SID)
- Enter Mode (DP_SID, Alternate Mode offset)
- Exit Mode (DP_SID, Alternate Mode offset)

Process Outline

The host functions in its role as a DFP_U.

1 DFP_U detects presence of a device (CC pins) and connector orientation.

2 DFP_U applies 5V to VBUS and VCONN.

3 DFP_U discovers USB hub in the dock with DisplayPort and enumerates the hub and the attached USB devices (if any).

4 DFP_U uses PD to make power contract with the dock device.

   a DFP_U runs Discover SVIDs process.

   b Sends Discover SVIDs message.

5 Receives Response message with list of SVIDs for which the dock device has modes.

6 DFP_U runs Discover Modes process.

   a Sends Discover Modes VDM for each of the SVIDs previously discovered that it supports, including DP_SID.

   b Receives Response message with a list of modes associated with DP_SID.

   c If a USB dock with DisplayPort Mode is not discovered – remains in USB Mode, discovers the *USB Billboard r1.0* after tAMETimeout, and informs user – done.

   d Else.

7 DFP_U runs Enter Mode process.

   a Sends Enter Mode VDM with DP_SID and offset to mode.

   b Receives Response:

      1 Responder ACK – DFP_U is now attached to the USB dock with DisplayPort, running in DisplayPort Mode and USB Configuration.

      2 Responder NAK – DFP_U returns to the default *USB r3.1* Configuration.

8 DFP_U discovers Adaptor status, then waits until the Adaptor reports that it has a connected display.

Licensed to Jay Lin (jinsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

9 DFP_U runs the DisplayPort Configure process.

   a Places signals to be re-configured into the Safe state.

   b Sends a DisplayPort Configure command to the Adaptor.

   c Receives Responder ACK response.

   d Configures signals to use DisplayPort; host is now supporting native DisplayPort signaling on the dock's DisplayPort connector in combination with USB SuperSpeed signaling.

10 If the DFP_U receives a Status update from the device that requires a change in configuration, the host re-runs the Configure Process (possibly re-configuring the interface back into USB Configuration).

11 DFP_U remains in this mode until:

   a The host is removed from the dock.

   b Explicitly exited by an Exit Alternate Mode VDM.

   c Hard Reset on PD.

   d Power is removed from the host.

The USB dock with DisplayPort is required to follow and/or not interfere with ALL rules for normal PD operation.

Licensed to Jay Lin (linsp@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.

B Main Contribution History

Table B-1: Main Contributors to *DP Alt on USB Type-C v1.0*

Company	Name	Contribution
Advanced Micro Devices, Inc.	Jim Hunkins	Technical Content
Analogix Semiconductor, Inc.	Mehran Badii	Technical Content
	Greg Stewart	Technical Content
Apple, Inc.	Sree Anantharanan	Technical Content
	Bill Cornelius	Technical Content
	Will Ferry	Technical Content
	Derek Iwamoto	Technical Content
	Souvik Mukherjee	Technical Content
	Anil Pannikkat	Technical Content
	Bob Ridenour	Technical Content
	Colin Whitby-Strevens	Subgroup Document Editor
Broadcom Corporation	Chris Pasqualino	Technical Content
Chrontel, Inc.	David Tsai	Technical Content
Foxconn Electronics	Glenn Moore	Technical Content
	Steve Sedio	Technical Content
Google, Inc.	David Schneider	Technical Content
Intel Corporation	Simon Ellis	Subgroup Leader
	Mike Hamann	Technical Content
	Christine Krause	Technical Content
	Robert “Jamie” Johnston	Technical Content
I-PEX	Greg Young	Technical Content
JAE Electronics, Inc.	Mark Saubert	Technical Content
	Toshio Shimoyama	Technical Content
MegaChips Technology America	Alan Kobayashi	DisplayPort Task Group Chair, Technical Content
NVIDIA Corporation	David Stears	Technical Content
Parade Technologies, Ltd.	Craig Wiley	Subgroup Co-leader
ROHM Semiconductor	Vijendra Kuroodi	Technical Content
Semtech Corporation	Chi Tai Hong	Technical Content
TE Connectivity, Ltd.	Jim McGrath	Technical Content
	Scott Shuey	Technical Content
Texas Instruments, Inc.	Richard Hubbard	Technical Content
	Yoon Lee	Technical Content
VTM Group	Bob Crepps	Technical Content

Licensed to Jay Lin (jinsong@realtek.com)

Copyright © 2014 – 2015 Video Electronics Standards Association. All rights reserved.